National Comprehensive Cancer Network®

NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®)

Antiemesis

Version 2.2026 — June 24, 2026

NCCN Guidelines Navigator™

NCCN Guidelines for Patients®

NCCN recognizes the importance of clinical trials and encourages participation when applicable and available. Trials should be designed to maximize inclusiveness and broad representative enrollment.

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Find an NCCN Member Institution: https://www.nccn.org/home/member-institutions.

NCCN Categories of Evidence and Consensus: All recommendations are category 2A unless otherwise indicated.

See NCCN Categories of Evidence and Consensus.

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Updates in Version 2.2026 of the NCCN Guidelines for Antiemesis include:

PEDAE-12
• Footnote removed: Benzodiazepines should be avoided in patients ≤12 years of age unless being treated for curativeintent. (Also page PEDAE-15)

Updates in Version 1.2026 of the NCCN Guidelines for Antiemesis include:

General:
• Option removed throughout guidelines: Dolasetron 100 mg PO once

Principles of Emesis Control for the Patient with Cancer
AE-1
• Fourth bullet, new tertiary bullet added: Less than 7 hours of sleep the night before chemotherapy.
• Fifth bullet, new sub-bullet added: Supplements/herbal products.
• New footnote a added: Refer to the NCCN Distress Thermometer and Problem List, which includes social determinants of health. See NCCN Guidelines for Distress Management (DIS-A). (Also page AE-2)

Adult Anticancer Therapy-Induced Emesis
AE-3
• High emetic risk:
  ▸ Revised: Melphalan ~~≥140 mg/m²~~
  ▸ Moved from Moderate emetic risk: Busulfan
  ▸ Moved from Low emetic risk on AE-4: Thiotepa
• Moderate emetic risk:
  ▸ Removed:
    ◊ Amifostine >300 mg/m²
    ◊ Melphalan <140 mg/m²
  ▸ New agent added: Treosulfan
• Text under table revised: Table framework is based on the *clinical experience of Panel members and on the* emetogenicity classifications described in the following publications: . .(Also pages AE-4, AE-5, AE-9, AE-10, and AE-11)
• New footnote h added: These agents should be considered highly emetogenic chemotherapy (HEC) when used in the hematopoietic cell transplant (HCT) setting.
• Footnote k revised: Risk of emesis and nausea is highest with the initial infusion *and has been noted in those patients with an intact stomach. Adjusting or pausing the infusion along with the use of acid-suppressing prophylaxis may be needed.* Based upon patient tolerability, providers may consider a de-escalation of emetic prophylaxis with subsequent infusions. Shah MA, et al. Nat Med 2023;29:2133-2141; Shitara K, et al. Lancet 2023;401:1655-1668. Erratum in: Lancet 2023;402:290. Erratum in: Lancet 2024;403:30. *Klempner SJ, et al. ESMO Gastro Oncol 2025;7:100131. Narita Y, et al. Cancers (Basel) 2025;17:1996.*

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Updates in Version 1.2026 of the NCCN Guidelines for Antiemesis include:

AE-4
- Low emetic risk:
  - New agents added:
    - ◊ Afamitresgene autoleucel
    - ◊ Amivantamab and hyaluronidase-lpuj
    - ◊ Linvoseltamab-gcpt
    - ◊ Obecabtagene autoleucel
    - ◊ Telisotuzumab vedotin-tllv
    - ◊ Zanidatamab-hrii
    - ◊ Zenocutuzumab-zbco
  - Removed: Amifostine >300 mg/m^2

AE-5
- Minimal emetic risk, new agents added:
  - Cosibelimab-ipdl
  - Pembrolizumab and berahyaluronidase alfa-pmph
  - Penpulimab-kcqx
- Footnote removed: For some low emetic risk agents, factors related to dosing schedule (particularly continuous dosing) and clinical experience suggest routine premedication is not required. An individualized approach is appropriate for whether to premedicate each dose or prescribe antiemetics as needed (PRN).

AE-7A
- Footnote removed: Especially for patients with anticipatory or anxiety-related breakthrough nausea, may consider adding lorazepam 0.5–1 mg PO or IV or sublingual (SL) every 6 hours PRN on days 1–4. Use the lowest effective dose and dosage interval possible. May be administered with or without H2 blocker or PPI if patient exhibits reflux symptoms. See Principles of Emesis Control for the Patient With Cancer (AE-1) and Pharmacologic Considerations for Antiemetic Prescribing (AE-B 3 of 4). (Also for pages AE-8 and AE-16)
- Footnotes revised:
  - Footnote s: Data suggest that a 2.5-mg dose of olanzapine may be efficacious, especially when used as part of a 4-drug regimen. Olanzapine doses of 2.5–5 mg may be less effective than 10 mg when used as part of a 3-drug regimen. Consider lower doses, especially for patients who are older or who are ~~over sedated~~ *at high risk for oversedation*. . .
  - Footnote y: Emerging data and clinical practice suggest dexamethasone doses may be individualized. Higher doses may be considered, especially when an NK1 RA is not given concomitantly. *For patients receiving MEC or HEC, especially those patients with few identifiable CINV risk factors or who are intolerant to corticosteroids, the use of* ~~L~~*l*ower doses, given for shorter durations, *limiting the administration to Day 1,* or even elimination of dexamethasone on Day 1 *and/*or subsequent days (for delayed nausea and emesis prevention) may be acceptable ~~based on patient characteristics~~ *and may not compromise nausea or antiemesis control*. . .
  - Footnote bb: A 3-drug prophylactic regimen (E or F) is recommended for select patients with additional patient-related risk factors (AE-1), are receiving an agent that may be considered highly emetogenic in certain patients (AE-3, footnote j), or for whom previous treatment with a corticosteroid + 5-HT3 RA alone *(option D)* was ineffective.
- Footnote z added to this page: Use of corticosteroid premedications should be avoided with cellular therapies. Clinicians may wish to consider a dexamethasone-sparing approach with ICI therapy as well. See Pharmacologic Considerations for Antiemetic Prescribing (AE-B). (Also page AE-15)

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Updates in Version 1.2026 of the NCCN Guidelines for Antiemesis include:

AE-8
- Following Low:
  - Top statement revised: Start before anticancer therapy *and continue daily on each day of anticancer therapy* (order does not imply preference):~~-Repeat daily for multiday doses of anticancer therapy~~
  - Bullet removed: Metoclopramide 10–20 mg PO/IV once or
- Last column revised: Breakthrough Treatment ~~for Anticancer Therapy-Induced Nausea/Vomiting~~ (AE-13)

AE-9
- New page created: Adult Oral Anticancer Agents – Emetogenic Potential and Emesis Prevention, Moderate to High Emetic Risk (Prophylaxis Required)
  - Moderate to high emetic risk (≥30% frequency of emesis): Prophylaxis required on days of PO anticancer agent administration table moved to this page.
  - Following Moderate to high emetic risk, new column added: Prophylaxis required.
    - ◊ Following Prophylaxis required, column revised: Start before anticancer therapy and continue daily *on each day of anticancer therapy* (order does notimply preference):

AE-10
- New page created: Adult Oral Anticancer Agents – Emetogenic Potential and Emesis Prevention, Moderate to High emetic risk (PRN Dosing Initially Appropriate)
  - Moderate to high emetic risk (≥30% frequency of emesis): As needed (PRN) dosing is initially appropriate on days of PO anticancer agent administration table moved to this page.
    - ◊ New agents added:
      - – Avutometinib
      - – Defactinib
      - – Mirdametinib
      - – Sunvozertinib
      - – Taletrectinib
  - Following ~~Minimal to low emetic risk and~~ Moderate to high ~~(PRN dosing initially appropriate)~~ *emetic risk*, column revised: PRN ~~recommended~~ *dosing initially appropriate*.
    - ◊ Following PRN *dosing initally appropriate*:
      - – Column revised: ~~Nausea/vomiting (~~Breakthrough Treatment *frequently required* ~~for Anticancer Therapy-Induced Nausea/Vomiting [~~*(*AE-13*)*~~] and consider escalating to a higher level of antiemetic prevention [ie, Moderate-High and consider prophylaxis] for the next cycle)~~
        - ▪ Following Breakthrough Treatment frequently required, column revised: *Begin prophylactic dosing before future doses of* anticancer therapy and continue daily *on each day of anticancer therapy* (order does not imply preference):
      - – New column added: Breakthrough Treatment not frequently required.
        - ▪ New option added: Continue PRN dosing.

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AE-11
- Minimal to low emetic risk, new agents added:
  - Ensartinib
  - Imlunestrant
  - Inavolisib
  - Lazertinib
  - Revumenib
  - Vimseltinib
  - Vorasidenib
  - Zongertinib

AE-12
- New page added: Adult Oral Anticancer Agents – Emesis Prevention, Minimal to Low Emetic Risk

AE-13
- Following Any nausea/vomiting:
  - New bullets added:
    - ◊ Anticholinergics
    - ◊ Dopamine-receptor antagonists
  - Sub-bullet revised: Dexamethasone ~~12~~ 4–8 mg PO/IV daily
- Footnote removed: When not used as part of the acute and delayed emesis prevention regimen.
- New footnote ll added: While no clinically significant drug-drug interactions with short-term use of antiemetics have emerged to date in randomized clinical trials, the Panel would discourage combining multiple dopamine-receptor antagonists as this could increase the risk of extrapyramidal reactions. When used for breakthrough CINV, a single drug from this class could be utilized.
- Footnote mm revised: Especially for patients ~~with anticipatory or~~ *where* anxiety *may be contributing to* ~~-related~~ breakthrough ~~nausea~~ *CINV*~~, may consider adding lorazepam 0.5–1 mg PO or IV or SL every 6 hours PRN on days 1–4. Use the lowest effective dose and dosage interval possible. May be administered with or without H2 blocker or PPI if patient exhibits reflux symptoms~~. . .

AE-14
- Moderate emetic risk, Agent, first bullet revised: ~~Iobenguane iodine-131~~ *I-131 MIBG (Iodine-131 Metaiodobenzylguanidine)*

AE-15
- Following Anticancer therapy and RT (including TBI), column revised: See emesis prevention for chemotherapy-induced nausea/vomiting*:*
  - *Parenteral Emetic Risk:* ~~(~~High [*(*AE-6*)*], Moderate [*(*AE-7*)*], Low and Minimal [*(*AE-8*)*]~~, and~~
  - Oral *Emetic Risk: Moderate to High (Prophylaxis Required [AE-9] and PRN Dosing Initially Appropriate [AE-10]), Minimal to Low (AE-12)*

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Updates in Version 1.2026 of the NCCN Guidelines for Antiemesis include:

AE-16
- Following Anticipatory nausea/vomiting, last bullet revised: Olanzapine 2.5–5 mg PO ~~as a single dose the night~~ *once at bedtime prior to the first dose of* ~~before~~ anticancer therapy ~~begins~~.
- Far right column revised: ~~Emesis Prevention~~ *See emesis prevention for chemotherapy-induced nausea/vomiting:*
  - *Parenteral Emetic Risk: High (AE-6), Moderate (AE-7), Low and Minimal (AE-8)*
  - *Oral Emetic Risk: Moderate to High (Prophylaxis Required [AE-9] and PRN Dosing Initially Appropriate [AE-10]), Minimal to Low (AE-12)* ~~and~~
  - Breakthrough Treatment ~~for Adult Anticancer Therapy-Induced Nausea/Vomiting (Antiemesis Table of Contents~~ *AE-13*).

Principles of Managing Multiday Emetogenic Chemotherapy Regimens

AE-A (2 of 3)
- Corticosteroids, first bullet revised: *When using corticosteroids as part of an antiemetic regimen,* dexamethasone should be administered once daily in the morning . .
- NK1 RAs, new bullet added: Lower-level evidence exists for using netupitant/palonosetron weekly with weekly cisplatin.

AE-A (3 of 3)
- New reference 5 added: Zhao Y, Yang Y, Gao F, et al. A multicenter, randomized, double-blind, placebo-controlled, phase 3 trial of olanzapine plus triple antiemetic regimen for the prevention of multiday highly emetogenic chemotherapy-induced nausea and vomiting (OFFER study). EClinicalMedicine. 2022;55:101771.

Pharmacologic Considerations for Antiemetic Prescribing

AE-B (1 of 6)
- 5-HT3 RAs:
  - First bullet revised: The U.S. Food and Drug Administration (FDA) recommends a maximum of 16 mg for a single dose of IV ondansetron to prevent prolongation of the QT interval of the electrocardiogram (ECG). ~~Dolasetron may increase the QT interval in a dose-dependent fashion.~~ *Clinically significant prolongation of the QT interval has not been reported with palonosetron.*
  - Fourth bullet, first sub-bullet revised: ~~After receiving Palonosetron, Granisetron transdermal patch, or extended-release injection, breakthrough 5-HT3 RAs play a limited role in the delayed infusion period and breakthrough antiemetic should focus on a different mechanism of action.~~ *If palonosetron, granisetron transdermal patch, or granisetron extended-release injection was administered on Day 1 for acute prophylaxis, antiemetics other than 5-HT3 RAs should be used for the prophylaxis of delayed phase CINV or the treatment of breakthrough CINV.*
- New heading added: Anticholinergics
  - First bullet, first sub-bullet revised: *Scopolamine causes central nervous system (*CNS*)* depression; use caution in patients at risk for falls (eg, those who are older, debilitated, frail).

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Updates in Version 1.2026 of the NCCN Guidelines for Antiemesis include:

AE-B (2 of 6)

- New heading added: Atypical antipsychotic:
  - First bullet:
    - ◊ New sub-bullets added:
      - – Olanzapine blocks multiple receptor types, including serotonin, histamine, muscarinic, alpha-adrenergic, and dopamine. In clinical trials, when olanzapine was used as part of a prophylactic regimen for CINV, no clinically significant drug-drug interactions emerged when a dopamine-receptor antagonist was additionally used as a breakthrough agent. See AE-B 4 of 7.
      - – Olanzapine may counteract the effects of dopamine-receptor agonist drugs such as pramipexole, ropinirole, carbidopa-levodopa, and others.
    - ◊ Fifth sub-bullet revised: *Olanzapine causes* CNS depression; use olanzapine with caution or consider a lower dose in patients at risk for falls (eg, those who are older, debilitated, frail) or at risk for orthostatic hypotension.
- Benzodiazepines,
  - First bullet revised: *Lorazepam causes* CNS depression; use caution in patients at risk for falls (eg, those who are older, debilitated, frail) ~~or in patients at risk for dependence~~.
  - Fourth bullet:
    - ◊ Sub-bullet removed: For patients with anticipatory or anxiety-related breakthrough nausea, may consider adding lorazepam 0.5–1 mg PO or IV or SL every 6 hours PRN on Days 1–4.
    - ◊ New sub-bullet added: Consider to prevent anticipatory CINV (AE-16) or where anxiety may be contributing to breakthrough CINV (AE-13).
  - Bullet removed: Benzodiazepines should be avoided in patients ≤12 years of age unless being treated for curative intent.

AE-B (3 of 6)

- Cannabinoid, first bullet revised: *Dronabinol causes* CNS depression; use caution in patients at risk for falls (eg, those who are older, debilitated, frail).
- Corticosteroids, fifth bullet:
  - First sub-bullet revised: *Emerging data and clinical practice suggest dexamethasone doses may be individualized. Higher doses may be considered, especially when an NK1 RA is not given concomitantly.* For patients receiving MEC or HEC, especially those patients with few identifiable CINV risk factors or who are intolerant to corticosteroids, *the use of lower doses, given for shorter durations,* limiting the administration *to Day 1, or even elimination* of dexamethasone ~~to on~~ Day 1 *and/or subsequent days (for delayed nausea and emesis prevention)* ~~only is an option that may not be associated with a significant reduction in antiemetic control~~ *may be acceptable and may not compromise nausea or antiemesis control. If dexamethasone is eliminated on Day 1 or subsequent days, consider the use of an antiemetic combination containing a 5-HT3 RA, NK1 RA, and olanzapine on Day 1, and olanzapine for delayed CINV. See Discussion.*
  - Sub-bullet removed: If patients cannot tolerate dexamethasone, consider replacing with olanzapine.

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Updates in Version 1.2026 of the NCCN Guidelines for Antiemesis include:

AE-B (4 of 6)

- New heading added: Dopamine-Receptor Antagonists
  - New bullets added:
    - ◊ While no clinically significant drug-drug interactions with short-term use of antiemetics have emerged to date in randomized clinical trials, the Panel would discourage combining multiple dopamine-receptor antagonists as this could increase the risk of extrapyramidal reactions. When used for breakthrough CINV (AE-13), a single drug from this class could be utilized.
    - ◊ Dopamine-receptor antagonists may counteract the effects of dopamine-receptor agonist drugs such as pramipexole, ropinirole, carbidopa-levodopa, and others.
    - ◊ Use caution in patients at risk for falls (eg, those who are older, debilitated, frail).
    - ◊ Monitor for dystonic reactions
  - Haloperidol, sub-bullets removed:
    - ◊ CNS depression: use caution in patients at risk for falls (eg, those who are older, debilitated, frail).
    - ◊ Monitor for dystonic reactions.
  - Metoclopramide:
    - ◊ Sub-bullet removed: Use caution in patients at risk for falls (eg, those who are older, debilitated, frail).
    - ◊ Second sub-bullet revised: May cause tardive dyskinesia (TD)~~; monitor for dystonic reactions~~.
      - – First tertiary bullet revised: The risk increases with increasing cumulative dose and duration of treatment*, and with concurrent multiple antidopaminergic agents*.
  - Phenothiazines:
    - ◊ Sub-bullets removed:
      - – CNS depression: use caution in patients at risk for falls (eg, those who are older, debilitated, frail).
      - – Monitor for dystonic reactions.
    - ◊ First sub-bullet revised: When administered parenterally, promethazine may cause severe tissue injury*;* ~~and~~ these routes were removed from the guidelines previously. Only PO or rectal administration of promethazine is recommended *in adult patients. IV administration via a central line only is a recommended option for pediatric patients ≥2 years with appropriate dilution.*
    - ◊ Fourth sub-bullet, tertiary bullet revised: ~~Promethazine~~ *Prochlorperazine* has more ~~histamine blockade than~~ *dopamine-blocking activity and less anticholinergic activity compared to* ~~prochlorperazine~~ *promethazine*; ~~and is~~ therefore, *prochlorperazine's antiemetic activity is associated with less sedation* ~~more sedating~~.

Continued

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Updates in Version 1.2026 of the NCCN Guidelines for Antiemesis include:

AE-B (6 of 6)
- New references added:
  - Reference 1: Celio L, Niger M, Ricchini F, Agustoni F. Palonosetron in the prevention of chemotherapy-induced nausea and vomiting: an evidence-based review of safety, efficacy, and place in therapy. Core Evid 2015;10:75-87.
  - Reference 19: Radhakrishnan V, Venkatakrishnan K, Perumal Kalaiyarasi J, et al. Dexamethasone free antiemetic prophylaxis for highly emetogenic chemotherapy: A double-blind, phase III randomized controlled trial (CINV POD study). JCO Glob Oncol 2024;10:e2300301.
  - Reference 22: Reilly JG, Ayis SA, Ferrier IN, et al. QTc-interval abnormalities and psychotropic drug therapy in psychiatric patients. Lancet 2000;355:1048-1052.
  - Reference 23: Ernst AA, Weiss SJ, Park S, et al. Prochlorperazine versus promethazine for uncomplicated nausea and vomiting in the emergency department: a randomized, double-blind clinical trial. Ann Emerg Med 2000;36:89-94.

Principles for Managing Breakthrough Emesis

AE-C (1 of 2)
- Bullets revised:
  - Fifth bullet: Multiple concurrent agents *from a different drug class*, perhaps in alternating schedules or by alternating routes, may be necessary. ~~Dopamine antagonists (eg, phenothiazines, olanzapine, metoclopramide, haloperidol), corticosteroids, *cannabinoids,* and agents such as lorazepam may be required.~~
  - Eight bullet, fifth sub-bullet: Add other concomitant antiemetics *from a different drug class* ~~(eg, dopamine antagonists such as metoclopramide or haloperidol),~~ if applicable.
  - Ninth bullet: Consider ~~antacid~~ *acid-suppressive* therapy *(H_2 blocker or PPI)* if patient has dyspepsia ~~(H_2 blocker or PPI)~~.
  - Tenth bullet: Topical capsaicin may be considered as an adjunct option*, in addition to guideline compliant antiemetic regimens,* to reduce delayed and extended CINV.
- New bullet added: Cannabinoids may be considered as an adjunct option, in addition to guideline-adherent antiemetic regimens, to prevent acute and delayed CINV

AE-C (2 of 2)
- New references added:
  - Reference 1: Grimison P, Mersiades A, Kirby A, et al. Oral cannabis extract for secondary prevention of chemotherapy-induced nausea and vomiting: Final results of a randomized, placebo-controlled, phase II/III trial. J Clin Oncol 2024;42:4040-4050.
  - Reference 2: Braun IM, Bohlke K, Abrams DI, et al. Cannabis and cannabinoids in adults with cancer: ASCO guideline. J Clin Oncol 2024;42:1575-1593.

Pediatric Antiemesis

PEDAE-1 through PEDAE-15
- New pediatric antiemesis recommendations added.

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PRINCIPLES OF EMESIS CONTROL FOR THE PATIENT WITH CANCER[a]

- Prevention of nausea/vomiting is the goal.
  - The risk of nausea/vomiting (acute ≤24 hours vs. delayed nausea >24 hours) for persons receiving anticancer agents of high and moderate emetic risk lasts for at least 3 days for high and 2 days for moderate after the last dose of anticancer agents. Patients need to be protected throughout the full period of risk.
- Oral (PO) and parenteral serotonin receptor antagonists (5-HT3 RAs) have equivalent efficacy when used at the appropriate doses and intervals.
- Consider the toxicity of the specific antiemetic(s). See Pharmacologic Considerations for Antiemetic Prescribing (AE-B).
- Choice of antiemetic(s) used should be based on the emetic risk of the therapy, prior experience with antiemetics, and patient factors. Continuous infusion may make an agent less emetogenic. The emetic risk is expected to be the same for biosimilars as for the parent compound unless otherwise noted.
  - Patient risk factors for anticancer agent-induced nausea/vomiting include:
    - ◊ Younger age
    - ◊ Female sex
    - ◊ Previous history of anticancer agent-induced nausea and vomiting (chemotherapy-induced nausea and vomiting [CINV])
    - ◊ Little or no previous alcohol use
    - ◊ Prone to motion sickness
    - ◊ History of morning sickness during pregnancy
    - ◊ Anxiety/high pretreatment expectation of nausea
    - ◊ Less than 7 hours of sleep the night before chemotherapy
- There are other potential causes of nausea or emesis in patients with cancer. These may include:
  - Partial or complete bowel obstruction
  - Vestibular dysfunction
  - Brain metastases
  - Electrolyte imbalance: hypercalcemia, hyperglycemia, or hyponatremia
  - Uremia
  - Concomitant drug treatments, including opioids
  - Supplements/herbal products
  - Gastroparesis: tumor or chemotherapy (eg, vincristine) induced or other causes (eg, diabetes)
  - Excessive secretions (eg, seen in patients with head and neck cancers)
  - Malignant ascites
  - Psychophysiologic:
    - ◊ Anxiety
    - ◊ Anticipatory nausea/vomiting
  - Cannabinoid hyperemesis syndrome
  - Rapid opioid withdrawal
  - Pancreatitis
  - Dysmotility
  - Concomitant radiation therapy (RT), especially total body irradiation (TBI) and RT directed at the abdomen or brain

[a] Refer to the NCCN Distress Thermometer and Problem List, which includes social determinants of health. See NCCN Guidelines for Distress Management (DIS-A).

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PRINCIPLES OF EMESIS CONTROL FOR THE PATIENT WITH CANCER[a]

- For uses of antiemetics for nausea/vomiting that are not related to radiation and/or anticancer therapy, see NCCN Guidelines for Palliative Care.
- For multi-drug regimens, select antiemetic therapy based on the drug with the highest emetic risk. See Adult Parenteral Anticancer Agents – Emetogenic Potential (AE-3 through AE-5), and see Adult Oral Anticancer Agents – Emetogenic Potential (AE-9).
- Antiemetic regimens added to a patient's anticancer agents may have a potential risk for drug-drug interactions. However, no clinically significant drug-drug interactions have emerged to date in randomized clinical trials of anticancer agents with antiemetics. The Panel feels, given a short duration of use (<4 days; not chronic use) of these prophylactic antiemetic regimens, they would not result in clinically relevant interactions with anticancer agents. However, in all situations where medications are prescribed, clinicians must balance the benefits and risks for each patient.
- Consider using an H_2 blocker or proton pump inhibitor (PPI) to prevent dyspepsia, which can mimic nausea.
- Lifestyle measures may help to alleviate nausea/vomiting, such as eating small frequent meals, choosing healthful foods, controlling the amount of food consumed, and eating food at room temperature. A dietary consult may also be useful. See NCI's "Eating Hints: Before, During, and After Cancer Treatment" (https://www.cancer.gov/publications/patient-education/eating-hints).
- While anticancer agents or RT-induced nausea and vomiting can significantly impact a patient's quality of life and lead to poor outcomes, providers must be aware of the potential for the overuse of prophylactic antiemetics, especially for anticancer therapy with minimal and low emetic risks, which may expose the patient to potential adverse effects from antiemetic drugs and pose an undue economic burden. Guideline adherence is always encouraged.
- When planning an antiemetic regimen, the health literacy of the patient must be considered, taking into account sociocultural differences. Effective provider-patient communication may improve patient satisfaction, adherence, safety, and outcomes. Clinicians need to identify language and literacy barriers and provide appropriate resources (eg, printed material, medication calendars, interpreter services) to help whenever possible.
- The Panel acknowledges that many patients are interested in complementary and alternative medicines (CAM) to treat their cancer or cancer treatment-related side effects. To date, however, there are no high-level, large, randomized clinical trials demonstrating superior benefit for these types of therapies in place of conventional antiemetic drug therapy as recommended in this guideline for the prevention or treatment of CINV in adult patients. Patients who wish to supplement their conventional antiemetic therapy with CAM should consult their provider or pharmacist to ensure safety and appropriateness.
- The Panel recognizes that high-dose chemotherapy in the setting of hematopoietic cell transplant (HCT) is a unique and challenging situation where patients often have multiple risk factors for CINV in addition to intensive chemotherapy. As the majority of conditioning regimens used in the HCT setting are highly emetogenic, the Panel would encourage clinicians to refer to AE-6 as a basis for developing a personalized plan for CINV prevention.
- The Panel is aware of the emerging clinical controversy that dexamethasone, with its broad immunosuppressive effect, could blunt the effect of immune checkpoint inhibitors (ICIs). To date, there are no randomized clinical trials that demonstrate the detriment of using dexamethasone as a prophylactic antiemetic when ICIs are used in combination with chemotherapy. The Panel recognizes some recent clinical trials that have not allowed dexamethasone when used solely as an antiemetic, and other publications that suggest a negative impact on outcomes when steroids are used with ICIs. Clinicians may wish to consider a dexamethasone-sparing approach (AE-B 2 of 6), or use alternative antiemetic agents, such as olanzapine for the purposes of delayed CINV prophylaxis, until more definitive data become available.[b-f]

[a] Refer to the NCCN Distress Thermometer and Problem List, which includes social determinants of health. See NCCN Guidelines for Distress Management (DIS-A).
[b] Petrelli F, et al. Cancers (Basel) 2020;12:546.
[c] Celio L. EMJ 2022;7:60-67.
[d] Tieri J, et al. J Oncol Pharm Pract 2023;29:1661-1666.
[e] Janowitz T, et al. Oncologist 2021;26:269-273.
[f] Paz-Ares L, et al. N Engl J Med 2018;379:2040-2051.

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ADULT PARENTERAL ANTICANCER AGENTS – EMETOGENIC POTENTIAL
HIGH AND MODERATE EMETIC RISK

LEVEL	AGENT		
High emetic risk (>90% frequency of emesis)[g]	• AC combination defined as any chemotherapy regimen that contains an anthracycline and Cyclophosphamide • Busulfan[h] • Carboplatin AUC ≥4 • Carmustine >250 mg/m^2 • Cisplatin • Cyclophosphamide >1500 mg/m^2	• Dacarbazine • Datopotamab deruxtecan-dlnk • Doxorubicin ≥60 mg/m^2 • Epirubicin >90 mg/m^2 • Fam-trastuzumab deruxtecan-nxki • Ifosfamide ≥2 g/m^2 per dose • Mechlorethamine • Melphalan[h]	• Sacituzumab govitecan-hziy • Streptozocin • Thiotepa[h] • Zolbetuximab-clzb[k]
Moderate emetic risk (>30%–90% frequency of emesis)[g]	• Aldesleukin >12–15 million IU/m^2 or 600,000 IU/kg[i] • Bendamustine • Carboplatin[j] AUC <4 • Carmustine[j] ≤250 mg/m^2 • Clofarabine • Cyclophosphamide[j] ≤1500 mg/m^2 • Cytarabine >200 mg/m^2 • Dactinomycin[j] • Daunorubicin[j] • Dinutuximab • Doxorubicin[j] <60 mg/m^2 • Dual-drug liposomal encapsulation of cytarabine and daunorubicin	• Epirubicin[j] ≤90 mg/m^2 • Idarubicin[j] • Ifosfamide[j] <2 g/m^2 per dose • Irinotecan[j] • Irinotecan (liposomal) • Lurbinectedin • Methotrexate[j] ≥250 mg/m^2 • Mirvetuximab soravtansine-gynx • Naxitamab-gqgk • Oxaliplatin[j]	• Romidepsin • Temozolomide • Trabectedin[j] • Treosulfan

Table framework is based on the clinical experience of Panel members and on the emetogenicity classifications described in the following publications: Hesketh PJ, et al. J Clin Oncol 1997;15:103-109; Grunberg SM, et al. Support Care Cancer 2011;19:S43-S47.

[g] Proportion of patients who experience emesis in the absence of effective antiemetic prophylaxis.
[h] These agents should be considered highly emetogenic chemotherapy (HEC) when used in the hematopoietic cell transplant (HCT) setting.
[i] The dose of aldesleukin is administered 3 to 24 hours after lifileucel.
[j] These agents may be highly emetogenic in certain patients.
[k] Risk of emesis and nausea is highest with the initial infusion and has been noted in those patients with an intact stomach. Adjusting or pausing the infusion along with the use of acid-suppressing prophylaxis may be needed. Based upon patient tolerability, providers may consider a de-escalation of emetic prophylaxis with subsequent infusions. Shah MA, et al. Nat Med 2023;29:2133-2141; Shitara K, et al. Lancet 2023;401:1655-1668. Erratum in: Lancet 2023;402:290. Erratum in: Lancet 2024;403:30; Klempner SJ, et al. ESMO Gastro Oncol 2025;7:100131; Narita Y, et al. Cancers (Basel) 2025;17:1996.

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ADULT PARENTERAL ANTICANCER AGENTS – EMETOGENIC POTENTIAL
LOW EMETIC RISK

LEVEL	AGENT		
Low emetic risk (10%–30% frequency of emesis)[g,l]	• Ado-trastuzumab emtansine • Afamitresgene autoleucel[m] • Aldesleukin ≤12 million IU/m^2 • Amivantamab-vmjw • Amivantamab and hyaluronidase-lpuj • Arsenic trioxide • Axicabtagene ciloleucel[m] • Azacitidine • Belinostat • Brentuximab vedotin • Brexucabtagene autoleucel[m] • Cabazitaxel • Carfilzomib • Ciltacabtagene autoleucel[m] • Copanlisib • Cytarabine (low dose) 100 mg/m^2 – 200 mg/m^2 • Docetaxel • Doxorubicin (liposomal) • Elranatamab-bcmm • Enfortumab vedotin-ejfv • Epcoritamab-bysp • Eribulin • Etoposide	• Floxuridine • 5-Fluorouracil • Gemcitabine • Gemtuzumab ozogamicin • Idecabtagene vicleucel[m] • Inotuzumab ozogamicin • Isatuximab-irfc • Ixabepilone • Lifileucel[m] • Linvoseltamab-gcpt • Lisocabtagene maraleucel[m] • Loncastuximab tesirine-lpyl • Methotrexate >50 mg/m^2 – <250 mg/m^2 • Mitomycin • Mitomycin pyelocalyceal solution • Mitoxantrone • Mogamulizumab-kpkc • Mosunetuzumab-axgb • Necitumumab • Obecabtagene autoleucel[m] • Omacetaxine	• Paclitaxel • Paclitaxel-albumin • Pemetrexed • Pentostatin • Polatuzumab vedotin-piig • Pralatrexate • Tafasitamab-cxix • Tagraxofusp-erzs • Talimogene laherparepvec • Tebentafusp-tebn • Telisotuzumab vedotin-tllv • Tisagenlecleucel[m] • Tisotumab vedotin-tftv • Topotecan • Zanidatamab-hrii • Zenocutuzumab-zbco • Ziv-aflibercept

Table framework is based on the clinical experience of Panel members and on the emetogenicity classifications described in the following publications: Hesketh PJ, et al. J Clin Oncol 1997;15:103-109; Grunberg SM, et al. Support Care Cancer 2011;19:S43-S47.

[g] Proportion of patients who experience emesis in the absence of effective antiemetic prophylaxis.
[l] For some low emetic risk agents, factors related to dosing schedule (particularly continuous dosing) and clinical experience suggest routine premedication is not required. An individualized approach is appropriate for whether to premedicate each dose or prescribe antiemetics as needed (PRN).
[m] Corticosteroid antiemetic premedication should be avoided for 3–5 days prior to and 90 days after chimeric antigen receptor (CAR) and tumor-infiltrating lymphocyte (TIL) T-cell therapies. Upon disease progression, corticosteroids may be resumed if needed. Antiemetic regimens used during lymphodepleting chemotherapy regimens should also employ a corticosteroid-sparing approach to antiemetic prophylaxis.

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ADULT PARENTERAL ANTICANCER AGENTS – EMETOGENIC POTENTIAL
MINIMAL EMETIC RISK

LEVEL	AGENT			
Minimal emetic risk (<10% frequency of emesis)[g]	• Alemtuzumab • Asparaginase[n] • Atezolizumab • Atezolizumab and hyaluronidase-tqjs • Avelumab • Betibeglogene autotemcel[m] • Bevacizumab • Bleomycin • Blinatumomab • Bortezomib • Cemiplimab-rwlc • Cetuximab • Cladribine • Cosibelimab-ipdl • Cytarabine <100 mg/m^2 • Daratumumab • Daratumumab and hyaluronidase-fihj • Decitabine • Degarelix • Dexrazoxane • Dostarlimab-gxly • Durvalumab	• Elotuzumab • Fludarabine • Fulvestrant • Glofitamab-gxbm • Goserelin • Histrelin • Imetelstat • Ipilimumab • Lanreotide • Leuprolide • Lovotibeglogene autotemcel[m] • Luspatercept-aamt • Margetuximab-cmkb • Methotrexate ≤50 mg/m^2 • Nelarabine • Nivolumab • Nivolumab and hyaluronidase-nvhy • Nivolumab/relatlimab-rmbw	• Obinutuzumab • Octreotide • Ofatumumab • Panitumumab • Pembrolizumab • Pembrolizumab and Berahyaluronidase alfa-pmph • Penpulimab-kcqx • Pertuzumab • Pertuzumab/trastuzumab and hyaluronidase-zzxf • Ramucirumab • Retifanlimab-dlwr • Rituximab • Rituximab and hyaluronidase • Siltuximab • Sirolimus-albumin	• Talquetamab-tgvs • Tarlatamab-dlle • Teclistamab-cqyv • Temsirolimus • Tislelizumab-jsgr • Toripalimab-tpzi • Trastuzumab • Trastuzumab and hyaluronidase-oysk • Tremelimumab-actl • Triptorelin • Valrubicin • Vinblastine • Vincristine • Vincristine (liposomal) • Vinorelbine

Table framework is based on the clinical experience of Panel members and on the emetogenicity classifications described in the following publications: Hesketh PJ, et al. J Clin Oncol 1997;15:103-109; Grunberg SM, et al. Support Care Cancer 2011;19:S43-S47.

[g] Proportion of patients who experience emesis in the absence of effective antiemetic prophylaxis.
[m] Corticosteroid antiemetic premedication should be avoided for 3–5 days prior to and 90 days after CAR and TIL T-cell therapies. Upon disease progression, corticosteroids may be resumed if needed. Antiemetic regimens used during lymphodepleting chemotherapy regimens should also employ a corticosteroid-sparing approach to antiemetic prophylaxis.
[n] Asparaginase includes calaspargase pegol-mknl, pegasparaginase, asparaginase erwinia chrysanthemi, and asparaginase erwinia chrysanthemi (recombinant)-rywm.

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ADULT PARENTERAL ANTICANCER AGENTS[o] – ACUTE AND DELAYED EMESIS PREVENTION
HIGH EMETIC RISK

DAY 1: Select treatment option A, B, or C **All treatment options are category 1 and should be started before anticancer therapy[p,q,r]**	**DAYS 2, 3, and 4:**
Treatment option A (preferred), use the following combination: 1. Olanzapine 2.5–10 mg PO once daily (bedtime)[s] 2. Neurokinin-1 receptor antagonist (NK1 RA) (choose one): ◊ Aprepitant 125 mg PO once ◊ Aprepitant injectable emulsion 130 mg intravenous (IV) once[t] ◊ Fosaprepitant 150 mg IV once ◊ Fosnetupitant 235 mg/Palonosetron 0.25 mg (available as fixed combination product only) IV once ◊ Netupitant 300 mg/Palonosetron 0.5 mg (available as fixed combination product only) PO once ◊ Rolapitant 180 mg PO once[u] 3. 5-HT3 RA (choose one):[v,w] ◊ Granisetron 10 mg subcutaneous (SQ) once,[x] or 2 mg PO once, or 0.01 mg/kg (max 1 mg) IV once, or 3.1 mg/24-h transdermal patch applied 24–48 h prior to first dose of anticancer therapy ◊ Ondansetron 16–24 mg PO once, or 8–16 mg IV once ◊ Palonosetron 0.25 mg IV once 4. Dexamethasone 12 mg PO/IV once[y,z]	**Treatment option A:** • Olanzapine 2.5–10 mg PO daily (bedtime) on Days 2, 3, and 4[s] • Aprepitant 80 mg PO daily on Days 2 and 3 (if Aprepitant PO is used on Day 1) • ± Dexamethasone 8 mg PO/IV daily on Days 2, 3, and 4[y,z]
Treatment option B, use the following combination:[aa] 1. Olanzapine 2.5–10 mg PO once daily (bedtime)[s] 2. Palonosetron 0.25 mg IV once 3. Dexamethasone 12 mg PO/IV once[y,z]	**Treatment option B:** • Olanzapine 2.5–10 mg PO daily (bedtime) on Days 2, 3, and 4[s]
Treatment option C, use the following combination:[aa] 1. NK1 RA (choose one): ◊ Aprepitant 125 mg PO once ◊ Aprepitant injectable emulsion 130 mg IV once[t] ◊ Fosaprepitant 150 mg IV once ◊ Fosnetupitant 235 mg/Palonosetron 0.25 mg (available as fixed combination product only) IV once ◊ Netupitant 300 mg/Palonosetron 0.5 mg (available as fixed combination product only) PO once ◊ Rolapitant 180 mg PO once[u] 2. 5-HT3 RA (choose one):[v,w] ◊ Granisetron 10 mg SQ once,[x] or 2 mg PO once, or 0.01 mg/kg (max 1 mg) IV once, or 3.1 mg/24-h transdermal patch applied 24–48 h prior to first dose of anticancer therapy ◊ Ondansetron 16–24 mg PO once, or 8–16 mg IV once ◊ Palonosetron 0.25 mg IV once 3. Dexamethasone 12 mg PO/IV once[y,z]	**Treatment option C:** • Aprepitant 80 mg PO daily on Days 2 and 3 (if Aprepitant PO is used on Day 1) • Dexamethasone 8 mg PO/IV daily on Days 2, 3, and 4[y,z]

[z] Use of corticosteroid premedications should be avoided with cellular therapies. Clinicians may wish to consider a dexamethasone-sparing approach with ICI therapy as well. See Pharmacologic Considerations for Antiemetic Prescribing (AE-B).

Footnotes on AE-7A

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ADULT PARENTERAL ANTICANCER AGENTS[o] – ACUTE AND DELAYED EMESIS PREVENTION
MODERATE EMETIC RISK

DAY 1: Select treatment option D, E, or F All treatment options are category 1 and should be started before anticancer therapy[p,q,r]	DAYS 2 and 3:
Treatment option D, use the following combination: 1. 5-HT3 RA (choose one): ◊ Palonosetron 0.25 mg IV once (preferred) ◊ Granisetron 10 mg SQ once[x] (preferred), or 2 mg PO once, or 0.01 mg/kg (max 1 mg) IV once, or 3.1 mg/24-h transdermal patch applied 24–48 h prior to first dose of anticancer therapy ◊ Ondansetron 16–24 mg PO once, or 8–16 mg IV once 2. Dexamethasone 12 mg PO/IV once[y,z]	**Treatment option D:** • Dexamethasone 8 mg PO/IV daily on Days 2 and 3[y,z] OR • 5-HT3 RA monotherapy:[dd] ◊ Granisetron 1–2 mg (total dose) PO daily or 0.01 mg/kg (max 1 mg) IV daily on Days 2 and 3 ◊ Ondansetron 8 mg PO twice daily or 16 mg PO daily or 8–16 mg IV daily on Days 2 and 3
Treatment option E, use the following combination:[bb,cc] 1. Olanzapine 2.5–10 mg PO once daily (bedtime)[s] 2. Palonosetron 0.25 mg IV once 3. Dexamethasone 12 mg PO/IV once[y,z]	**Treatment option E:** • Olanzapine 2.5–10 mg PO daily (bedtime) on Days 2 and 3[s]
Treatment option F, use the following combination:[bb,cc] 1. NK1 RA (choose one): ◊ Aprepitant 125 mg PO once ◊ Aprepitant injectable emulsion 130 mg IV once[t] ◊ Fosaprepitant 150 mg IV once ◊ Fosnetupitant 235 mg/Palonosetron 0.25 mg (available as fixed combination product only) IV once ◊ Netupitant 300 mg/Palonosetron 0.5 mg (available as fixed combination product only) PO once ◊ Rolapitant 180 mg PO once[u] 2. 5-HT3 RA (choose one):[v,w] ◊ Granisetron 10 mg SQ once,[x] or 2 mg PO once, or 0.01 mg/kg (max 1 mg) IV once, or 3.1 mg/24-h transdermal patch applied 24–48 h prior to first dose of anticancer therapy ◊ Ondansetron 16–24 mg PO once, or 8–16 mg IV once ◊ Palonosetron 0.25 mg IV once 3. Dexamethasone 12 mg PO/IV once[y,z]	**Treatment option F:** • Aprepitant 80 mg PO daily on Days 2 and 3 (if aprepitant PO used on Day 1) • ± Dexamethasone 8 mg PO/IV daily on Days 2 and 3[y,z]

[z] Use of corticosteroid premedications should be avoided with cellular therapies. Clinicians may wish to consider a dexamethasone-sparing approach with ICI therapy as well. See Pharmacologic Considerations for Antiemetic Prescribing (AE-B).

Footnotes on AE-7A

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FOOTNOTES FOR ADULT PARENTERAL ANTICANCER AGENTS ACUTE AND DELAYED EMESIS PREVENTION

[o] Adult Parenteral Anticancer Agents – Emetogenic Potential (AE-3).
[p] Antiemetic regimens should be chosen based on the drug with the highest emetic risk as well as patient-specific risk factors.
[q] Principles of Managing Multiday Emetogenic Chemotherapy Regimens (AE-A).
[r] Pharmacologic Considerations for Antiemetic Prescribing (AE-B).
[s] Data suggest that a 2.5-mg dose of olanzapine may be efficacious, especially when used as part of a 4-drug regimen. Olanzapine doses of 2.5–5 mg may be less effective than 10 mg when used as part of a 3-drug regimen. Consider lower doses, especially for patients who are older or who are at high risk for oversedation. Hashimoto H, et al. Lancet Oncol 2020;21:242-249; Bajpai J, et al. Lancet Oncol 2024;25:246-254. See Pharmacologic Considerations for Antiemetic Prescribing (AE-B).
[t] Aprepitant injectable emulsion is a unique formulation of aprepitant and is NOT interchangeable with the IV formulation of fosaprepitant.
[u] Rolapitant has an extended half-life and should not be administered at less than 2-week intervals.
[v] If netupitant/palonosetron or fosnetupitant/palonosetron fixed combination product is used, no further 5-HT3 RA is required.
[w] When used in combination with an NK1 RA, there is no preferred 5-HT3 RA. See Principles of Managing Multiday Emetogenic Chemotherapy Regimens (AE-A).
[x] Granisetron extended-release injection is a unique formulation of granisetron using a polymer-based drug delivery system. This formulation is specifically intended for SQ administration and is NOT interchangeable with the IV formulation. Granisetron extended-release injection has an extended half-life and should not be administered at less than 1-week intervals.
[y] Emerging data and clinical practice suggest dexamethasone doses may be individualized. Higher doses may be considered, especially when an NK1 RA is not given concomitantly. For patients receiving MEC or HEC, especially those patients with few identifiable CINV risk factors or who are intolerant to corticosteroids, the use of lower doses, given for shorter durations, limiting the administration to Day 1 or even elimination of dexamethasone on Day 1 and/or subsequent days (for delayed nausea and emesis prevention) may be acceptable and may not compromise nausea or antiemesis control. If dexamethasone is eliminated on Day 1 or subsequent days, consider the use of an antiemetic combination containing a 5-HT3 RA, NK1 RA, and olanzapine on Day 1, and olanzapine for delayed CINV. See Discussion. Radhakrishnan V, et al. JCO Glob Oncol 2024;10:e2300301.
[z] Use of corticosteroid premedications should be avoided with cellular therapies. Clinicians may wish to consider a dexamethasone-sparing approach with ICI therapy as well. See Pharmacologic Considerations for Antiemetic Prescribing (AE-B).
[aa] If not used previously, consider escalating to a 4-drug regimen (option A) if emesis occurred during a previous cycle of anticancer therapy with a 3-drug regimen (olanzapine-containing regimen B or E or NK1 RA-containing regimen C or F). Olanzapine-containing regimens may be useful for patients with severe nausea. See Principles for Managing Breakthrough Emesis (AE-C).
[bb] A 3-drug prophylactic regimen (E or F) is recommended for select patients with additional patient-related risk factors (AE-1), who are receiving an agent that may be considered highly emetogenic in certain patients (AE-3, footnote j), or for whom previous treatment with a corticosteroid + 5-HT3 RA alone (option D) was ineffective.
[cc] For select patients with additional patient-related risk factors (AE-1) or for whom previous treatment with a 3-drug prophylactic regimen (E or F) was ineffective, or who are receiving moderately emetogenic chemotherapy (MEC) known to be higher risk, a 4-drug regimen (see option A on AE-6) may be considered. Ostwal V, et al. JAMA Netw Open 2024;7:e2426076. Some regimens listed as moderately emetogenic may be highly emetogenic in certain patients (AE-3).
[dd] No further 5-HT3 therapy is required if palonosetron or granisetron extended-release injection is administered, or if granisetron transdermal patch is applied, on Day 1.

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ADULT PARENTERAL ANTICANCER AGENTS[o] – EMESIS PREVENTION
LOW AND MINIMAL EMETIC RISK

Low →

Start before anticancer therapy and continue daily on each day of anticancer therapy (order does not imply preference):[p,q,r]
- **5-HT3 RA[p,q] (select one):**
  - **Granisetron 1–2 mg (total dose) PO once**
  - **Ondansetron 8–16 mg PO once**

or
- **Dexamethasone 8–12 mg PO/IV once[z]**

or
- **Prochlorperazine 10 mg PO/IV once**

→ **Breakthrough Treatment (AE-13)**

Minimal → **No routine prophylaxis** →

[o] Adult Parenteral Anticancer Agents - Emetogenic Potential (AE-3).
[p] Antiemetic regimens should be chosen based on the drug with the highest emetic risk as well as patient-specific risk factors.
[q] Principles of Managing Multiday Emetogenic Chemotherapy Regimens (AE-A).
[r] Pharmacologic Considerations for Antiemetic Prescribing (AE-B).
[z] Use of corticosteroid premedications should be avoided with cellular therapies. Clinicians may wish to consider a dexamethasone-sparing approach with ICI therapy as well. See Pharmacologic Considerations for Antiemetic Prescribing (AE-B).

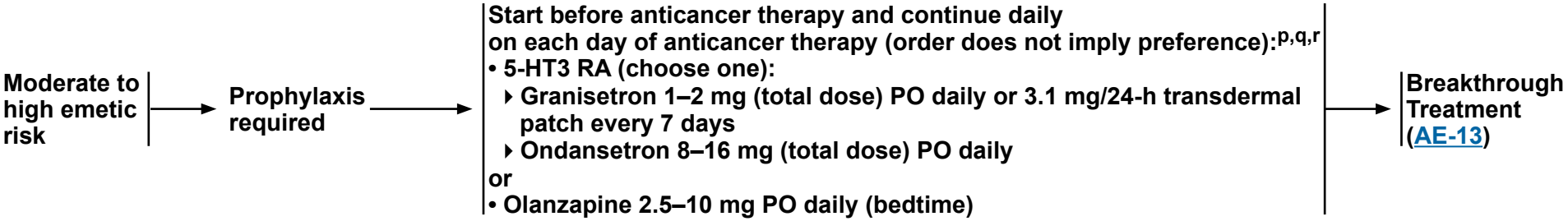
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ADULT ORAL ANTICANCER AGENTS – EMETOGENIC POTENTIAL AND EMESIS PREVENTION[ee]
MODERATE TO HIGH EMETIC RISK (PROPHYLAXIS REQUIRED)

LEVEL	AGENT		
Moderate to high emetic risk[g] (≥30% frequency of emesis): Prophylaxis required on days of PO anticancer agent administration	• Azacitidine[ff] • Busulfan ≥4 mg/day • Ceritinib • Cyclophosphamide ≥100 mg/m^2/day	• Fedratinib • Lomustine (single day) • Midostaurin • Mitotane	• Selinexor[gg] • Temozolomide >75 mg/m^2/day • Temozolomide ≤75 mg/m^2/day with concurrent RT

Table framework is based on the clinical experience of Panel members and on the emetogenicity classifications described in the following publications: Hesketh PJ, et al. J Clin Oncol 1997;15:103-109; Grunberg SM, et al. Support Care Cancer 2011;19:S43-S47.



[g] Proportion of patients who experience emesis in the absence of effective antiemetic prophylaxis.
[p] Antiemetic regimens should be chosen based on the drug with the highest emetic risk as well as patient-specific risk factors.
[q] Principles of Managing Multiday Emetogenic Chemotherapy Regimens (AE-A).
[r] Pharmacologic Considerations for Antiemetic Prescribing (AE-B).
[ee] These antiemetic recommendations apply to PO anticancer therapy only. When combined with IV agents in a combination anticancer therapy regimen, the antiemetic recommendations for the agent with the highest level of emetogenicity should be followed. If multiple PO agents are combined, emetic risk may increase and require prophylaxis.
[ff] If patient is tolerating after first 2 cycles, may change to PRN dosing. Wei AH, et al. N Engl J Med 2020;383:2526-2537.
[gg] Emerging data and clinical practice suggest adding low-dose olanzapine and/or NK1 RA or 5-HT3 RA for nausea prevention.

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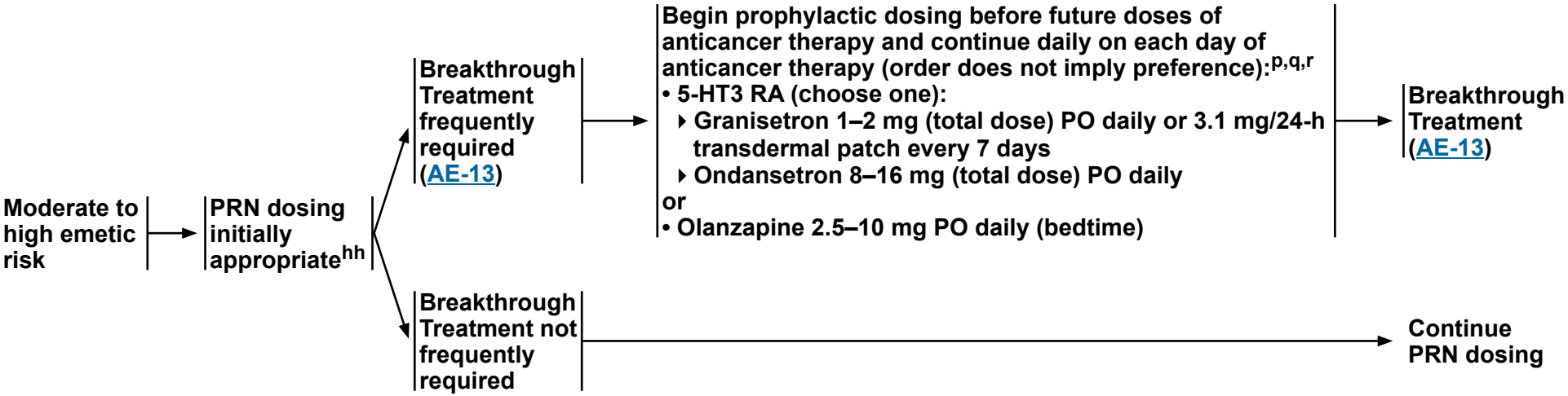
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ADULT ORAL ANTICANCER AGENTS – EMETOGENIC POTENTIAL AND EMESIS PREVENTION[ee]
MODERATE TO HIGH EMETIC RISK (PRN DOSING INITIALLY APPROPRIATE)

LEVEL	AGENT				
Moderate to high emetic risk[g] (≥30% frequency of emesis): As needed (PRN) dosing is initially appropriate on days of PO anticancer agent administration	• Abemaciclib • Adagrasib • Avapritinib • Avutometinib • Binimetinib • Bosutinib >400 mg/day	• Cabozantinib • Crizotinib • Dabrafenib • Defactinib • Elacestrant • Enasidenib	• Encorafenib • Estramustine • Etoposide • Imatinib >400 mg/day • Lenvatinib >12 mg/day • Mirdametinib	• Niraparib • Olaparib • Procarbazine • Rucaparib • Sunvozertinib • Taletrectinib	• Tovorafenib • Trifluridine/tipiracil

Table framework is based on the clinical experience of Panel members and on the emetogenicity classifications described in the following publications: Hesketh PJ, et al. J Clin Oncol 1997;15:103-109; Grunberg SM, et al. Support Care Cancer 2011;19:S43-S47.



[g] Proportion of patients who experience emesis in the absence of effective antiemetic prophylaxis.
[p] Antiemetic regimens should be chosen based on the drug with the highest emetic risk as well as patient-specific risk factors.
[q] Principles of Managing Multiday Emetogenic Chemotherapy Regimens (AE-A).
[r] Pharmacologic Considerations for Antiemetic Prescribing (AE-B).
[ee] These antiemetic recommendations apply to PO anticancer therapy only. When combined with IV agents in a combination anticancer therapy regimen, the antiemetic recommendations for the agent with the highest level of emetogenicity should be followed. If multiple PO agents are combined, emetic risk may increase and require prophylaxis.
[hh] For some moderate to high emetic risk agents, factors related to dosing schedule (particularly continuous dosing for prolonged periods), and clinical experience suggest routine premedication is not required. An individualized approach is appropriate for whether to premedicate each dose or prescribe antiemetics PRN.

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ADULT ORAL ANTICANCER AGENTS – EMETOGENIC POTENTIAL
MINIMAL TO LOW EMETIC RISK

LEVEL	AGENT				
Minimal to low emetic risk[g] (<30% frequency of emesis)	• Abiraterone • Acalabrutinib • Afatinib • Alectinib • Alpelisib • Anastrozole • Apalutamide • Asciminib • Axitinib • Belzutifan • Bexarotene • Bicalutamide • Bosutinib ≤400 mg/day • Brigatinib • Busulfan <4 mg/day • Capecitabine • Capivasertib • Capmatinib • Chlorambucil • Cobimetinib • Cyclophosphamide <100 mg/m^2/day • Dacomitinib • Darolutamide • Dasatinib • Decitabine and cedazuridine • Duvelisib	• Eflornithine • Ensartinib • Entrectinib • Enzalutamide • Erdafitinib • Erlotinib • Everolimus • Exemestane • Fludarabine • Flutamide • Fruquintinib • Futibatinib • Gefitinib • Gilteritinib • Glasdegib • Hydroxyurea • Ibrutinib • Idelalisib • Imatinib ≤400 mg/day • Imlunestrant • Inavolisib • Ivosidenib • Ixazomib • Lapatinib • Larotrectinib • Lazertinib • Lenalidomide • Lenvatinib ≤12 mg/day • Letrozole • Lorlatinib	• Megestrol • Melphalan • Mercaptopurine • Methotrexate • Momelotinib • Neratinib • Nilotinib • Nilutamide • Nirogacestat • Olutasidenib • Osimertinib • Pacritinib • Palbociclib • Pazopanib • Pemigatinib • Pexidartinib • Pirtobrutinib • Pomalidomide • Ponatinib • Pralsetinib • Quizartinib • Regorafenib • Relugolix • Repotrectinib • Revumenib • Ribociclib • Ripretinib • Ruxolitinib	• Selpercatinib • Sonidegib • Sorafenib • Sotorasib • Sunitinib • Talazoparib tosylate • Tamoxifen • Tazemetostat • Temozolomide ≤75 mg/m^2/day • Tepotinib • Thalidomide • Thioguanine • Tivozanib • Topotecan • Toremifene • Trametinib • Tretinoin • Tucatinib	• Vandetanib • Vemurafenib • Venetoclax • Vimseltinib • Vismodegib • Vorasidenib • Vorinostat • Zanubrutinib • Zongertinib

Table framework is based on the clinical experience of Panel members and on the emetogenicity classifications described in the following publications: Hesketh PJ, et al. J Clin Oncol 1997;15:103-109; Grunberg SM, et al. Support Care Cancer 2011;19:S43-S47.

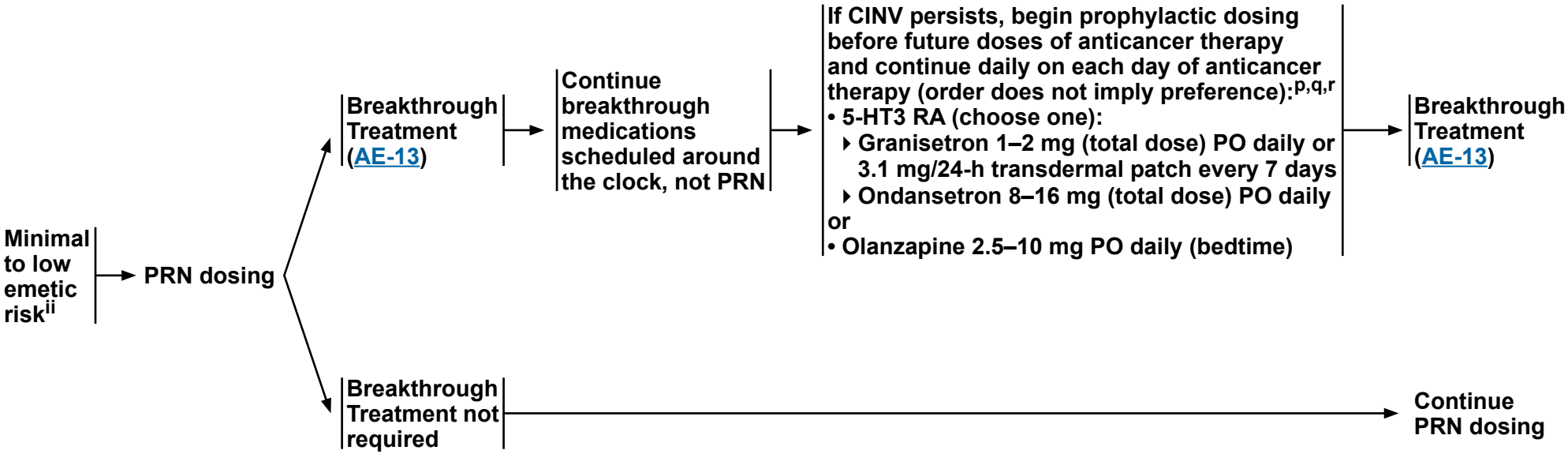
[g] Proportion of patients who experience emesis in the absence of effective antiemetic prophylaxis.

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ADULT ORAL ANTICANCER AGENTS – EMESIS PREVENTION[ee]
MINIMAL TO LOW EMETIC RISK



[p] Antiemetic regimens should be chosen based on the drug with the highest emetic risk as well as patient-specific risk factors.
[q] Principles of Managing Multiday Emetogenic Chemotherapy Regimens (AE-A).
[r] Pharmacologic Considerations for Antiemetic Prescribing (AE-B).
[ee] These antiemetic recommendations apply to PO anticancer therapy only. When combined with IV agents in a combination anticancer therapy regimen, the antiemetic recommendations for the agent with the highest level of emetogenicity should be followed. If multiple PO agents are combined, emetic risk may increase and require prophylaxis.
[ii] Adult Oral Anticancer Agents - Minimal to Low Emetogenic Potential (AE-11).

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BREAKTHROUGH TREATMENT FOR ADULT ANTICANCER THERAPY-INDUCED NAUSEA/VOMITING[q,jj]

Any nausea/vomiting →

The general principle of breakthrough treatment is to add one agent from a different drug class to the current regimen.[r]

- Atypical antipsychotic:
  - Olanzapine 2.5–10 mg PO daily (bedtime) (category 1, preferred)
- Cannabinoid:
  - Dronabinol capsules 5–10 mg, or dronabinol PO solution 2.1–4.2 mg/m², PO 3–4 times daily[kk]
- Anticholinergics:
  - Scopolamine 1.5 mg transdermal patch 1 patch every 72 h
- Dopamine-receptor antagonists:[ll]
  - Haloperidol 0.5–2 mg PO/IV every 4–6 h
  - Metoclopramide 10–20 mg PO/IV every 4–6 h
  - Phenothiazine:
    - ◊ Prochlorperazine 25 mg suppository (supp) PR every 12 h or 10 mg PO/IV every 6 h
    - ◊ Promethazine 25 mg supp PR every 6 h or 12.5–25 mg PO every 4–6 h
- 5-HT3 RA:
  - Granisetron 1–2 mg PO daily or 1 mg PO twice a day (BID) or 0.01 mg/kg (maximum 1 mg) IV daily or 3.1 mg/24-h transdermal patch every 7 days
  - Ondansetron 8 mg PO/IV every 8–12 h (16–24 mg total daily dose)
- Corticosteroid:
  - Dexamethasone 4–8 mg PO/IV daily[z]
- Benzodiazepine:[mm]
  - Lorazepam 0.5–1 mg PO/sublingual (SL)/IV every 6 h

RESPONSE

Nausea and vomiting controlled → Continue breakthrough medications, on a schedule, not PRN

Nausea and/or vomiting uncontrolled → Re-evaluate and consider dose adjustments and/or sequentially add one agent from a different drug class

SUBSEQUENT CYCLES

Consider changing antiemetic therapy to higher level primary treatment for next cycle

[q] Principles of Managing Multiday Emetogenic Chemotherapy Regimens (AE-A).

[r] Pharmacologic Considerations for Antiemetic Prescribing (AE-B).

[z] Use of corticosteroid premedications should be avoided with cellular therapies. Clinicians may wish to consider a dexamethasone-sparing approach with ICI therapy as well. See Pharmacologic Considerations for Antiemetic Prescribing (AE-B).

[jj] Principles for Managing Breakthrough Emesis (AE-C).

[kk] Dronabinol PO solution has greater PO bioavailability than dronabinol capsules; 2.1 mg PO solution = 2.5 mg capsules.

[ll] While no clinically significant drug-drug interactions with short-term use of antiemetics have emerged to date in randomized clinical trials, the Panel would discourage combining multiple dopamine-receptor antagonists as this could increase the risk of extrapyramidal reactions. When used for breakthrough CINV, a single drug from this class could be utilized.

[mm] Especially for patients where anxiety may be contributing to breakthrough CINV. See Principles of Emesis Control for the Patient with Cancer (AE-1) and Pharmacologic Considerations for Antiemetic Prescribing (AE-B).

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ADULT RADIOPHARMACEUTICAL ANTICANCER AGENTS – EMETOGENIC POTENTIAL MODERATE, LOW, AND MINIMAL EMETIC RISK

LEVEL	AGENT
Moderate emetic risk (30%–90% frequency of emesis)	• I-131 MIBG (Iodine-131 Metaiodobenzylguanidine) • Lutetium Lu-177 dotatate[nn,oo,pp]
Low emetic risk (10%–30% frequency of emesis)	• Lutetium Lu-177 vipivotide tetraxetan
Minimal emetic risk (<10% frequency of emesis)	• Radium-223 dichloride • Sodium iodide I-131 • Strontium-89 • Yttrium-90 ibritumomab tiuxetan • Yttrium-90 microspheres

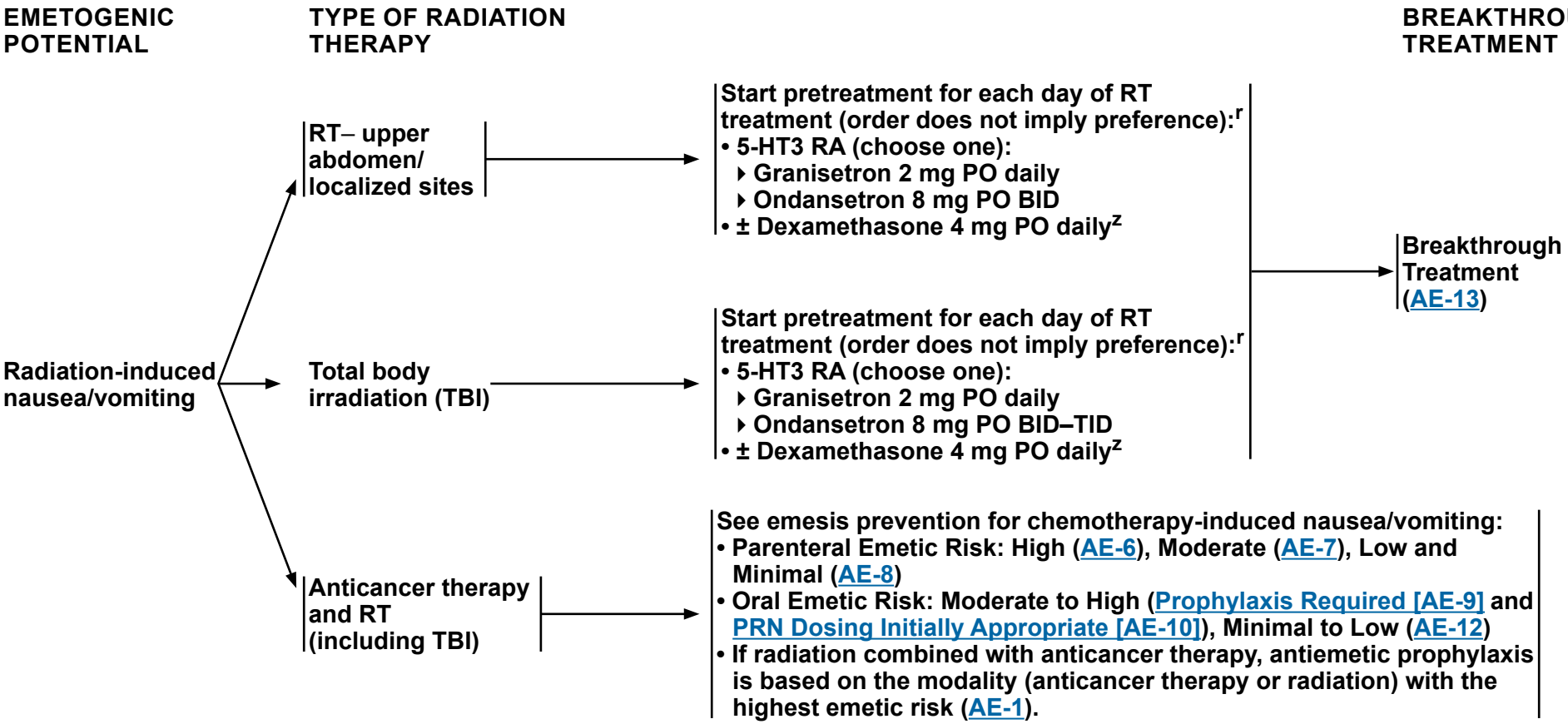
[nn] Nausea associated with therapy derives largely from the amino acid infusion accompanying Lu-177 dotatate treatment; variation in emetogenicity is reported between various compounded and commercially available amino acid products.
[oo] Antiemetics should be administered 30 minutes prior to the start of amino acid infusion.
[pp] Glucocorticoids can induce down-regulation of subtype 2 somatostatin receptors (SST2). The manufacturer recommends avoiding repeated administration of “high-doses” of glucocorticoids during Lu-177 dotatate treatment. The Panel suggests avoiding glucocorticoids as antiemetic prophylaxis, as other agents (5-HT3 RAs and NK1 RAs) are available options. Pivonello R, et al. Front Endocrinol (Lausanne) 2022;12:793262.

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ADULT RADIATION-INDUCED EMESIS – PREVENTION/TREATMENT



[r] Pharmacologic Considerations for Antiemetic Prescribing (AE-B).
[z] Use of corticosteroid premedications should be avoided with cellular therapies. Clinicians may wish to consider a dexamethasone-sparing approach with ICI therapy as well. See Pharmacologic Considerations for Antiemetic Prescribing (AE-B).

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ADULT ANTICIPATORY EMESIS – PREVENTION/TREATMENT

Anticipatory nausea/vomiting →

- **Prevention is key:**
  - **Use optimal antiemetic therapy during every cycle of treatment**
  - **Avoid strong smells that may precipitate symptoms**
- **Behavioral therapy:**
  - **Relaxation/systematic desensitization**
  - **Hypnosis**
  - **Relaxation exercises**
    - ◊ **Guided imagery**
    - ◊ **Progressive muscle relaxation (PMR)**
    - ◊ **Biofeedback**
    - ◊ **Music therapy**
  - **Cognitive distraction**
  - **Yoga (if approved by physician)**
- **Acupuncture/acupressure**
- **Consider anxiolytic therapy:**[r]
  - **Lorazepam 0.5–1 mg PO beginning on the night before treatment and then repeated the next day 1–2 hours before anticancer therapy begins**

  or
  - **Olanzapine 2.5–5 mg PO once at bedtime prior to first dose of anticancer therapy**

→ **See emesis prevention for chemotherapy-induced nausea/vomiting:**

- **Parenteral Emetic Risk: High (AE-6), Moderate (AE-7), Low and Minimal (AE-8)**
- **Oral Emetic Risk: Moderate to High (Prophylaxis Required [AE-9] and PRN Dosing Initially Appropriate [AE-10]), Minimal to Low Emetic Risk (AE-12)**
- **Breakthrough Treatment (AE-13)**

[r] Pharmacologic Considerations for Antiemetic Prescribing (AE-B).

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PRINCIPLES OF MANAGING MULTIDAY EMETOGENIC CHEMOTHERAPY REGIMENS

Summary

- Patients receiving multiday chemotherapy are at risk for both acute and delayed nausea/vomiting based on the emetogenic potential of the individual chemotherapy agents administered on any given day and their sequence. It is therefore difficult to recommend a specific antiemetic regimen for each day, especially since acute and delayed emesis may overlap after the initial day of chemotherapy until the last day of chemotherapy.
- After chemotherapy administration concludes, the period of risk for delayed emesis also depends on the specific regimen and the emetogenic potential of the last chemotherapy agent administered in the regimen.
- Practical issues also need to be considered when designing the antiemetic regimen, taking into account the administration setting (eg, inpatient vs. outpatient); preferred route of administration (parenteral, PO, or transdermal); duration of action of the 5-HT3 RA, NK1 RA, and appropriate associated dosing intervals; tolerability of daily antiemetics (eg, corticosteroids); adherence issues; and individual risk factors.
- The Panel acknowledges that evidence is lacking to support every clinical scenario. Decisions should be individualized for each chemotherapy regimen and each patient. Extensive knowledge of the available clinical data, pharmacology, pharmacodynamics, pharmacokinetics of antiemetics and chemotherapy, and experience with patients (regarding tolerability and efficacy) are all paramount to successfully implementing these guidelines into clinical practice.

References on AE-A 3 of 3

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PRINCIPLES OF MANAGING MULTIDAY EMETOGENIC CHEMOTHERAPY REGIMENS

General Principles

5-HT3 RAs:
- A 5-HT3 RA should be administered prior to the first (and subsequent) doses of moderately emetogenic chemotherapy (MEC) or highly emetogenic chemotherapy (HEC). The frequency or need for repeated administration of the 5-HT3 RA depends on the agent chosen and its mode of administration (parenteral/PO/transdermal).
- Palonosetron:
  - A single IV palonosetron dose of 0.25 mg may be sufficient prior to the start of a 3-day chemotherapy regimen instead of multiple daily doses of another PO or IV 5-HT3 RA.
  - Repeat dosing of palonosetron 0.25 mg IV at 48–72 hours appears to be safe and effective.
  - In terms of efficacy, limited data are available for multiday dosing.[1-3]

Atypical antipsychotics:
- If olanzapine is being used prophylactically as part of the antiemetic regimen, it may be used once daily (prior to chemotherapy or at bedtime) and continued for 2–3 days after chemotherapy for regimens that are likely to cause significant delayed emesis.[4,5]

Corticosteroids:
- When using corticosteroids as part of an antiemetic regimen, dexamethasone should be administered once daily in the morning (either PO or IV) for MEC or HEC, then continued for 2 to 3 days after chemotherapy for regimens that are likely to cause significant delayed emesis.
- Dexamethasone dose may be modified or omitted when the chemotherapy regimen already includes a corticosteroid.

NK1 RAs:
- NK1 RAs may be used for multiday chemotherapy regimens likely to be MEC or HEC and associated with significant risk for delayed nausea and emesis.
- If the PO aprepitant regimen is chosen, limited data exist to support administration of aprepitant on Days 4 and 5 after multiday chemotherapy.
- Data from a small phase III randomized study support the use of aprepitant (125 mg Day 3, 80 mg Days 4–7) with 5-HT3 RA (Days 1–5) and dexamethasone (20 mg Days 1 and 2) in patients with germ cell cancers treated with a 5-day cisplatin-based chemotherapy.[6]
- Several other single-arm, non-randomized studies in MEC and HEC support the use of aprepitant prior to each dose of multiday chemotherapy and for up to 2 days after (up to 7 days total of aprepitant).[7-9]
- Data from two phase II studies support multiday dosing of netupitant/palonosetron fixed combination product in association with BEAM (carmustine, etoposide, and melphalan)/FEAM (fotemustine, etoposide, cytarabine, and melphalan) conditioning regimens prior to HCT.[10,11]
- Lower-level evidence exists for using netupitant/palonosetron weekly with weekly cisplatin.
- An open-label randomized controlled study supports multiday dosing of fosaprepitant.[12]
- Studies investigating repeat dosing of aprepitant injectable emulsion, fosnetupitant, and rolapitant are not available.

References on AE-A 3 of 3

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PRINCIPLES OF MANAGING MULTIDAY EMETOGENIC CHEMOTHERAPY REGIMENS
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PHARMACOLOGIC CONSIDERATIONS FOR ANTIEMETIC PRESCRIBING

To ensure safe and effective treatment with antiemetic therapy, develop a treatment plan with the patient that includes medication access, screening of concomitant medications, goals of therapy, instructions for proper use and side effect management, and adherence assessment. Many of the antiemetic agents contained within this guideline may have multiple potential drug-drug or drug-disease interactions, especially with chronic use. Review the patient medical profile and drug package insert for specific interactions and recommendations.

5-HT3 RAs:

- The U.S. Food and Drug Administration (FDA) recommends a maximum of 16 mg for a single dose of IV ondansetron to prevent prolongation of the QT interval of the electrocardiogram (ECG). Clinically significant prolongation of the QT interval has not been observed with palonosetron.[1]
- Granisetron extended-release injection is a unique formulation of granisetron using a polymer-based drug delivery system. This formulation is specifically intended for SQ administration and is NOT interchangeable with the IV formulation. Granisetron extended-release injection has an extended half-life and should not be administered at less than 1-week intervals.
  - A single SQ dose of 10 mg was found to be non-inferior to a single IV dose of palonosetron 0.25 mg for the prevention of acute and delayed CINV following MEC or HEC when both are used in combination with dexamethasone.[2]
  - A single SQ dose of 10 mg was found to be superior to a single IV dose of ondansetron for the prevention of delayed CINV following HEC when both are used in combination with fosaprepitant and dexamethasone.[3]
- When palonosetron or granisetron extended-release injection is used as part of an antiemetic regimen that does NOT contain an NK1 RA or olanzapine, palonosetron, or granisetron extended-release injection are the preferred 5-HT3 RAs for MEC (AE-7).[2,4]
- Clinical pearl:
  - If palonosetron, granisetron transdermal patch, or granisetron extended-release injection was administered on Day 1 for acute prophylaxis, antiemetics other than 5-HT3 RAs should be used for the prophylaxis of delayed phase CINV or the treatment of breakthrough CINV.
  - Non-sedating; most common side effects are headache and constipation. Optimal effects are seen with scheduled administration, not PRN use. Educate patients regarding constipation and its management.
  - Patients who are known CYP2D6 ultrarapid metabolizers may experience decreased antiemetic response with ondansetron. Consider selecting an alternative 5-HT3.[5,6]

Anticholinergics:

- Scopolamine:
  - Scopolamine causes central nervous system (CNS) depression; use caution in patients at risk for falls (eg, those who are older, debilitated, frail).
  - Clinical pearl:
    - ◊ Consider using when positional changes, movement, or excessive secretions are triggering episodes of nausea/vomiting.

References on AE-B 6 of 6

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PHARMACOLOGIC CONSIDERATIONS FOR ANTIEMETIC PRESCRIBING

Atypical Antipsychotic:

- Olanzapine:
  - Olanzapine blocks multiple receptor types, including serotonin, histamine, muscarinic, alpha-adrenergic, and dopamine. In clinical trials, when olanzapine was used as part of a prophylactic regimen for CINV, no clinically significant drug-drug interactions emerged when a dopamine-receptor antagonist was additionally used as a breakthrough agent. See AE-B 4 of 7.
  - Olanzapine may counteract the effects of dopamine-receptor agonist drugs such as pramipexole, ropinirole, carbidopa-levodopa, and others.
  - Monitor for dystonic reactions.[a]
  - At low doses, studies suggest olanzapine impact on QT interval is unlikely to be clinically relevant; consider monitoring ECG in patients with other significant risk factors for QT prolongation.[7,8]
  - Olanzapine causes CNS depression; use olanzapine with caution or consider a lower dose in patients at risk for falls (eg, those who are older, debilitated, frail) or at risk for orthostatic hypotension.
  - Clinical pearl:
    - ◊ Consider a dose of 2.5–5 mg if the previously administered 5- to 10-mg dose caused excessive sedation.
    - ◊ Consider increasing dose to 5–10 mg if the previously administered 2.5- to 5-mg dose was ineffective.
    - ◊ Data suggest that sedation is most notable on Day 2 and improves over time.
    - ◊ Olanzapine may stimulate appetite; may consider 2.5 mg of olanzapine for patients experiencing concurrent chemotherapy-related anorexia.[9]
    - ◊ Bedtime administration is recommended when possible due to sedation.

Benzodiazepines:

- Lorazepam causes CNS depression; use caution in patients at risk for falls (eg, those who are older, debilitated, frail)
- Clinicians should assess patient risk for substance use disorder and addiction. Continued use may lead to dependence.
- Clinical pearl:
  - Use caution in patients receiving opioids due to increased risk of respiratory depression.
  - Consider to prevent anticipatory CINV (AE-16) or where anxiety may be contributing to breakthrough CINV (AE-13).

[a] Use diphenhydramine 25–50 mg PO/IV either every 4 or every 6 h for dystonic reactions. If allergic to diphenhydramine, use benztropine at 1–2 mg IV or intramuscularly (IM) x 1 dose, followed by PO dose of 1–2 mg daily or BID if needed. May consider using amantadine 100 mg BID–TID as treatment of drug-induced dystonic reactions for those patients intolerant of anticholinergic medications.

References on AE-B 6 of 6

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PHARMACOLOGIC CONSIDERATIONS FOR ANTIEMETIC PRESCRIBING

Cannabinoid:
- Dronabinol causes CNS depression; use caution in patients at risk for falls (eg, those who are older, debilitated, frail).
- Clinical pearl:
  - May stimulate appetite. To minimize adverse effects, consider starting with lower doses (especially in those who are older or are cannabis [colloquially known as marijuana]-naïve) and titrate to effect.
  - Excessive cannabinoid use can lead to cannabinoid hyperemesis.

Corticosteroids:
- Side effects associated with prolonged dexamethasone administration should be carefully considered.
- Dexamethasone may increase serum glucose; consider monitoring prior to therapy and as clinically indicated. Use with caution in patients with diabetes mellitus.
- Dexamethasone may cause dyspepsia. Take with food to minimize dyspepsia. Consider acid-blocking therapy with H_2 antagonist or PPI as clinically indicated.
- Dexamethasone may cause hiccups.[10-12]
- Dexamethasone-sparing strategies:
  - Emerging data and clinical practice suggest dexamethasone doses may be individualized. Higher doses may be considered, especially when an NK1 RA is not given concomitantly. For patients receiving MEC or HEC, especially those patients with few identifiable CINV risk factors or who are intolerant to corticosteroids, the use of lower doses, given for shorter durations, limiting the administration to Day 1, or even elimination of dexamethasone on Day 1 and/or subsequent days (for delayed nausea and emesis prevention) may be acceptable and may not compromise nausea or antiemesis control.[13-19] If dexamethasone is eliminated on Day 1 or subsequent days, consider the use of an antiemetic combination containing a 5-HT3 RA, NK1 RA, and olanzapine on Day 1, and olanzapine for delayed CINV. See Discussion.
- Clinical pearl:
  - For patients suffering from extended delayed CINV, consider extending the course of delayed dexamethasone as clinically appropriate. Consider AM dosing to minimize insomnia.
  - Corticosteroid antiemetic premedication should be avoided for 3–5 days prior to and 90 days after chimeric antigen receptor (CAR) T-cell therapies. Upon disease progression, corticosteroids may be resumed if needed.

References on AE-B 6 of 6

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PHARMACOLOGIC CONSIDERATIONS FOR ANTIEMETIC PRESCRIBING

Dopamine-Receptor Antagonists:

- General:
  - While no clinically significant drug-drug interactions with short-term use of antiemetics have emerged to date in randomized clinical trials, the Panel would discourage combining multiple dopamine-receptor antagonists as this could increase the risk of extrapyramidal reactions. When used for breakthrough CINV (AE-13), a single drug from this class could be utilized.
  - Dopamine-receptor antagonists may counteract the effects of dopamine-receptor agonist drugs such as pramipexole, ropinirole, carbidopa-levodopa, and others.
  - Use caution in patients at risk for falls (eg, those who are older, debilitated, frail).
  - Monitor for dystonic reactions.[a]
- Haloperidol:
  - May increase the QT interval of the ECG, but some data suggest that lower PO doses do not contribute to QT interval prolongation.[b,20,21]
  - Clinical pearl:
    - ◊ Generally, lower doses of haloperidol (AE-13) are required to produce an antiemetic effect than what is required for an antipsychotic effect.
- Metoclopramide:
  - May increase the QT interval of the ECG.[b,22]
  - May cause tardive dyskinesia (TD).
    - ◊ The risk increases with increasing cumulative dose and duration of treatment, and with concurrent multiple antidopaminergic agents.
    - ◊ The FDA recommends short-term use (<12 weeks) for metoclopramide given risk for TD with longer use.
  - Clinical pearl:
    - ◊ Metoclopramide increases gut motility and can be utilized to manage gastroparesis.
- Phenothiazines:
  - When administered parenterally, promethazine may cause severe tissue injury; these routes were removed from the guidelines previously. Only PO or rectal administration of promethazine is recommended in adult patients. IV administration via a central line is only a recommended option for pediatric patients ≥2 years of age with appropriate dilution.
  - Clinical pearl:
    - ◊ Prochlorperazine has more dopamine-blocking activity and less anticholinergic activity compared to promethazine; therefore, prochlorperazine's antiemetic activity is associated with less sedation.[23]

[a] Use diphenhydramine 25–50 mg PO/IV either every 4 or every 6 h for dystonic reactions. If allergic to diphenhydramine, use benztropine at 1–2 mg IV or IM x 1 dose, followed by PO dose of 1–2 mg daily or BID if needed. May consider using amantadine 100 mg BID–TID as treatment of drug-induced dystonic reactions for those patients intolerant of anticholinergic medications.
[b] Use caution and monitor ECG in patients with other risk factors for QT prolongation.

References on AE-B 6 of 6

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PHARMACOLOGIC CONSIDERATIONS FOR ANTIEMETIC PRESCRIBING

NK1 RAs:

- Aprepitant, aprepitant injectable emulsion, fosaprepitant, netupitant, and fosnetupitant inhibit the metabolism of dexamethasone, thus increasing dexamethasone serum levels when administered concomitantly. Rolapitant does not share this interaction with dexamethasone.
- Fosaprepitant contains polysorbate 80, which may be implicated in infusion hypersensitivity reactions. Aprepitant emulsion and fosnetupitant do not contain polysorbate 80.
- Rolapitant has an extended half-life and should not be administered at less than 2-week intervals.
- Although no head-to-head clinical trial data with a superiority endpoint have been published, emerging clinical evidence (non-inferiority endpoints) and anecdotal evidence suggest there may be differences in efficacy.
- Clinical pearl:
  - Place in therapy is for prevention of CINV, not treatment of CINV.
    - Largest benefit is seen in a delayed CINV setting.

References on AE-B 6 of 6

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PRINCIPLES FOR MANAGING BREAKTHROUGH EMESIS

- Breakthrough emesis presents a difficult situation, as correction of refractory ongoing nausea/vomiting is often challenging to reverse. It is generally far easier to prevent nausea/vomiting than it is to treat it.
- The general principle of breakthrough treatment is to give an additional agent from a different drug class. The choice of agent should be based on assessment of the current prevention strategies used. Some patients may require several agents utilizing differing mechanisms of action.
- Routine, around-the-clock administration rather than PRN dosing should be strongly considered.
- The PO route is not likely to be feasible due to ongoing vomiting; therefore, rectal or IV administration is often required.
- Multiple concurrent agents from a different drug class, perhaps in alternating schedules or by alternating routes, may be necessary.
- Ensure adequate hydration or fluid repletion, simultaneously checking and correcting any possible electrolyte abnormalities.
- Prior to administering the next cycle of anticancer therapy, the patient should be reassessed, with attention given to various possible non-anticancer therapy–related reasons for breakthrough emesis with the current cycle:
  - Brain metastases
  - Electrolyte abnormalities
  - Tumor infiltration of the bowel or other gastrointestinal abnormality
  - Other comorbidities
- Prior to the next cycle of anticancer therapy, reassess both the Day 1 and post-anticancer therapy antiemetic regimen, which did not protect the patient during the present cycle, and consider alternatives: (Suggestions are not in order of preference)
  - Add an NK1 RA if not previously included.
  - Consider changing from NK1 RA–containing regimens to an olanzapine-containing regimen, or vice versa.
  - Consider combining an NK1 RA regimen with olanzapine; see Adult Parenteral Anticancer Agents - Acute and Delayed Emesis Prevention - High Emetic Risk, option C (AE-6).
  - Consider changing to a different NK1 RA with a different pharmacokinetic/pharmacodynamic profile. Although no available head-to-head clinical trial data support this, anecdotal evidence suggests it may be helpful.
  - Add other concomitant antiemetics from a different drug class, if applicable.
  - Possibly adjust dose(s), either intensity or frequency, of the 5-HT3 RA. Based on the patient's experiences, the anticancer therapy regimen in question may be more emetogenic than generally classified (eg, Hesketh method).
  - Possibly switch to a different 5-HT3 RA. Although not necessarily likely to be effective, anecdotal and limited investigational trial data suggest it may sometimes be efficacious. 5-HT3 RAs have different pharmacokinetics/pharmacodynamics and different routes of metabolism that may account for different efficacy in certain populations.
  - If the goal of anticancer therapy is non-curative, consider other appropriate regimens, if any, that might be less emetogenic.
  - It may be beneficial to add an anxiolytic agent in combination with the antiemetic agents.
- Consider acid-suppressive therapy (H_2 blocker or PPI) if patient has dyspepsia.
- Topical capsaicin may be considered as an adjunct option, in addition to guideline-adherent antiemetic regimens, to reduce delayed and extended CINV.[a,1]
- Cannabinoids may be considered as an adjunct option, in addition to guideline-adherent antiemetic regimens, to prevent acute and delayed CINV.[2,3]

[a] Avoid application to sores, broken or irritated skin, infected skin, or prior RT sites. Patients with sensory dysfunction should proceed with caution.

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PEDIATRIC PARENTERAL ANTICANCER AGENTS[a] – EMETOGENIC POTENTIAL HIGH AND MODERATE EMETIC RISK

LEVEL	AGENT		
High emetic risk (>90% frequency of emesis)[b]	• Busulfan • Carboplatin ≥175 mg/m^2 • Carmustine >250 mg/m^2 • Cisplatin ≥12 mg/m^2 • Cyclophosphamide ≥1200 mg/m^2 • Cyclophosphamide + anthracycline (eg, Doxorubicin) • Cyclophosphamide + Etoposide • Cytarabine ≥3000 mg/m^2/day • Cytarabine ≥150 mg/m^2 + Daunorubicin • Cytarabine ≥300 mg/m^2 + Etoposide • Cytarabine ≥90 mg/m^2 + Methotrexate ≥150 mg/m^2	• Dacarbazine • Dactinomycin • Doxorubicin ≥30 mg/m^2 • Doxorubicin + Ifosfamide • Doxorubicin + Methotrexate ≥5 g/m^2 • Etoposide + Ifosfamide • Etoposide + Thiotepa	• Idarubicin ≥30 mg/m^2 • Ifosfamide ≥2 g/m^2 • Melphalan • Methotrexate ≥12 g/m^2 • Thiotepa ≥300 mg/m^2
Moderate emetic risk (>30%–90% frequency of emesis)[b]	• Aldesleukin >12–15 million IU/m^2 or 600,000 IU/kg • Arsenic trioxide • Azacitidine • Bendamustine • Carboplatin <175 mg/m^2 • Carmustine ≤250 mg/m^2 • Clofarabine • Cyclophosphamide <1200 mg/m^2 • Cytarabine >75 mg/m^2 • Cytarabine intrathecal	• Daunorubicin • Dinutuximab • Doxorubicin <30 mg/m^2 • Dual-drug liposomal encapsulation of Cytarabine and Daunorubicin • Gemtuzumab ozogamicin • Idarubicin <30 mg/m^2 • Ifosfamide <2 g/m^2 • Irinotecan	• Methotrexate ≥250 mg/m^2 • Methotrexate intrathecal • Naxitamab-gqgk • Oxaliplatin • Temozolomide • Trabectedin

Table framework is based on the clinical experience of Panel members and on the emetogenicity classifications described in the following publications: Paw Cho Sing E, Robinson PD, Flank J, et al. Classification of the acute emetogenicity of chemotherapy in pediatric patients: A clinical practice guideline. Pediatr Blood Cancer 2019;66:e27646. Erratum in: Pediatr Blood Cancer 2021;68:e28990; Children's Oncology Group. Guidelines on Chemotherapy-induced Nausea and Vomiting in Pediatric Cancer Patients. Version date 1.15.2024; Hesketh P, Kris M, Basch E, et al. Antiemetics: ASCO Guideline Update. J Clin Oncol 38:2782-2797. For medications not listed in pediatric tables, refer to adult data when no pediatric-specific information is available.

[a] mg/m^2 or g/m^2 refers to mg/m^2/dose or g/m^2/dose, except where otherwise noted.
[b] Proportion of patients who experience emesis in the absence of effective antiemetic prophylaxis.

Note: All recommendations are category 2A unless otherwise indicated.

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PEDIATRIC PARENTERAL ANTICANCER AGENTS[a] – EMETOGENIC POTENTIAL
LOW AND MINIMAL EMETIC RISK

LEVEL	AGENT		
Low emetic risk (10%–30% frequency of emesis)[b,c]	• Aldesleukin ≤12 million IU/m^2 • Blinatumomab • Brentuximab vedotin • Cabazitaxel • Carfilzomib • Docetaxel • Doxorubicin (liposomal)	• Etoposide • Etoposide phosphate • 5-Fluorouracil • Gemcitabine • Inotuzumab ozogamicin • Methotrexate <250 mg/m^2 • Mitoxantrone	• Paclitaxel • Paclitaxel-albumin • Pemetrexed • Tagraxofusp-erzs • Thiotepa <300 mg/m^2 • Tisagenlecleucel[e] • Topotecan
Minimal emetic risk (<10% frequency of emesis)[b]	• Alemtuzumab • Asparaginase[d] • Avelumab • Betibeglogene autotemcel[e] • Bevacizumab • Bleomycin • Bortezomib • Cetuximab • Cladribine • Daratumumab • Daratumumab and Hyaluronidase-fihj • Decitabine • Dexrazoxane	• Fludarabine • Ipilimumab • Lovotibeglogene autotemcel[e] • Nelarabine • Nivolumab • Nivolumab and Hyaluronidase-nvhy • Pembrolizumab • Pembrolizumab and Berahyaluronidase alfa-pmph • Pentostatin • Rituximab • Rituximab and Hyaluronidase	• Siltuximab • Temsirolimus • Trastuzumab • Trastuzumab and Hyaluronidase-oysk • Vinblastine • Vincristine • Vinorelbine

Table framework is based on the clinical experience of Panel members and on the emetogenicity classifications described in the following publications: Paw Cho Sing E, Robinson PD, Flank J, et al. Classification of the acute emetogenicity of chemotherapy in pediatric patients: A clinical practice guideline. Pediatr Blood Cancer 2019;66:e27646. Erratum in: Pediatr Blood Cancer 2021;68:e28990; Children's Oncology Group. Guidelines on Chemotherapy-induced Nausea and Vomiting in Pediatric Cancer Patients. Version date 1.15.2024; Hesketh P, Kris M, Basch E, et al. Antiemetics: ASCO Guideline Update. J Clin Oncol 38:2782-2797. For medications not listed in pediatric tables, refer to adult data when no pediatric-specific information is available.

[a] mg/m^2 or g/m^2 refers to mg/m^2/dose or g/m^2/dose, except where otherwise noted.
[b] Proportion of patients who experience emesis in the absence of effective antiemetic prophylaxis.
[c] For some low-emetic-risk agents, factors related to dosing schedule (particularly continuous dosing) and clinical experience suggest routine premedication is not required. An individualized approach is appropriate for whether to premedicate each dose or prescribe antiemetics PRN.
[d] Asparaginase includes calaspargase pegol-mknl, pegaspargase, asparaginase Erwinia chrysanthemi, and asparaginase Erwinia chrysanthemi (recombinant)-rywn.
[e] Corticosteroid antiemetic premedication should be avoided for 3–5 days prior to and 90 days after CAR and TIL T-cell therapies. Upon disease progression, corticosteroids may be resumed if needed. Antiemetic regimens used during lymphodepleting chemotherapy regimens should also employ a corticosteroid-sparing approach to antiemetic prophylaxis.

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PEDIATRIC PARENTERAL ANTICANCER AGENTS[f] – ACUTE AND DELAYED EMESIS PREVENTION
HIGH EMETIC RISK

Select treatment option A, B, C, or D
All treatment options should be started before anticancer therapy[g]

Treatment option A (children able to receive an NK1 RA and Dexamethasone), use the following combination:

1. 5-HT3 RA (choose one):
   - ◊ Palonosetron 0.02 mg/kg (max 1.5 mg) IV once if <17 y of age. If ≥17 y of age, 0.25 mg IV once (preferred)[h]
   - ◊ Granisetron 0.01–0.04 mg/kg IV once, or 0.04 mg/kg PO every 12–24 hours (max 2 mg/day)
   - ◊ Ondansetron 0.15 mg/kg or 5 mg/m^2 (max 16 mg/dose or 0.45 mg/kg/day) PO/IV every 8–24 hours, not to exceed 3 doses/day (max 24 mg/day)
     - – ± Ondansetron or Granisetron on all subsequent chemotherapy days if used in acute and delayed phase[i]
2. Dexamethasone PO/IV once[j,k,l]
   - ◊ ± Dexamethasone PO/IV daily if used in acute phase, on all subsequent chemotherapy days and for delayed phase

Consider adding in children ≥6 mo of age:

3. NK1 RA (choose one):
   - ◊ Aprepitant
     - – Age ≥6 mo and <12 y, weight <30 kg: 3 mg/kg PO once (max 125 mg)
     - – Age ≥12 y or weight ≥30 kg: 125 mg PO once
     - – 2 mg/kg PO daily on Days 2 and 3 (max 80 mg)
   - ◊ Aprepitant injectable emulsion IV once (3-day regimen)[m]
     - – Age ≥6 y and <12 y: 3 mg/kg IV once (max 100 mg)
     - – Age ≥12 y: 100 mg IV once
     - – 2 mg/kg PO daily on Days 2 and 3 (max 80 mg)
   - ◊ Fosaprepitant single-dose regimen
     - – Age ≥6 mo and <2 y, weight ≥6 kg: 5 mg/kg IV once (max 150 mg)
     - – Age ≥2 y and <12 y: 4 mg/kg IV once (max 150 mg)
     - – Age ≥12 y: 150 mg IV once
   - ◊ Fosaprepitant 3-day regimen
     - – Age ≥6 mo and <2 y, weight ≥6 kg: 3 mg/kg IV once (max 115 mg)
     - – Age ≥2 y and <12 y: 3 mg/kg IV once (max 115 mg)
     - – Age ≥12 y: 115 mg IV once
     - – 2 mg/kg IV daily on Days 2 and 3 (max 80 mg)

Consider adding in children ≥3 y of age:

4. Olanzapine 0.05–0.2 mg/kg (2.5–10 mg) PO once at bedtime prior to first dose of anticancer therapy[n]
   - ◊ ± Olanzapine 0.05–0.2 mg/kg (2.5–10 mg) PO daily (bedtime) on Days 2, 3, and 4

[j] Use of corticosteroid premedications should be avoided with cellular therapies. Clinicians may wish to consider a dexamethasone-sparing approach with ICI therapy as well. See Pharmacologic Considerations for Antiemetic Prescribing (AE-B).

Footnotes on PEDAE-7A

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PEDIATRIC PARENTERAL ANTICANCER AGENTS[f] – ACUTE AND DELAYED EMESIS PREVENTION
HIGH EMETIC RISK

Select treatment option A, B, C, or D
All treatment options should be started before anticancer therapy[g]

Treatment option B (children unable to receive an NK1 RA), use the following combination:

1. 5-HT3 RA (choose one):
   - ◊ Palonosetron 0.02 mg/kg (max 1.5 mg) IV once if <17 y of age. If ≥17 y of age, 0.25 mg IV once (preferred)[h]
   - ◊ Granisetron 0.01–0.04 mg/kg IV once, or 0.04 mg/kg PO every 12–24 hours (max 2 mg/day)
   - ◊ Ondansetron 0.15 mg/kg or 5 mg/m^2 (max 16 mg/dose or 0.45 mg/kg/day) PO/IV every 8–24 hours, not to exceed 3 doses/day (max 24 mg/day)
     - – ± Ondansetron or Granisetron on all subsequent chemotherapy days if used in acute and delayed phase[i]
2. Dexamethasone PO/IV once[j,k,l]
   - ◊ Dexamethasone PO/IV daily if used in acute phase, on all subsequent chemotherapy days and for delayed phase

Consider adding in children ≥3 y of age:

3. Olanzapine 0.05–0.2 mg/kg (2.5–10 mg) PO once at bedtime or prior to first dose of anticancer therapy[n]
   - ◊ ± Olanzapine 0.05–0.2 mg/kg (2.5–10 mg) PO daily (bedtime) on Days 2, 3, and 4

Treatment option C (children unable to receive an NK1 RA and Dexamethasone), use the following combination:

1. 5-HT3 RA (choose one):
   - ◊ Palonosetron 0.02 mg/kg (max 1.5 mg) IV once if <17 y of age. If ≥17 y of age, 0.25 mg IV once (preferred)[h]
   - ◊ Granisetron 0.01–0.04 mg/kg IV once, or 0.04 mg/kg PO every 12–24 hours (max 2 mg/day)
   - ◊ Ondansetron 0.15 mg/kg or 5 mg/m^2 (max 16 mg/dose or 0.45 mg/kg/day) PO/IV every 8–24 hours, not to exceed 3 doses/day (max 24 mg/day)
     - – ± Ondansetron or Granisetron on all subsequent chemotherapy days if used in acute and delayed phase[i]

Consider adding in children ≥3 y of age:

2. Olanzapine 0.05–0.2 mg/kg (2.5–10 mg) PO once at bedtime prior to first dose of anticancer therapy[n]
   - ◊ ± Olanzapine 0.05–0.2 mg/kg (2.5–10 mg) PO daily (bedtime) on Days 2, 3, and 4

[j] Use of corticosteroid premedications should be avoided with cellular therapies. Clinicians may wish to consider a dexamethasone-sparing approach with ICI therapy as well. See Pharmacologic Considerations for Antiemetic Prescribing (AE-B).

Footnotes on PEDAE-7A

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PEDIATRIC PARENTERAL ANTICANCER AGENTS[f] – ACUTE AND DELAYED EMESIS PREVENTION HIGH EMETIC RISK

Select treatment option A, B, C, or D
All treatment options should be started before anticancer therapy[g]

Treatment option D (children unable to receive Dexamethasone), use the following combination:

1. 5-HT3 RA (choose one):
   - ◊ Palonosetron 0.02 mg/kg (max 1.5 mg) IV once if <17 y of age. If ≥17 y of age, 0.25 mg IV once (preferred)[h]
   - ◊ Granisetron 0.01–0.04 mg/kg IV once, or 0.04 mg/kg PO every 12–24 hours (max 2 mg/day)
   - ◊ Ondansetron 0.15 mg/kg or 5 mg/m^2 (max 16 mg/dose or 0.45 mg/kg/day) PO/IV every 8–24 hours, not to exceed 3 doses/day (max 24 mg/day)
     - – ± Ondansetron or Granisetron on all subsequent chemotherapy days if used in acute and delayed phase[i]

Consider adding in children ≥6 mo of age:

2. NK1 RA (choose one):
   - ◊ Aprepitant
     - – Age ≥6 mo and <12 y, weight <30 kg: 3 mg/kg PO once (max 125 mg)
     - – Age ≥12 y or weight ≥30 kg: 125 mg PO once
     - – 2 mg/kg PO daily on Days 2 and 3 (max 80 mg)
   - ◊ Aprepitant injectable emulsion IV once (3-day regimen)[m]
     - – Age ≥6 y and <12 y: 3 mg/kg IV once (max 100 mg)
     - – Age ≥12 y: 100 mg IV once
     - – 2 mg/kg PO daily on Days 2 and 3 (max 80 mg)
   - ◊ Fosaprepitant single-dose regimen
     - – Age ≥6 mo and <2 y, weight ≥6 kg: 5 mg/kg IV once (max 150 mg)
     - – Age ≥2 y and <12 y: 4 mg/kg IV once (max 150 mg)
     - – Age ≥12 y: 150 mg IV once
   - ◊ Fosaprepitant 3-day regimen
     - – Age ≥6 mo and <2 y, weight ≥6 kg: 3 mg/kg IV once (max 115 mg)
     - – Age ≥2 y and <12 y: 3 mg/kg IV once (max 115 mg)
     - – Age ≥12 y: 115 mg IV once
     - – 2 mg/kg IV daily on Days 2 and 3 (max 80 mg)

Consider adding in children ≥3 y of age:

3. Olanzapine 0.05–0.2 mg/kg (2.5–10 mg) PO once at bedtime prior to first dose of anticancer therapy[n]
   - ◊ ± Olanzapine 0.05–0.2 mg/kg (2.5–10 mg) PO daily (bedtime) on Days 2, 3, and 4

Footnotes on PEDAE-7A

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PEDIATRIC PARENTERAL ANTICANCER AGENTS[f] – ACUTE AND DELAYED EMESIS PREVENTION
MODERATE EMETIC RISK

Select treatment option E, F, or G
All treatment options should be started before anticancer therapy[g]

Treatment option E (children unable to receive NK1 RA), use the following combination:

1. 5-HT3 RA (choose one):
   - ◊ Granisetron 0.01–0.04 mg/kg IV once, or 0.04 mg/kg PO every 12–24 hours (max 2 mg/day)
   - ◊ Ondansetron 0.15 mg/kg or 5 mg/m^2 (max 16 mg/dose or 0.45 mg/kg/day) PO/IV every 8–24 hours, not to exceed 3 doses/day (max 24 mg/day)
     - – ± Ondansetron or Granisetron on all subsequent chemotherapy days if used in acute and delayed phase[i]
   - ◊ Palonosetron 0.02 mg/kg (max 1.5 mg) IV once if <17 y of age. If ≥17 y of age, 0.25 mg IV once

2. Dexamethasone PO/IV once[j,k,l]
   - ◊ ± Dexamethasone PO/IV daily if used in acute phase, on all subsequent chemotherapy days and for delayed phase

Treatment option F (children unable to receive an NK1 RA and Dexamethasone), use the following combination:

1. 5-HT3 RA (choose one):
   - ◊ Granisetron 0.01–0.04 mg/kg IV once, or 0.04 mg/kg PO every 12–24 hours (max 2 mg/day)
   - ◊ Ondansetron 0.15 mg/kg or 5 mg/m^2 (max 16 mg/dose or 0.45 mg/kg/day) PO/IV every 8–24 hours, not to exceed 3 doses/day (max 24 mg/day)
     - – ± Ondansetron or Granisetron on all subsequent chemotherapy days if used in acute and delayed phase[i]
   - ◊ Palonosetron 0.02 mg/kg (max 1.5 mg) IV once if <17 y of age. If ≥17 y of age, 0.25 mg IV once

Consider adding in children ≥3 y of age:

2. Olanzapine 0.05–0.2 mg/kg (2.5–10 mg) PO once at bedtime prior to first dose of anticancer therapy[n]
   - ◊ ± Olanzapine 0.05–0.2 mg/kg (2.5–10 mg) PO daily (bedtime) on Days 2, 3, and 4

[j] Use of corticosteroid premedications should be avoided with cellular therapies. Clinicians may wish to consider a dexamethasone-sparing approach with ICI therapy as well. See Pharmacologic Considerations for Antiemetic Prescribing (AE-B).

Footnotes on PEDAE-7A

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PEDIATRIC PARENTERAL ANTICANCER AGENTS[f] – ACUTE AND DELAYED EMESIS PREVENTION
MODERATE EMETIC RISK

Select treatment option E, F, or G
All treatment options should be started before anticancer therapy[g]

Treatment option G (children unable to receive Dexamethasone), use the following combination:

1. 5-HT3 RA (choose one):
   - ◊ Granisetron 0.01–0.04 mg/kg IV once, or 0.04 mg/kg PO every 12–24 hours (max 2 mg/day)
   - ◊ Ondansetron 0.15 mg/kg or 5 mg/m^2 (max 16 mg/dose or 0.45 mg/kg/day) PO/IV every 8–24 hours, not to exceed 3 doses/day (max 24 mg/day)
     - – ± Ondansetron or Granisetron on all subsequent chemotherapy days if used in acute and delayed phase[i]
   - ◊ Palonosetron 0.02 mg/kg (max 1.5 mg) IV once if <17 y of age. If ≥17 y of age, 0.25 mg IV once

Consider adding in children ≥6 mo of age:

2. NK1 RA (choose one):
   - ◊ Aprepitant
     - – Age ≥6 mo and <12 y, weight <30 kg: 3 mg/kg PO once (max 125 mg)
     - – Age ≥12 y or weight ≥30 kg: 125 mg PO once
     - – 2 mg/kg PO daily on Days 2 and 3 (max 80 mg)
   - ◊ Aprepitant injectable emulsion IV once (3-day regimen)[m]
     - – Age ≥6 y and <12 y: 3 mg/kg IV once (max 100 mg)
     - – Age ≥12 y: 100 mg IV once
     - – 2 mg/kg PO daily on Days 2 and 3 (max 80 mg)
   - ◊ Fosaprepitant single-dose regimen
     - – Age ≥6 mo and <2 y, weight ≥6 kg: 5 mg/kg IV once (max 150 mg)
     - – Age ≥2 y and <12 y: 4 mg/kg IV once (max 150 mg)
     - – Age ≥12 y: 150 mg IV once
   - ◊ Fosaprepitant 3-day regimen
     - – Age ≥6 mo and <2 y, weight ≥6 kg: 3 mg/kg IV once (max 115 mg)
     - – Age ≥2 y and <12 y: 3 mg/kg IV once (max 115 mg)
     - – Age ≥12 y: 115 mg IV once
     - – 2 mg/kg IV daily on Days 2 and 3 (max 80 mg)

Footnotes on PEDAE-7A

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FOOTNOTES FOR PEDIATRIC PARENTERAL ANTICANCER AGENTS ACUTE AND DELAYED EMESIS PREVENTION

[f] Pediatric Parenteral Anticancer Agents - Emetogenic Potential (PEDAE-1).
[g] Antiemetic regimens should be chosen based on the drug with the highest emetic risk as well as patient-specific risk factors.
[h] Palonosetron is the preferred 5-HT3 RA in the acute phase if there is a high risk of delayed-phase CINV and if unable to use an NK1 RA and/or dexamethasone.
[i] If palonosetron is given on Day 1, ondansetron or granisetron is not needed on subsequent days.
[j] Use of corticosteroid premedications should be avoided with cellular therapies. Clinicians may wish to consider a dexamethasone-sparing approach with ICI therapy as well. See Pharmacologic Considerations for Antiemetic Prescribing (AE-B).
[k] No optimal pediatric dexamethasone dose has been identified. Dexamethasone doses may be individualized. Consider doses of 5–10 mg/m^2/day once daily or divided. Dexamethasone should be reduced by 50% when given concomitantly with NK1 RAs.
[l] Dexamethasone should be given in the delayed phase if ondansetron or granisetron is given in the acute phase. Dexamethasone may be considered in the delayed phase if palonosetron is used in the acute phase.
[m] Aprepitant injectable emulsion is a unique formulation of aprepitant and is NOT interchangeable with the IV formulation of fosaprepitant.
[n] Pediatric randomized controlled trials evaluating olanzapine excluded patients <5 years of age. However, safety has been established among pediatric patients with mental health conditions; thus, caution is warranted in younger children.

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PEDIATRIC PARENTERAL ANTICANCER AGENTS[f] – EMESIS PREVENTION
LOW AND MINIMAL EMETIC RISK

Low →

Start before anticancer therapy and continue daily on each day of anticancer therapy (order does not imply preference):[g]
- **5-HT3 RA (select one):**
  - **Granisetron 0.01–0.04 mg/kg IV once, or 0.04 mg/kg PO every 12–24 hours (max 2 mg/day)**
  - **Ondansetron 0.15 mg/kg or 5 mg/m² (max 16 mg/dose or 0.45 mg/kg/day) PO/IV every 8–24 hours, not to exceed 3 doses/day (max 24 mg/day)**

or
- **Prochlorperazine 0.1 mg/kg PO/IV once (max 10 mg/dose)**

→ **Breakthrough Treatment (PEDAE-12)**

Minimal → **No routine prophylaxis** →

[f] Pediatric Parenteral Anticancer Agents - Emetogenic Potential (PEDAE-1).
[g] Antiemetic regimens should be chosen based on the drug with the highest emetic risk as well as patient-specific risk factors.

Note: All recommendations are category 2A unless otherwise indicated.

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PEDIATRIC ORAL ANTICANCER AGENTS – EMETOGENIC POTENTIAL AND EMESIS PREVENTION[o]
MODERATE TO HIGH EMETIC RISK (PROPHYLAXIS REQUIRED)

LEVEL	AGENT
Moderate to high emetic risk[b] (≥30% frequency of emesis): Prophylaxis required on days of PO anticancer agent administration	• Azacitidine • Cyclophosphamide ≥100 mg/m^2/day • Lomustine (single day) • Midostaurin • Selinexor • Temozolomide >75 mg/m^2/day • Temozolomide ≤75 mg/m^2/day with concurrent RT

Table framework is based on the clinical experience of Panel members and on the emetogenicity classifications described in the following publications: Paw Cho Sing E, Robinson PD, Flank J, et al. Classification of the acute emetogenicity of chemotherapy in pediatric patients: A clinical practice guideline. Pediatr Blood Cancer 2019;66:e27646. Erratum in: Pediatr Blood Cancer 2021;68:e28990; Children's Oncology Group. Guidelines on Chemotherapy-induced Nausea and Vomiting in Pediatric Cancer Patients. Version date 1.15.2024; Hesketh P, Kris M, Basch E, et al. Antiemetics: ASCO Guideline Update. J Clin Oncol 38:2782-2797. For medications not listed in pediatric tables, refer to adult data when no pediatric-specific information is available.

Moderate to high emetic risk → **Prophylaxis required** →

Start before anticancer therapy and continue daily on each day of anticancer therapy (order does not imply preference):[g]
- **5-HT3 RA (choose one):**
  - **Granisetron 0.01–0.04 mg/kg IV once or 0.04 mg/kg PO every 12–24 hours (max 2 mg/day)**
  - **Ondansetron 0.15 mg/kg or 5 mg/m^2 (max 16 mg/dose or 0.45 mg/kg/day) PO/IV every 8–24 hours, not to exceed 3 doses/day (max 24 mg/day)**

or
- **Olanzapine 0.05–0.2 mg/kg (2.5–10 mg) PO daily (bedtime)**

→ **Breakthrough Treatment (PEDAE-12)**

[b] Proportion of patients who experience emesis in the absence of effective antiemetic prophylaxis.
[g] Antiemetic regimens should be chosen based on the drug with the highest emetic risk as well as patient-specific risk factors.
[o] These antiemetic recommendations apply to PO anticancer therapy only. When combined with IV agents in a combination anticancer therapy regimen, the antiemetic recommendations for the agent with the highest level of emetogenicity should be followed. If multiple PO agents are combined, emetic risk may increase and require prophylaxis.

Note: All recommendations are category 2A unless otherwise indicated.

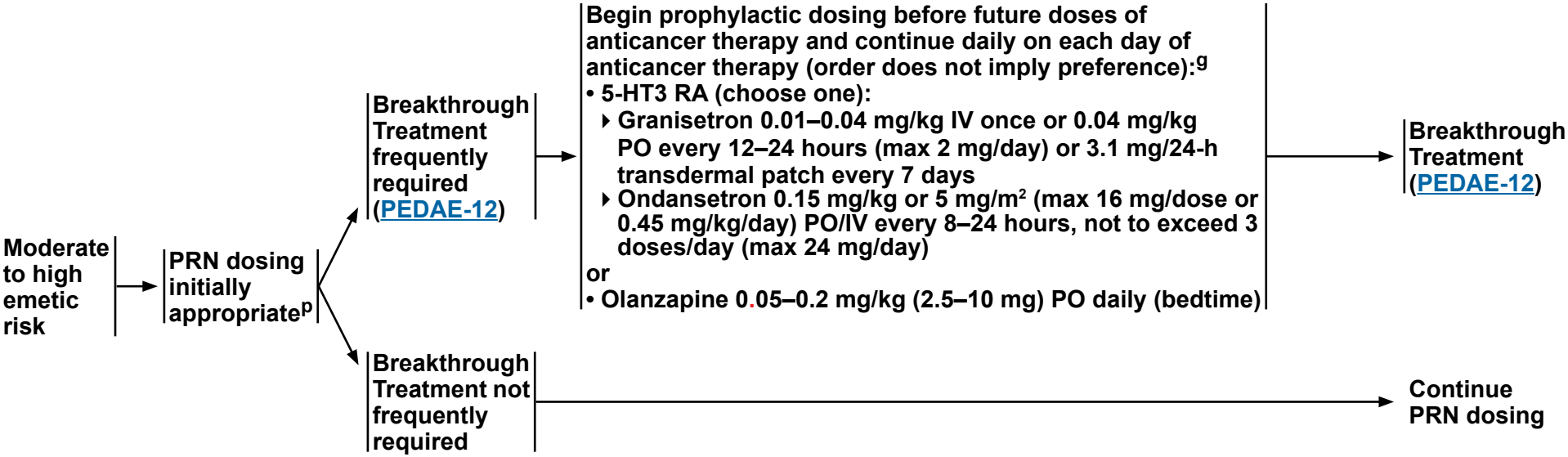
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PEDIATRIC ORAL ANTICANCER AGENTS – EMETOGENIC POTENTIAL AND EMESIS PREVENTION[o]
MODERATE TO HIGH EMETIC RISK (PRN DOSING INITIALLY APPROPRIATE)

LEVEL	AGENT		
Moderate to high emetic risk[b,o] (≥30% frequency of emesis): As needed (PRN) dosing is initially appropriate on days of PO anticancer agent administration	• Abemaciclib • Cabozantinib • Crizotinib • Dabrafenib	• Enasidenib • Etoposide • Imatinib >400 mg/day • Lenvatinib >12 mg/day	• Procarbazine • Tovorafenib

Table framework is based on the clinical experience of Panel members and on the emetogenicity classifications described in the following publications: Paw Cho Sing E, Robinson PD, Flank J, et al. Classification of the acute emetogenicity of chemotherapy in pediatric patients: A clinical practice guideline. Pediatr Blood Cancer 2019;66:e27646. Erratum in: Pediatr Blood Cancer 2021;68:e28990; Children's Oncology Group. Guidelines on Chemotherapy-induced Nausea and Vomiting in Pediatric Cancer Patients. Version date 1.15.2024; Hesketh P, Kris M, Basch E, et al. Antiemetics: ASCO Guideline Update. J Clin Oncol 38:2782-2797. For medications not listed in pediatric tables, refer to adult data when no pediatric-specific information is available.



[b] Proportion of patients who experience emesis in the absence of effective antiemetic prophylaxis.
[g] Antiemetic regimens should be chosen based on the drug with the highest emetic risk as well as patient-specific risk factors.
[o] These antiemetic recommendations apply to PO anticancer therapy only. When combined with IV agents in a combination anticancer therapy regimen, the antiemetic recommendations for the agent with the highest level of emetogenicity should be followed. If multiple PO agents are combined, emetic risk may increase and require prophylaxis.
[p] For some moderate to high emetic risk agents, factors related to dosing schedule (particularly continuous dosing for prolonged periods), and clinical experience suggest routine premedication is not required. An individualized approach is appropriate for whether to premedicate each dose or prescribe antiemetics PRN.

Note: All recommendations are category 2A unless otherwise indicated.

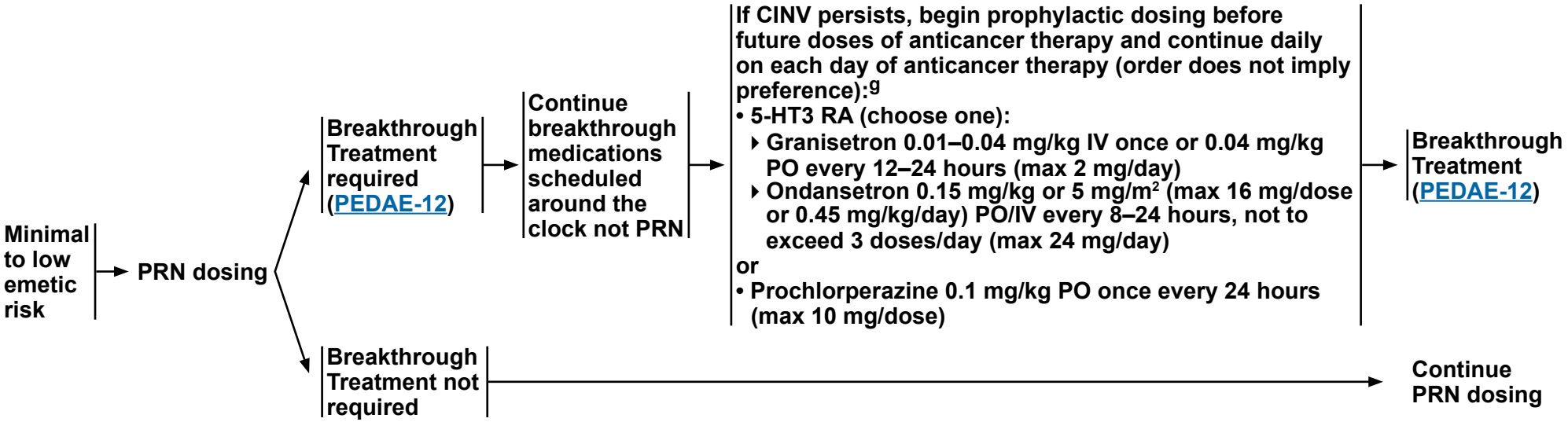
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PEDIATRIC ORAL ANTICANCER AGENTS – EMETOGENIC POTENTIAL AND EMESIS PREVENTION[o]
MINIMAL TO LOW EMETIC RISK

LEVEL	AGENT			
Minimal to low emetic risk[b] (<30% frequency of emesis)	• Cobimetinib • Cyclophosphamide <100 mg/m^2/day • Dasatinib • Entrectinib • Everolimus • Gilteritinib • Hydroxyurea	• Ibrutinib • Imatinib ≤400 mg/day • Ivosidenib • Larotrectinib • Lenalidomide • Lenvatinib ≤12 mg/day • Lorlatinib	• Mercaptopurine • Methotrexate • Nilotinib • Pazopanib • Ponatinib • Ruxolitinib • Selpercatinib • Sorafenib	• Temozolomide ≤75 mg/m^2/day • Thioguanine • Trametinib • Tretinoin • Venetoclax • Vorinostat

Table framework is based on the clinical experience of Panel members and on the emetogenicity classifications described in the following publications: Paw Cho Sing E, Robinson PD, Flank J, et al. Classification of the acute emetogenicity of chemotherapy in pediatric patients: A clinical practice guideline. Pediatr Blood Cancer 2019;66:e27646. Erratum in: Pediatr Blood Cancer 2021;68:e28990; Children's Oncology Group. Guidelines on Chemotherapy-induced Nausea and Vomiting in Pediatric Cancer Patients. Version date 1.15.2024; Hesketh P, Kris M, Basch E, et al. Antiemetics: ASCO Guideline Update. J Clin Oncol 38:2782-2797. For medications not listed in pediatric tables, refer to adult data when no pediatric-specific information is available.



[b] Proportion of patients who experience emesis in the absence of effective antiemetic prophylaxis.

[g] Antiemetic regimens should be chosen based on the drug with the highest emetic risk as well as patient-specific risk factors.

[o] These antiemetic recommendations apply to PO anticancer therapy only. When combined with IV agents in a combination anticancer therapy regimen, the antiemetic recommendations for the agent with the highest level of emetogenicity should be followed. If multiple PO agents are combined, emetic risk may increase and require prophylaxis.

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BREAKTHROUGH TREATMENT FOR PEDIATRIC ANTICANCER THERAPY-INDUCED NAUSEA/VOMITING

Any nausea/ vomiting →

The general principle of breakthrough treatment is to add one agent from a different drug class to the current regimen.

- **Atypical antipsychotic:**
  - **Olanzapine 0.05–0.2 mg/kg (2.5–10 mg) PO daily (bedtime)[q]**
- **Cannabinoid:**
  - **Dronabinol 2.5–10 mg PO every 6–8 hours[r]**
- **Anticholinergics:**
  - **Scopolamine 1.5 mg transdermal patch 0.5–1 patch[s] every 72 hours**
- **Dopamine-receptor antagonists:[t]**
  - **Metoclopramide 0.1–0.25 mg/kg PO/IV every 6 hours (max 10 mg/dose)**
  - **Phenothiazine:**
    - ◊ **Prochlorperazine 0.1 mg/kg PO/IV every 6 hours (max 10 mg/dose)**
    - ◊ **Promethazine 0.25–0.5 mg/kg (6.25–25 mg/dose) PO/IV every 6 hours (only for patients ≥2 years of age)**
- **5-HT3 RA:**
  - **Granisetron 0.01–0.04 mg/kg IV once or 0.04 mg/kg PO every 12–24 hours (max 2 mg/day)**
  - **Ondansetron 0.15 mg/kg or 5 mg/m² (max 16 mg/dose or 0.45 mg/kg/day) PO/IV every 8–24 hours, not to exceed 3 doses/day (max 24 mg/day)**
- **Corticosteroid:**
  - **Dexamethasone PO/IV every 12–24 hours[j,k]**
- **Benzodiazepine:**
  - **Lorazepam 0.025–0.05 mg/kg/dose PO/IV every 6 hours (max 2 mg/dose)[u]**

RESPONSE

Nausea and vomiting controlled → **Continue breakthrough medications, on a schedule, not PRN**

Nausea and/ or vomiting uncontrolled → **Re-evaluate and consider dose adjustments and/or sequentially add one agent from a different drug class**

SUBSEQUENT CYCLES

Consider changing antiemetic therapy to higher level primary treatment for next cycle

[j] Use of corticosteroid premedications should be avoided with cellular therapies. Clinicians may wish to consider a dexamethasone-sparing approach with ICI therapy as well. See Pharmacologic Considerations for Antiemetic Prescribing (AE-B).

[k] No optimal pediatric dexamethasone dose has been identified. Dexamethasone doses may be individualized. Consider doses of 5–10 mg/m²/day once daily or divided. Dexamethasone should be reduced by 50% when given concomitantly with NK1 RAs.

[q] When not used as part of the acute and delayed emesis prevention regimen.

[r] Dronabinol PO solution has greater PO bioavailability than dronabinol capsules; 2.1 mg PO solution = 2.5 mg capsules.

[s] Do not cut patches. If 1/2 patch is needed, remove only 1/2 of the backing prior to application. Cover with occlusive dressing to keep in place.

[t] While no clinically significant drug-drug interactions with short-term use of antiemetics have emerged to date in randomized clinical trials, the Panel would discourage combining multiple dopamine-receptor antagonists as this could increase the risk of extrapyramidal reactions. When used for breakthrough CINV, a single drug from this class could be utilized.

[u] Especially for patients where anxiety may be contributing to breakthrough CINV.

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PEDIATRIC RADIOPHARMACEUTICAL ANTICANCER AGENTS – EMETOGENIC POTENTIAL
MODERATE AND MINIMAL EMETIC RISK

LEVEL	AGENT
Moderate emetic risk (>30%–90% frequency of emesis)	• I-131 MIBG (Iodine-131 Metaiodobenzylguanidine) • Lutetium Lu-177 dotatate[v,w,x]
Minimal emetic risk (<10% frequency of emesis)	• Sodium iodide I-131 • Yttrium-90 microspheres

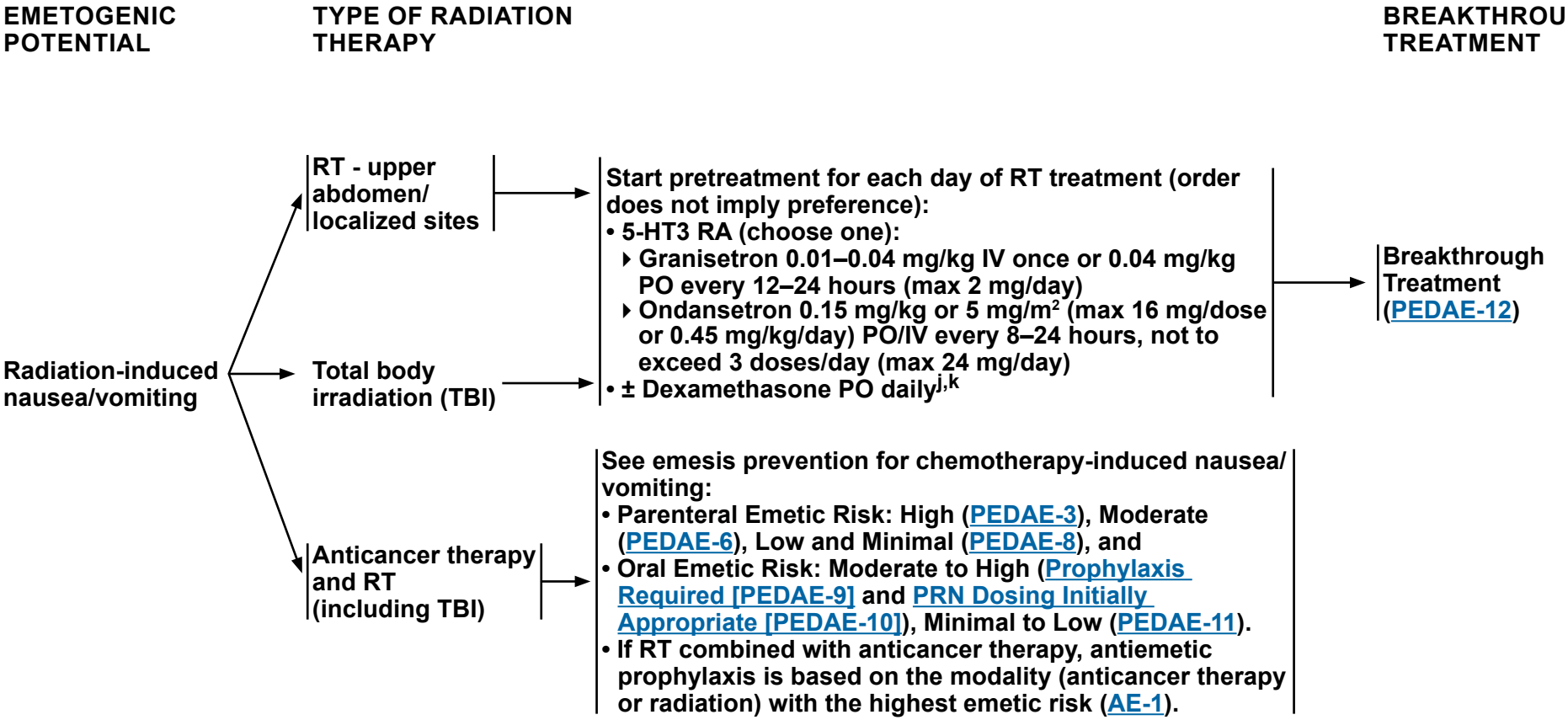
[v] Nausea associated with therapy derives largely from the amino acid infusion accompanying Lu-177 dotatate treatment; variation in emetogenicity is reported between various compounded and commercially available amino acid products.
[w] Antiemetics should be administered 30 minutes prior to the start of amino acid infusion.
[x] Glucocorticoids can induce down-regulation of SST2. The manufacturer recommends avoiding repeated administration of “high-doses” of glucocorticoids during Lu-177 dotatate treatment. The Panel suggests avoiding glucocorticoids as antiemetic prophylaxis, as other agents (5-HT3 RAs and NK1 RAs) are available options. Pivonello R, et al. Front Endocrinol (Lausanne) 2022;12:793262.

Note: All recommendations are category 2A unless otherwise indicated.

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PEDIATRIC RADIATION-INDUCED EMESIS – PREVENTION/TREATMENT



[j] Use of corticosteroid premedications should be avoided with cellular therapies. Clinicians may wish to consider a dexamethasone-sparing approach with ICI therapy as well. See Pharmacologic Considerations for Antiemetic Prescribing (AE-B).

[k] No optimal pediatric dexamethasone dose has been identified. Dexamethasone doses may be individualized. Consider doses of 5–10 mg/m²/day once daily or divided. Dexamethasone should be reduced by 50% when given concomitantly with NK1 RAs.

Note: All recommendations are category 2A unless otherwise indicated.

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PEDIATRIC ANTICIPATORY EMESIS – PREVENTION/TREATMENT

Anticipatory nausea/vomiting →

- **Prevention is key:**
  - **Use optimal antiemetic therapy during every cycle of treatment**
  - **Avoid strong smells that may precipitate symptoms**
- **Behavioral therapy:**
  - **Relaxation/systematic desensitization**
  - **Hypnosis**
  - **Relaxation exercises**
    - ◊ **Guided imagery**
    - ◊ **PMR**
    - ◊ **Biofeedback**
    - ◊ **Music therapy**
  - **Cognitive distraction**
  - **Yoga (if approved by physician)**
- **Acupuncture/acupressure**
- **Consider anxiolytic therapy:**
  - **Lorazepam 0.025–0.05 mg/kg/dose PO beginning on the night before treatment and then repeated the next day and/or continued PO/IV every 6 hours before anticancer therapy begins (max 2 mg/dose)[u]**

  or
  - **Olanzapine 0.05–0.2 mg/kg (2.5–10 mg) PO once at bedtime prior to first dose of anticancer therapy[n]**

→ **See emesis prevention for chemotherapy-induced nausea/vomiting:**
- **Parenteral Emetic Risk: High (PEDAE-3), Moderate (PEDAE-6), Low and Minimal (PEDAE-8)**
- **Oral Emetic Risk: Moderate to High (Prophylaxis Required [PEDAE-9] and PRN Dosing Initially Appropriate [PEDAE-10]), Minimal to Low (PEDAE-11)**
- **Breakthrough Treatment PEDAE-12)**

[n] Pediatric randomized controlled trials evaluating olanzapine excluded patients <5 years of age. However, safety has been established among pediatric patients with mental health conditions; thus, caution is warranted in younger children.
[u] Especially for patients where anxiety may be contributing to breakthrough CINV.

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ABBREVIATIONS

AUC	area under the curve
CAM	complementary and alternative medicine
CAR	chimeric antigen receptor
CINV	chemotherapy-induced nausea and vomiting
CNS	central nervous system
ECG	electrocardiogram
EPS	extrapyramidal symptom
HCT	hematopoietic cell transplant
HEC	highly emetogenic chemotherapy
ICI	immune checkpoint inhibitor
MEC	moderately emetogenic chemotherapy
PMR	progressive muscle relaxation
PPI	proton pump inhibitor
PRN	as needed
RA	receptor antagonist
SST2	subtype 2 somatostatin receptor
TBI	total body irradiation
TD	tardive dyskinesia
TIL	tumor-infiltrating lymphocyte

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NCCN Categories of Evidence and Consensus	
Category 1	Based upon high-level evidence (≥1 randomized phase 3 trials or high-quality, robust meta-analyses), there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2A	Based upon lower-level evidence, there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2B	Based upon lower-level evidence, there is NCCN consensus (≥50%, but <85% support of the Panel) that the intervention is appropriate.
Category 3	Based upon any level of evidence, there is major NCCN disagreement that the intervention is appropriate.

All recommendations are category 2A unless otherwise indicated.

NCCN Categories of Preference	
Preferred	Interventions that are based on superior efficacy, safety, and evidence; and, when appropriate, affordability.
Other recommended	Other interventions that may be somewhat less efficacious, more toxic, or based on less mature data; or significantly less affordable for similar outcomes.
Useful in certain circumstances	Other interventions that may be used for selected patient populations (defined with recommendation).

All recommendations are considered appropriate.

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Discussion

This discussion corresponds to the NCCN Guidelines for Antiemesis. Last updated on May 12, 2025.

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Discussion update in progress

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Overview

Vomiting (emesis) and nausea caused by anticancer agents and/or radiation therapy (RT) can significantly affect a patient's quality of life (QOL), leading to non-adherence to treatment with further anticancer agents and/or RT.[1,2] In addition, nausea and/or vomiting can result in dehydration, metabolic imbalances, degeneration of self-care and functional ability, nutrient depletion, anorexia, decline of the patient's performance status and mental status, wound dehiscence, esophageal tears, and withdrawal from potentially useful or curative anticancer treatment.[3-6] Anticancer agents include chemotherapy, targeted therapy, and immunotherapy, which will all be referred to as *anticancer agents* throughout this Discussion text.

The incidence and severity of nausea and/or vomiting in patients receiving anticancer agents and/or RT is affected by numerous factors, including: 1) the specific therapeutic agents used; 2) dosage of the agents; 3) schedule and route of administration of the agents; 4) target of the RT (eg, whole body, upper abdomen); and 5) individual patient variability (eg, younger age; female sex; prior anticancer agents; history of little or no alcohol use, morning sickness, motion sickness, anxiety).[7,8] More than 90% of patients receiving highly emetogenic chemotherapy (HEC) will have episodes of vomiting. However, if patients receive prophylactic (preventive) antiemetic regimens before treatment with HEC, then only approximately 30% of these patients will vomit.[7,9,10] Although vomiting can often be prevented or substantially decreased by using prophylactic antiemetic regimens, nausea is harder to control.[11-14]

The NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) for Antiemesis are intended to provide an overview of the treatment principles for preventing anticancer agent- or RT-induced nausea and/or vomiting, and recommendations for antiemetic prophylaxis according to the emetogenic potential of anticancer agents. By definition, the NCCN Guidelines® cannot incorporate all possible clinical variations and are not intended to replace good clinical judgment or individualization of treatments.

The NCCN Guidelines also provide specific category designations for all interventions in the guidelines, which are based on evidence from the biomedical literature and consensus among the Panel members. In contrast to other NCCN Guidelines in which most of the recommendations are category 2A, many of the recommendations for antiemetic regimens are category 1, reflecting the large number of randomized controlled trials (RCTs) that have focused on antiemetic management.

Discussion
Update in progress

Guidelines Update Methodology

The complete details of the Development and Update of the NCCN Guidelines are available at www.NCCN.org.

Literature Search Criteria

Prior to the update of the NCCN Guidelines for Antiemesis, an electronic search of the PubMed database was performed to obtain key literature in Antiemesis published since the previous Guidelines update, using the search terms: chemotherapy-induced nausea vomiting, antiemetics chemotherapy, antiemetic regimens, and antiemesis. The PubMed database was chosen because it remains the most widely used resource for medical literature and indexes peer-reviewed biomedical literature.[15]

The search results were narrowed by selecting studies in humans published in English. Results were confined to the following article types: Clinical Trial, Phase 2; Clinical Trial, Phase 3; Clinical Trial, Phase 4; Guideline; Meta-Analysis; Practice Guideline; Randomized Controlled Trial; Systematic Reviews; and Validation Studies. The data from key PubMed articles as well as articles from additional sources deemed as relevant to these guidelines as discussed by the Panel during the

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Guidelines update have been included in this version of the Discussion section. Recommendations for which high-level evidence is lacking are based on the Panel's review of lower-level evidence and expert opinion.

Sensitive/Inclusive Language Usage

NCCN Guidelines strive to use language that advances the goals of equity, inclusion, and representation.[16] NCCN Guidelines endeavor to use language that is person-first; not stigmatizing; anti-racist, anti-classist, anti-misogynist, anti-ageist, anti-ableist, and anti-weight-biased; and inclusive of individuals of all sexual orientations and gender identities. NCCN Guidelines incorporate non-gendered language, instead focusing on organ-specific recommendations. This language is both more accurate and more inclusive and can help fully address the needs of individuals of all sexual orientations and gender identities. NCCN Guidelines will continue to use the terms men, women, female, and male when citing statistics, recommendations, or data from organizations or sources that do not use inclusive terms. Most studies do not report how sex and gender data are collected and use these terms interchangeably or inconsistently. If sources do not differentiate gender from sex assigned at birth or organs present, the information is presumed to predominantly represent cisgender individuals. NCCN encourages researchers to collect more specific data in future studies and organizations to use more inclusive and accurate language in their future analyses.

Pathophysiology of Emesis

Vomiting results from stimulation of a multistep reflex pathway controlled by the brain.[1,7,17] Vomiting is triggered by afferent impulses to the vomiting center (located in the medulla) from the chemoreceptor trigger zone, pharynx and gastrointestinal (GI) tract (via vagal afferent fibers), and cerebral cortex. Vomiting occurs when efferent impulses are sent from the vomiting center to the salivation center, abdominal muscles, respiratory center, and cranial nerves.[18]

The chemoreceptor trigger zone, vomiting center, and GI tract have many neurotransmitter receptors. Activation of these receptors by chemotherapeutic agents or their metabolites may be responsible for anticancer agent–induced emesis. The principal neuroreceptors involved in the emetic response are the serotonin (5-hydroxytryptamine [5-HT3]) and dopamine receptors; 5-HT3 receptors are associated with acute emesis via a peripheral pathway.[1,19,20] Other neuroreceptors involved in emesis include acetylcholine, corticosteroid, histamine, cannabinoid, opioid, and neurokinin-1 (NK1) receptors, which are located in the vomiting and vestibular centers of the brain.[21] NK1 receptors are associated with delayed emesis via a central pathway.[1]

Antiemetics can block different neuronal pathways, exert their effects at different points during the course of emesis, or behave synergistically with other antiemetics to potentiate an antiemetic effect. When used at a certain concentration, each antiemetic agent predominantly blocks one receptor type. Olanzapine is the exception in that it acts on multiple receptors involved in the emetic pathway.[22] A final common pathway for emesis has yet to be identified. No single agent can be expected to provide complete protection from the various emetic phases of anticancer agents. Therefore, prophylactic antiemetic regimens for HEC and moderately emetogenic chemotherapy (MEC) include two to four antiemetics that block different receptors.

Nausea

With use of effective antiemetic regimens, patients receiving emetogenic anticancer agents often experience more nausea than vomiting.[11-13,23-25] Vomiting and nausea are related; however, they may occur via different mechanisms.[26,27] In general, younger patients are more likely to have nausea than older patients. Younger females receiving anticancer agents for breast cancer are more prone to nausea than other populations.[14] Delayed nausea is more common than acute nausea, is often more

Discussion Update in progress

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severe, and tends to be resistant to treatment (see *Delayed Nausea* in this Discussion).[25]

Types of Nausea and/or Vomiting

Chemotherapy-Induced Nausea and/or Vomiting

Nausea and/or vomiting induced by anticancer agents has traditionally been referred to as chemotherapy-induced nausea and/or vomiting (CINV); it is commonly classified as acute, delayed, anticipatory, breakthrough, or refractory.

Acute-Onset Nausea and/or Vomiting

Acute-onset nausea and/or vomiting usually occurs within a few minutes to several hours after administration of certain anticancer agents and commonly resolves within the first 24 hours. The intensity of acute-onset emesis generally peaks after 5 to 6 hours. Factors that influence acute emesis include type and dosage of the emetogenic agent, history of nausea and/or vomiting, environment in which anticancer agents are administered, and efficacy of the antiemetic regimen.[28] The occurrence of acute emesis is increased in younger (<50 years) females with a history of no or low ethanol use, motion sickness, or morning sickness.

Delayed-Onset CINV

Delayed-onset CINV develops in patients more than 24 hours after anticancer agent administration.[29,30] It occurs commonly with the administration of HEC, such as cisplatin, carboplatin, cyclophosphamide, and/or anthracyclines. For cisplatin, emesis reaches its maximal intensity 48 to 72 hours after administration and can last for 6 to 7 days.

Anticipatory CINV

Anticipatory CINV occurs before patients receive their next treatment with anticancer agents. Because it is primarily considered a conditioned response, anticipatory emesis typically occurs after a previous negative experience with anticancer agents. The incidence of anticipatory CINV ranges from 18% to 57%, and nausea is more common than vomiting.[31,32] Younger patients may be more susceptible to anticipatory nausea and/or vomiting, because they generally receive more aggressive anticancer agents and, overall, have poorer emesis control than older patients.[33]

Breakthrough CINV

Breakthrough CINV refers to nausea and/or vomiting that occurs despite prophylactic antiemesis treatment and/or requires rescue with antiemetics.[34]

Refractory CINV

Refractory CINV refers to nausea and/or vomiting that occurs during subsequent treatment cycles when antiemetic prophylaxis and/or rescue has not been effective in earlier cycles.[35]

Causes of Nausea and/or Vomiting

As enumerated in the guideline, aside from CINV, the differential diagnoses of nausea and/or vomiting in patients with cancer include: partial or complete bowel obstruction, vestibular dysfunction, brain metastases, electrolyte imbalances (ie, hypercalcemia, hyperglycemia, hyponatremia), uremia, gastroparesis, excess secretions (eg, such as those seen in patients with head and neck cancer), malignant ascites, cannabinoid hyperemesis syndrome,[36] rapid opioid withdrawal,[37] dysmotility, concomitant RT, and pancreatitis, as well as psychophysiologic causes. A thorough evaluation of the patient is thus

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necessary to rule out alternative causes of nausea and/or vomiting and decide on the most appropriate intervention.

Radiation-Induced Nausea and/or Vomiting

Patients receiving total body RT have the greatest likelihood of developing nausea and/or vomiting (>90% emesis); those receiving upper abdominal RT are at moderate risk of emesis (30%–90%).[34,38-40] The GI tract (specifically, the small intestine) contains rapidly dividing cells that are particularly sensitive to RT. In addition, the potential for nausea and/or vomiting increases with larger daily fractional doses of RT, larger total doses, and larger amounts of irradiated tissue. Total body irradiation (TBI), when given before hematopoietic cell transplantation (HCT), commonly induces nausea and/or vomiting.[34,38,41]

Emetogenicity of Anticancer Agents

The frequency of anticancer agent–induced emesis depends primarily on the emetogenic potential of the specific chemotherapeutic agents used. Several classifications have been developed to define the emetogenicity of anticancer agents; however, none have been universally accepted.[18,42-45]

Hesketh and colleagues developed a classification of the acute emetogenicity of anticancer chemotherapeutic agents and developed an algorithm to define the emetogenicity of combination chemotherapeutic regimens.[9] The classification was updated by Grunberg and colleagues; it divides chemotherapeutic agents into four levels according to the percentage of patients who experience acute emesis when they do not receive antiemetic prophylaxis.[46] This classification is used in these NCCN Guidelines and is updated each year by the NCCN Panel with recently introduced drugs.

The NCCN Guidelines currently outline antiemetic treatment using four categories of emetogenic potential for parenteral agents, which correspond to the Hesketh/Grunberg classification as follows:

- High emetic risk – >90% of patients experience acute emesis
- Moderate emetic risk – >30% to 90% of patients experience acute emesis
- Low emetic risk – 10% to 30% of patients experience acute emesis
- Minimal emetic risk – <10% of patients experience acute emesis

In addition, the NCCN Guidelines attempt to define antiemetic regimens for particular anticancer agents that cover the entire duration of time a patient is at risk for nausea and/or vomiting. Panel members were concerned that some patients may not receive adequate prophylaxis for delayed emesis; therefore, the NCCN Guidelines incorporate a dosing schedule that covers both acute and delayed emesis into single algorithms for HEC and MEC. The NCCN Panel has also categorized the emetogenic potential of oral anticancer agents.[46] In the current version, the Panel added a section on the emetogenic potential of available radiopharmaceutical anticancer agents in the NCCN Guidelines.

Clinicians should avoid overuse of antiemetics, especially in settings where the anticancer agents are of minimal or low emetic risk, to avoid exposing patients to adverse effects from antiemetics, to decrease possible drug-drug interactions, and to prevent unnecessary expense (see *Principles of Emesis Control for the Patient with Cancer* in the algorithm).[38,47,48] Routine antiemetic premedication may not be required for continuous dosing of some low emetic risk parenteral agents or some moderate to high emetic risk oral agents; an individualized approach is appropriate in these settings. If clinicians use the emetogenic classification of anticancer agents in the NCCN Guidelines, this will decrease unnecessary prescribing of antiemetics.

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Types of Antiemetic Therapies

In general, to provide maximal protection against anticancer agent–induced emesis, antiemetic therapy should be initiated before treatment with anticancer agents. Antiemetic therapy should also be continued for the same length of time as the duration of the anticancer agents being used. However, daily use of certain antiemetics, such as dexamethasone, may not be recommended for some anticancer agents that are taken long-term on a regular basis, such as the oral anticancer agents of moderate/high emetic risk (see *Oral Anticancer Agents - Emesis Prevention* in the algorithm). Antiemetics can be administered by the oral (PO), sublingual, rectal, intravenous (IV), intramuscular, subcutaneous, or transdermal route. Parenteral and PO 5-HT3 antagonists have equivalent efficacy when used at the appropriate doses.[10,49] However, subcutaneous granisetron extended-release injection and IV granisetron are not interchangeable; the subcutaneous formulation should not be given via IV route and vice versa. Aprepitant injectable emulsion and IV fosaprepitant are also not interchangeable. The dosing is different for all of these formulations. For patients at risk for CINV or unable to swallow or digest tablets because of emesis, non-PO antiemetics are recommended.

Although studies may show antiemetics to be equally effective on a population basis, individual patients may have different responses to treatment. Therefore, some antiemetic options may be based on a patient's individual experience. Patients may be at risk for drug-drug interactions if they are receiving anticancer agents along with antiemetic regimens; clinicians should balance benefit and risk for each patient. Many drug-drug interactions between antiemetics and anticancer agents occur with chronic dosing and are often not clinically relevant with short-term use of prophylactic antiemetic regimens, as shown by the lack of clinically significant drug-drug interactions in randomized trials of anticancer agents along with antiemetic regimens.[50]

Serotonin (5-HT3) Antagonists

Ondansetron, Granisetron, and Dolasetron

All of the 5-HT3 antagonists—dolasetron mesylate, granisetron, ondansetron, and palonosetron—have been shown to be effective in controlling acute nausea and/or vomiting associated with anticancer agents.[51-67] Ondansetron, granisetron, and dolasetron mesylate are first-generation 5-HT3 antagonists. Many clinical trials have compared ondansetron, granisetron, dolasetron mesylate, and palonosetron. These trials have used various doses, routes, and schedules of administration.[68-85] A meta-analysis found no difference in efficacy between the first-generation 5-HT3 antagonists.[86] Another meta-analysis of studies comparing ondansetron with granisetron has also confirmed the similar efficacy of these first-generation 5-HT3 antagonists in controlling acute and delayed nausea and vomiting, with similar safety profiles between these agents.[87] The *CYP2D6* gene encodes an enzyme involved in the metabolism of 5-HT3 agonists, such as ondansetron, affecting its efficacy in the prevention of postoperative nausea and vomiting.[88,89] For patients with ultrarapid metabolism of *CYP2D6*, ondansetron may have a lower antiemetic response and the NCCN Panel recommends the use of an alternative 5-HT3 regimen.

A meta-analysis of RCTs comparing palonosetron with the first-generation 5-HT3 antagonists reported that palonosetron was significantly more effective in preventing acute and delayed nausea and vomiting for both HEC and MEC; most patients receiving MEC actually received anthracycline and cyclophosphamide (AC regimens).[90] AC regimens are classified as HEC, although they were previously classified as MEC.[38,91] Palonosetron is preferred for MEC if the regimen does not contain an NK1 receptor antagonist (RA) or olanzapine (see *Palonosetron* in this Discussion).[69] Similar to palonosetron, the Panel also recommends subcutaneous granisetron extended-release injection as a preferred 5-HT3 antagonist option when used with dexamethasone in antiemetic regimens

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that do not contain an NK1 RA or olanzapine (see *Principles of Managing Multiday Emetogenic Chemotherapy Regimens* in the algorithm).[92]

Ondansetron, granisetron, and dolasetron are effective in preventing acute emesis but appear to be less effective for delayed emesis. A meta-analysis of RCTs found that adding a 5-HT3 antagonist to dexamethasone did not improve the antiemetic effect of dexamethasone for preventing delayed emesis.[93] Another study found that 5-HT3 antagonists (except for palonosetron, which was not studied) were not more effective than prochlorperazine in preventing delayed emesis.[25] A single dose of IV palonosetron appears to be effective for preventing both delayed and acute emesis.

The NCCN Guidelines recommend IV palonosetron as a preferred 5-HT3 antagonist for MEC when used with dexamethasone but without an NK1 RA or olanzapine (see *Principles of Managing Multiday Emetogenic Chemotherapy Regimens* in the algorithm).[69] Several studies[94-97] have evaluated the efficacy of a three-drug combination regimen with palonosetron, dexamethasone, and NK1 RAs as prophylaxis in patients receiving MEC (see *Neurokinin-1 Receptor Antagonists* in this Discussion). However, these studies do not provide evidence that a single dose of palonosetron is better than a single dose of a first-generation 5-HT3 antagonist when using an NK1-antagonist–containing regimen for MEC.

A phase 3 trial assessed subcutaneous granisetron extended-release injection versus IV palonosetron in a two-drug regimen with dexamethasone for patients receiving HEC or MEC.[92] Two doses of subcutaneous granisetron extended-release injection were assessed: 5 mg and 10 mg. The data showed that subcutaneous granisetron extended-release injection is not inferior to IV palonosetron in preventing acute and delayed CINV after either HEC or MEC. For patients receiving HEC, acute complete responses (CRs) for the 5- or 10-mg granisetron doses were 77.7% (-12.1, 6.1) and 81.3% (-8.2, 9.3), respectively, compared with 80.7% for those receiving a 0.25-mg dose of IV palonosetron. For patients receiving MEC, acute CRs for 5 mg or 10 mg of subcutaneous granisetron were 74.8% (−9.8, 9.3) and 76.9% (−7.5, 11.4), respectively, compared with 75.0% for palonosetron. The U.S. Food and Drug Administration (FDA) approved the use of a 10-mg dose of subcutaneous granisetron extended-release injection when used in antiemetic regimens for MEC or AC combination anticancer agent regimens. Based on this trial and the FDA approval, the NCCN Panel recommends IV palonosetron or subcutaneous granisetron extended-release injection as preferred 5-HT3 antagonists for MEC when used with dexamethasone in antiemetic regimens that do not contain an NK1 RA or olanzapine. The Panel does not recommend these two-drug antiemetic regimens for HEC. The Panel recommends for HEC either a four-drug preferred regimen, which includes olanzapine and an NK1 RA, or three-drug antiemetic regimens, which include an NK1 RA or olanzapine.

MAGIC, a phase 3 randomized trial, assessed a single dose of subcutaneous granisetron extended-release injection compared with a single dose of IV ondansetron in a three-drug regimen with dexamethasone and fosaprepitant for patients receiving HEC.[98,99] The data show that the regimen containing granisetron extended-release injection improved the CR rate (no emesis and rescue medication) for delayed-phase CINV (24–120 hours) compared with the ondansetron regimen ($P = .014$). This was the first published trial that compared a single dose of two different 5-HT3 antagonists when used in combination with dexamethasone and an NK1 RA. As a result, granisetron extended-release injection was the first FDA-approved 5-HT3 antagonist indicated for the prevention of delayed CINV associated with AC anticancer agents. When administered subcutaneously, granisetron extended-release injection is effective for ≥5 days.

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The NCCN Panel recommends a 10-mg dose of subcutaneous granisetron extended-release injection on day 1 only for patients receiving either HEC or MEC when used in the antiemetic regimens based on the MAGIC trial, the trial comparing dexamethasone with either palonosetron or subcutaneous granisetron, and the FDA approval.[92,98,99] It is important to note that granisetron extended-release injection is a unique formulation of granisetron using a polymer-based drug delivery system. This formulation is specifically intended for subcutaneous administration and is NOT interchangeable with the IV formulation; the subcutaneous formulation should not be injected and vice versa. Subcutaneous granisetron extended-release injection has an extended half-life and should not be administered at less than 1-week intervals.

Ondansetron and granisetron can be delivered via PO/IV route; granisetron extended-release injection is administered subcutaneously. Note that use of IV dolasetron or 32 mg of ondansetron is no longer recommended for the prevention of nausea and/or vomiting, because each has been associated with an increased risk for cardiac arrhythmias (see *Cardiac Side Effects* in this Discussion).[100-103] Ondansetron PO administration poses less of a risk for cardiac arrhythmias than IV administration.[102] Dolasetron PO is still recommended.

A phase 3 randomized trial compared the granisetron transdermal patch to PO granisetron in patients receiving either HEC or MEC. The patch contains 3.1 mg of granisetron every 24 hours and is applied approximately 24 to 48 hours before the first dose of anticancer agents; the maximum duration of the patch is 7 days. The patch proved non-inferior to repeat dosing of the oral antiemetic granisetron over 3 to 5 days.[13,104] A phase 4 trial assessed an antiemetic regimen containing the transdermal granisetron patch versus a palonosetron regimen for patients receiving MEC; transdermal granisetron was not inferior to palonosetron in preventing nausea and vomiting in the acute stage.[105] The NCCN Panel recommends the granisetron transdermal patch as a 5-HT3 option when used as part of recommended antiemetic regimens for patients receiving either HEC or MEC based on clinical trial data and the FDA approval.[13,104,105] No further 5-HT3 therapy is required on days 2 to 3 if a granisetron transdermal patch is applied on day 1 or if palonosetron or granisetron extended-release injection is given on day 1.

The addition of dexamethasone improves the efficacy of antiemetic regimens containing 5-HT3 antagonists (see *Dexamethasone* in this Discussion). However, dexamethasone is associated with side effects, such as insomnia. When dexamethasone is used with palonosetron for MEC, a randomized trial suggests that the dose of dexamethasone can be decreased to 8 mg on day 1 and also eliminated on days 2 to 3.[106]

Cardiac Side Effects

Ondansetron, granisetron, and dolasetron have been associated with an increased risk for developing abnormal electrical activity of the heart that is detectable on electrocardiogram (ECG), including prolongation of electrocardiographic intervals, such as PR or QT intervals.[102,103,107-114] Although the ECG changes can be reversible and asymptomatic, abnormal activity can also result in potentially fatal cardiac arrhythmias (including torsade de pointes) in some instances.[102] Patients who may be particularly at risk for developing torsade de pointes include those with congenital long QT syndrome or other underlying cardiac diseases, congestive heart failure, bradycardia, those with electrolyte abnormalities (eg, hypokalemia, hypomagnesemia), and those taking other medications that can lead to QT prolongation.[103,111,115] A single IV dose of 32 mg of ondansetron is no longer recommended based on FDA review of clinical data suggesting prolongation of the QT interval of the ECG at this dose.[100-102] The FDA recommends a maximum single IV dose of 16 mg of ondansetron given once on the first day; the dose recommendations for ondansetron are 16 to 24 mg PO given once on the first day.[101] IV

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dolasetron is no longer recommended for the prevention of nausea and vomiting, because it has been associated with an increased risk for cardiac arrhythmias.[102,103]

Palonosetron

Palonosetron is a 5-HT3 antagonist with an approximately 100-fold higher binding affinity for the 5-HT3 receptor compared to ondansetron, granisetron, and dolasetron. Palonosetron has a half-life of approximately 40 hours, which is significantly longer than other commercially available 5-HT3 antagonists.[53] Data suggest that palonosetron is associated with prolonged inhibition of the 5-HT3 receptor and thus differs from ondansetron, granisetron, and dolasetron.[116,117] By suppressing cross talk between 5-HT3 and NK1 signaling pathways, palonosetron may indirectly inhibit substance P.

Several randomized phase 3 trials have assessed the efficacy of palonosetron compared with other 5-HT3 antagonists in preventing emesis associated with both MEC and HEC regimens, particularly for delayed emesis.[68-71] In these studies, the primary efficacy endpoint was CR, defined as having no emesis and no rescue treatments. In a study in patients receiving MEC (N = 563 evaluable), a single dose of palonosetron (0.25 mg IV) was found to be superior to a single dose of ondansetron (32 mg IV) in preventing both acute (CR rate, 81% vs. 69%; $P < .01$) and delayed emesis (CR rate, 74% vs. 55%; $P < .01$); no concomitant corticosteroids were given in this study.[71] The safety and side-effect profiles of palonosetron were indistinguishable from the control 5-HT3 antagonists (ondansetron and dolasetron). Note that the FDA recommends a maximum of 16 mg for a single dose of IV ondansetron.[102]

A phase 3 randomized trial compared palonosetron with ondansetron in patients receiving HEC (N = 667), and most patients (67%) received dexamethasone on day 1 of antiemetic therapy; NK1 RAs were not used in this trial.[68] Among this subgroup of patients who received concomitant dexamethasone (n = 447), palonosetron (0.25 mg IV) was similar to ondansetron (32 mg IV) in preventing acute emesis (CR rate, 65% vs. 56%); however, palonosetron was significantly more effective in preventing delayed emesis (CR rate, 41% vs. 25%; $P = .021$).

Another phase 3 randomized trial in patients treated with HEC (N = 1114 evaluable) compared a single dose of palonosetron (at a higher dose of 0.75 mg IV) with a single dose of granisetron (40 mcg/kg IV), both in combination with dexamethasone; NK1 RAs were not used in this trial. Palonosetron showed similar activity to granisetron in preventing acute emesis (CR rate, 75% vs. 73%) and superior activity in preventing delayed emesis (CR rate, 57% vs. 44.5%; $P < .0001$).[69] A meta-analysis of 24 RCTs assessed whether palonosetron was more efficacious than the other 5-HT3 antagonists. Although palonosetron seems to be more efficacious and safe than other 5-HT3 RAs and was statistically superior in 10 of 19 endpoints, overall the authors suggest that palonosetron should generally not be the preferred 5-HT3 antagonist.[118] The NCCN Panel does not recommend palonosetron as the preferred 5-HT3 antagonist in regimens for HEC, because an NK1 RA was not used in these studies and it is unknown if a single dose of palonosetron would be superior to a single dose of granisetron in the presence of an NK1 RA.[68-71,118]

As previously mentioned, the NCCN Panel recommends either palonosetron or subcutaneous granisetron extended-release injection as preferred 5-HT3 antagonists for MEC when used with dexamethasone in two-drug antiemetic regimens that do not contain an NK1 RA or olanzapine (see *Ondansetron, Granisetron, and Dolasetron* in this Discussion and *Principles of Managing Multiday Emetogenic Chemotherapy Regimens* in the algorithm).[92] Palonosetron (0.25 mg IV) is FDA approved as a single dose on day 1 for the prevention of acute and delayed nausea and vomiting associated with MEC and for the prevention of acute nausea and vomiting associated with HEC. No further 5-HT3

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therapy is required for MEC on days 2 to 3 if palonosetron or granisetron extended-release injection is given on day 1 or if granisetron transdermal patch is applied on day 1.

IV palonosetron is superior to other first-generation 5-HT3 antagonists in preventing delayed nausea.[24,68-71] Repeat dosing of palonosetron on days 2 or 3 after anticancer agents is likely to be safe. However, in the setting of multiday anticancer agents, limited data are available to recommend multiday dosing with palonosetron (see *Principles of Managing Multiday Emetogenic Chemotherapy Regimens* in the algorithm).[119]

Neurokinin-1 Receptor Antagonists

For patients receiving HEC and MEC, the NCCN Panel recommends several options for prophylactic antiemetic regimens based on clinical trial data and FDA approvals, including: 1) NK1 RA-containing regimens, which are discussed in this section; and 2) olanzapine-containing regimens. NK1 RA regimens include PO aprepitant, IV fosaprepitant, PO rolapitant, PO netupitant, or IV fosnetupitant. Although no head-to-head clinical trial data with a superiority endpoint have been published, emerging clinical evidence based on non-inferiority endpoints as well as anecdotal evidence suggest that there may be differences in efficacy.[120,121] It is important to note that PO netupitant (or IV fosnetupitant) is only available in combination with palonosetron (NEPA); netupitant is not available as a single agent.

A prophylactic two-drug regimen of one of the 5-HT3 antagonists plus dexamethasone is recommended for MEC but not HEC. However, a prophylactic three-drug antiemetic regimen that includes either an NK1 RA or olanzapine is recommended for select patients receiving MEC who have additional risk factors or for whom previous treatment with the two-drug regimen was ineffective. These additional risk factors include younger age; female sex; anxiety and/or high pretreatment expectation of nausea and/or vomiting; and history of CINV, motion sickness, morning sickness during pregnancy, and little or no alcohol use.[122] For select patients, including those with additional patient-related risk factors or for whom previous treatment with a 3-drug prophylactic regimen was ineffective, or who are receiving MEC known to be of higher risk in some patients, a 4-drug regimen may be considered. [123] For select patients with additional patient-related risk factors (see *Principles of Emesis Control for the Patient With Cancer* in the algorithm) or for whom previous treatment with a 3-drug prophylactic regimen (E or F) was ineffective, or who are receiving MEC known to be higher risk, a 4-drug regimen (see *High Emetic Risk Parenteral Anticancer Agents — Acute and Delayed Emesis Prevention*, Treatment option A within the algorithm) may be considered. Patients receiving anticancer agents that are classified as moderate emetic risk but are at the higher end of the risk spectrum (eg, carboplatin, carmustine, cyclophosphamide, dactinomycin, daunorubicin, doxorubicin, epirubicin, idarubicin, ifosfamide, irinotecan, methotrexate, oxaliplatin, trabectedin) are at greater risk for emesis and may also need a three-drug prophylactic antiemetic regimen.

Aprepitant

Aprepitant selectively blocks the binding of substance P at the NK1 receptor in the central nervous system. Thus, it provides a different and complementary mechanism of action compared with other commercially available antiemetics. Aprepitant has been shown to augment the antiemetic activity of the 5-HT3 antagonists and the corticosteroid dexamethasone to prevent both acute and delayed cisplatin-induced emesis.[124-126] Most of the clinical trial data described in this Discussion are based on studies with PO aprepitant. Aprepitant injectable emulsion is a formulation of aprepitant that is approved by the FDA for HEC and MEC when used in combination with other antiemetic regimens.[127]

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Oral Aprepitant

A randomized phase 3 trial compared ondansetron 32 mg IV and PO dexamethasone with or without the addition of PO aprepitant in patients receiving emetogenic systemic therapy with high-dose cisplatin (N = 521 evaluable). The addition of PO aprepitant was significantly more effective than the two-drug regimen in controlling both acute (CR rate, 89% vs. 78%; $P < .001$) and delayed emesis (CR rate, 75% vs. 56%; $P < .001$).[125] Another similarly designed randomized phase 3 study (N = 523 evaluable) also showed a significant benefit of adding PO aprepitant to ondansetron and dexamethasone compared with the two-drug regimen alone for controlling both acute (CR rate, 83% vs. 68%; $P < .001$) and delayed emesis (CR rate, 68% vs. 47%; $P < .001$).[126] A pooled analysis of data combined from these two phase 3 trials found that the PO aprepitant regimen was particularly beneficial in improving CR rates for patients receiving concomitant emetogenic therapy with doxorubicin and cyclophosphamide (AC regimen) or cyclophosphamide, along with high-dose cisplatin therapy.[124]

A large meta-analysis (of 17 RCTs) evaluated outcomes with typical antiemetic therapy with or without PO aprepitant in patients receiving MEC or HEC. The addition of PO aprepitant was associated with significantly improved CR (no emetic episodes and no rescue medication) rate compared with control antiemetic therapy (72% vs. 54%; $P < .001$) during the overall timeframe from 0 to 120 hours after starting anticancer agents.[128] The significant increase in CR rate associated with PO aprepitant was observed for both the acute and delayed periods. A smaller meta-analysis (of seven RCTs) of patients receiving HEC found that PO aprepitant used alone or with control antiemetic therapy did not significantly increase protection from acute emesis or nausea; however, for delayed emesis and nausea, PO aprepitant was associated with significantly increased protection compared with control.[129] Based on data from three trials that reported on infectious complications, both PO aprepitant regimens and other antiemetic regimens were associated with a low rate of severe infections (6% vs. 2%; $P < .001$); the risk of febrile neutropenia or other hematologic toxicities was not increased.[128] A randomized phase 3 trial (N = 866) showed that an PO aprepitant regimen was more effective than a control antiemetic regimen in preventing vomiting in patients receiving HEC during 120 hours after initiation of anticancer agents (CR rate, 51% vs. 43%; $P = .015$); no delayed dexamethasone was used in this trial. However, approximately 40% of patients receiving either regimen still experienced significant nausea.[130] The PO aprepitant regimen included ondansetron and dexamethasone; the control antiemetic regimen included ondansetron and dexamethasone.

A three-drug antiemetic regimen with palonosetron, dexamethasone, and PO aprepitant has also been investigated in patients undergoing treatment with HEC. A phase 2 study in patients receiving HEC with cisplatin-containing regimens (N = 222) showed that the three-drug combination of palonosetron (0.25 mg IV day 1), PO aprepitant (125 mg day 1; 80 mg days 2–3), and dexamethasone (20 mg IV day 1; 4 mg PO days 2–3) resulted in a CR rate (no emetic episodes and no rescue medication) of 70% during the overall study period (0–120 hours).[96] In addition, 93% of patients had no emesis and 60% had no nausea during the study period. Constipation was the most commonly reported adverse event (39%).[96] A phase 2 study evaluated a higher dose of palonosetron (0.75 mg IV day 1) with PO aprepitant (125 mg day 1; 80 mg days 2 to 3) and dexamethasone (10 mg PO day 1; 8 mg PO days 2–4) in patients with lung cancer undergoing HEC (N = 63); the CR rate was 81% during the overall study period (0–120 hours).[97] The CR rates during the acute and delayed phases were 97% and 81%, respectively. In addition, 54% of patients had no nausea during the overall study period. Grade 1 or 2 constipation was the most commonly reported adverse event.[97]

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A phase 3 trial added PO aprepitant to a control antiemetic regimen of PO granisetron and PO dexamethasone in patients receiving MEC. The addition of aprepitant improved control of nausea, vomiting, and QOL compared with granisetron and dexamethasone.[131] A phase 2 study (N = 58) found that combining palonosetron (0.25 mg IV day 1), aprepitant (125 mg PO day 1; 80 mg days 2–3), and dexamethasone (12 mg PO day 1; 8 mg days 2–3) was effective in preventing both acute and delayed emesis and nausea when using various chemotherapeutic regimens (moderate to moderately highly emetogenic); 78% of patients had a CR (no emetic episodes and no rescue medication) during the overall timeframe, from 0 to 120 hours after initiation of emetogenic therapy.[94] A phase 2 study in patients with breast cancer (N = 41) receiving MEC also found that a single-day regimen of palonosetron (0.25 mg IV), aprepitant (285 mg PO), and dexamethasone (20 mg PO) was effective; 76% and 66% of patients had a CR during the acute and delayed phases, respectively.[95]

A randomized, double-blind, phase 3 trial compared the effectiveness of combining ondansetron (8 mg PO twice daily [BID] day 1), aprepitant (125 mg PO day 1; 80 mg days 2–3), and dexamethasone (12 mg PO day 1) versus control antiemetic therapy with ondansetron (8 mg PO BID days 1–3) and dexamethasone (20 mg PO day 1) in patients receiving MEC (N = 585).[132] Dexamethasone was only given on day 1 for both treatment groups. A significantly higher proportion of patients in the three-drug regimen with PO aprepitant had no vomiting compared with the control antiemetic regimen (76% vs. 62%; $P < .001$) during the overall timeframe from 0 to 120 hours after starting anticancer agents. In addition, the CR (no emetic episodes and no rescue medications) rate was significantly increased in the PO aprepitant group (69% vs. 56%; $P < .001$) during the overall time period. The significant improvement in antiemetic activity (with regard to no emesis as well as CR rate) in the PO aprepitant group was observed for both the acute and delayed phases. The three-drug regimen was well tolerated, and the incidence of adverse events was similar between treatment groups.[132] Oral aprepitant is FDA approved for the prevention of nausea and vomiting in patients receiving HEC (eg, cisplatin-containing) and MEC. The PO doses of aprepitant are 125 mg on day 1 (before anticancer agents) and then 80 mg on days 2 to 3 (after anticancer agents).[133]

The NCCN Panel recommends prophylactic PO aprepitant in combination with dexamethasone, a 5-HT3 RA, and with or without olanzapine (category 1) for acute and delayed emesis prevention for HEC and MEC based on clinical trial data and on FDA approvals.[124-126]

Fosaprepitant

Fosaprepitant dimeglumine is an IV version of aprepitant, which can be given on day 1 only; it is also FDA approved. A single dose of 150 mg IV fosaprepitant was shown to be non-inferior to the control antiemetic regimen with 3-day PO aprepitant in a randomized study.[134] As previously mentioned, IV fosaprepitant is NOT interchangeable with aprepitant injectable emulsion. IV fosaprepitant is given 30 minutes before anticancer agents on day 1 only, per the package insert. If a higher dose of fosaprepitant is used (150 mg IV) on day 1, then it is not necessary to give PO aprepitant on days 2 to 3.[135,136] Note that the dexamethasone dosing is slightly different on days 3 and 4 (8 mg PO/IV BID) when using the higher dose of fosaprepitant (150 mg IV) per the package insert. There are no studies showing efficacy or safety of chronic dosing with PO aprepitant. It is possible that the drug-drug interaction profile may change with chronic dosing.

The NCCN Panel recommends prophylactic fosaprepitant in combination with dexamethasone, a 5-HT3 RA, and with or without olanzapine (category 1) for acute and delayed emesis prevention for HEC and MEC based on clinical trial data and on FDA approvals.[134]

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Aprepitant Injectable Emulsion

IV fosaprepitant contains polysorbate 80 and other surfactants that may cause infusion-site reactions including pain, erythema, and swelling.[127,137,138] Aprepitant injectable emulsion is a formulation of aprepitant that does not contain polysorbate 80 and other surfactants. A phase 1 bioequivalence study (n = 100) compared IV fosaprepitant with aprepitant injectable emulsion.[127] Patients receiving aprepitant injectable emulsion had fewer treatment-emergent adverse effects compared with those receiving IV fosaprepitant (1% vs. 20%); all these adverse events resolved. Three patients receiving IV fosaprepitant had dyspnea. None of the patients had severe treatment-emergent adverse effects, serious adverse events, or resultant death. Aprepitant injectable emulsion was bioequivalent to IV fosaprepitant (bioequivalence bounds, 80%–125%).

The NCCN Panel recommends prophylactic aprepitant injectable emulsion in combination with dexamethasone, a 5-HT3 RA, and with or without olanzapine (category 1) for acute and delayed emesis prevention for HEC and MEC based on clinical trial data and on FDA approvals.[127] As previously mentioned, aprepitant injectable emulsion is not interchangeable with IV fosaprepitant.

Drug Interactions

Aprepitant is simultaneously a substrate, moderate inducer, and moderate inhibitor of cytochrome P450 enzyme 3A4 (CYP3A4); aprepitant also induces CYP2C9.[139] Thus, aprepitant can alter the metabolism of certain drugs and change their plasma concentrations (ie, areas under the curve [AUCs]). However, these interactions are more significant with orally administered forms of these drugs than with IV forms because of first-pass metabolism. Patients should not take PO aprepitant or aprepitant injectable emulsion with pimozide or astemizole; these combinations are contraindicated because they may cause serious or life-threatening reactions (see the aprepitant package inserts). Chemotherapeutic agents known to be metabolized by CYP3A4 include docetaxel, paclitaxel, etoposide, irinotecan, ifosfamide, imatinib, vinorelbine, vinblastine, and vincristine. In clinical trials, PO aprepitant was used concurrently with etoposide, vinorelbine, or paclitaxel; although anticancer agent doses were not adjusted for potential drug interactions in phase 3 trials, no observed adverse effect or decreased efficacy was observed; caution is urged when using any chemotherapeutic agent that is metabolized by CYP3A4. A systematic review also describes potential drug-drug interactions with aprepitant and fosaprepitant.[140] However, short-term use of antiemetics may not result in clinically relevant drug interactions.[50]

Aprepitant has been shown to interact with several non-chemotherapeutic drugs, including warfarin, dexamethasone, methylprednisolone, and PO contraceptives. Induction of warfarin metabolism by aprepitant may lead to clinically significant reductions in international normalized ratio (INR) values, particularly for patients on therapeutic (as compared to prophylactic) warfarin regimens. These changes, although brief in duration, may require increased patient monitoring. Aprepitant decreases the AUC for patients taking PO contraceptives; thus, other methods of birth control should be used during treatment with aprepitant and for 1 month after the last dose of aprepitant. Additionally, certain drugs can affect the AUCs of aprepitant. Concomitant administration with CYP3A4 inhibitors (eg, ketoconazole, itraconazole, erythromycin) may lead to increased aprepitant AUCs, whereas concomitant administration with CYP3A4 inducers (eg, carbamazepine, rifampin, phenytoin) may lead to decreased levels of aprepitant.

Netupitant (or Fosnetupitant) and Palonosetron (NEPA)

Netupitant is a highly selective NK1 RA that targets serotonin and substance P–mediated pathways involved in CINV. PO netupitant is combined with PO palonosetron (NEPA) in a single tablet, and netupitant is not available as a single agent; PO NEPA is approved by the FDA for

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the prevention of nausea and vomiting in patients receiving HEC and MEC based on several randomized trials.[141-144] IV fosnetupitant is combined with IV palonosetron (IV NEPA), and fosnetupitant is also not available as a single agent; IV NEPA is approved by the FDA for the prevention of nausea and vomiting in patients receiving HEC and other types of anticancer agents.

A randomized trial in patients receiving HEC assessed dexamethasone plus three dose levels of prophylactic PO NEPA compared with PO palonosetron plus dexamethasone.[141] The PO NEPA fixed-dose combination of 300 mg of netupitant decreased nausea and vomiting in the acute, delayed, and overall phases versus palonosetron alone. The CR for the NEPA300 arm was 89.6% versus 76.5% for the palonosetron arm ($P < .050$). A randomized phase 3 trial in patients receiving AC regimens assessed PO NEPA plus dexamethasone compared with palonosetron plus dexamethasone.[143] More patients in the PO NEPA arm had CR during the delayed phase compared with the control arm (76.9% vs. 69.5%; $P = .001$). In addition, patients in the PO NEPA arm also had more CRs in the overall phases (0–120 hours) (74.3% vs. 66.6%; $P = .001$) and acute phases (0–24 hours) (88.4% vs. 85.0%; $P = .047$).

A phase 3 randomized trial assessed a single dose of PO NEPA compared with a 3-day aprepitant/granisetron regimen in patients (n = 828) receiving HEC; all patients received PO dexamethasone on days 1 through 4.[121] The PO NEPA regimen was non-inferior to the aprepitant regimen (overall CR: PO NEPA, 73.8% vs. aprepitant/granisetron, 72.4%; 95% CI, -4.5%–7.5%). Similar rates were observed for both groups for no emesis (PO NEPA, 75.0% vs. aprepitant/granisetron, 74.0%; 95% CI, -4.8%–6.9%) and no significant nausea (PO NEPA, 75.7% vs. aprepitant/granisetron, 70.4%; 95% CI, -0.6%–11.4%).

The efficacy and safety of fosnetupitant versus fosaprepitant was evaluated in the phase 3 randomized CONSOLE trial (n = 785) in patients receiving HEC; all patients received palonosetron on day 1 and dexamethasone on days 1 through 4. The fosnetupitant regimen was non-inferior to the fosaprepitant regimen (overall CR: fosnetupitant 75.2% vs. fosaprepitant 71.0%; 95% CI, -2.1%–10.3%).[120] The safety profile was similar between the two regimens, although significantly fewer injection-site reactions were associated with fosnetupitant than fosaprepitant.

The safety and efficacy of a single dose of IV NEPA compared with PO NEPA on day 1 before receiving HEC was investigated in a randomized phase 3 study (n = 404); all patients also received PO dexamethasone on days 1 through 4.[145] The overall safety profile of IV NEPA was similar to that of PO NEPA in cycle 1 and throughout the trial. No injection site reactions were considered to be treatment-related; additionally, no anaphylactic reactions occurred with either IV or PO NEPA. The no emesis rates were similar (84.2% for IV NEPA, 88.6% for PO NEPA). Another randomized phase 3b study (n = 402) compared the safety and efficacy of IV NEPA versus PO NEPA on day 1 in patients with breast cancer (who were naïve to MEC or HEC) prior to receiving AC-based chemotherapy.[146] All patients also received 12 mg of PO dexamethasone before AC chemotherapy on day 1 only. The incidence and intensity of treatment-related adverse events were similar between the IV and PO NEPA treatment groups; no infusion-site reactions were deemed to be treatment-related in either group. The efficacy of IV and PO NEPA were comparable; high CR (no emesis and no emetic rescue) rates were observed in cycle 1 (overall [0–120 hours] 73.0% for IV NEPA, 77.3% for PO NEPA) and maintained over the subsequent cycles.

The NCCN Panel recommends prophylactic PO/IV NEPA in combination with dexamethasone and with or without olanzapine (category 1) for acute and delayed emesis prevention for HEC and MEC based on randomized trials, FDA approvals, and clinical experience.[121,141,143,145] Currently, IV

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NEPA is only FDA approved for HEC. However, the NCCN Panel recommends IV NEPA regimens for MEC based on clinical experience.

Similar to the other NK1 RAs (ie, PO aprepitant, IV fosaprepitant, PO rolapitant), netupitant and fosnetupitant improve control for delayed emesis compared with traditional antiemetic regimens. Netupitant and fosnetupitant inhibit CYP3A4; therefore, caution should be used with drugs that are metabolized by CYP3A4 to avoid drug interactions (see prescribing information). Concomitant use with certain agents that are strong inducers (eg, rifampin) of CYP3A4 is contraindicated. However, short-term use of antiemetics may not result in clinically relevant drug interactions.

Rolapitant

Oral rolapitant is another NK1 RA that is approved by the FDA for the prevention of nausea and vomiting in patients receiving HEC and other types of anticancer agents based on several phase 3 randomized trials.[147,148] In a phase 3 randomized trial assessing a prophylactic PO rolapitant-containing regimen for HEC, patients received 180 mg of PO rolapitant on day 1 only; all patients received granisetron (10 mcg/kg IV) and dexamethasone (20 mg PO) on day 1, and dexamethasone (8 mg PO) BID on days 2 to 4.[148] More patients receiving the PO rolapitant-containing regimen had CRs for prevention of delayed emesis compared with those receiving granisetron/dexamethasone alone (pooled studies: 382 [71%] vs. 322 [60%]; odds ratio [OR], 1.6; 95% CI, 1.3–2.1; *P* = .0001). For patients receiving HEC, the NCCN Panel recommends several prophylactic antiemetic regimens (category 1); a 5-HT3 antagonist, dexamethasone, and PO rolapitant regimen with or without olanzapine is recommended for acute and delayed emesis prevention based on the FDA approval and the phase 3 randomized trial.[148]

A randomized phase 3 trial assessed a prophylactic PO rolapitant-containing regimen for anticancer regimens previously considered to be MEC, which are now categorized as HEC by the NCCN Panel (ie, AC regimens and regimens containing carboplatin with an AUC ≥4). With the revised definition of HEC regimens, this trial contained mostly HEC and only some MEC regimens (18% and 14% of patients had non-AC regimens and non-carboplatin regimens).[91,147] Most patients also received granisetron (2 mg PO) and dexamethasone (20 mg PO) on day 1 followed by granisetron (2 mg PO) on days 2 to 3.[147] Significantly more patients receiving the PO rolapitant-containing regimen had CRs in the delayed phase than did those receiving granisetron/dexamethasone alone (475 [71%] vs. 410 [62%]; OR, 1.6; 95% CI, 1.2–2.0; *P* = .0002). For patients receiving MEC, the NCCN Panel recommends several prophylactic antiemetic regimens (category 1); a 5-HT3 antagonist/dexamethasone (category 1) with (or without) PO rolapitant is recommended for acute and delayed emesis prevention based on the FDA approval and the phase 3 randomized trial.[147] Although most of the clinical trial data for rolapitant are in patients receiving HEC, the NCCN Panel feels that prophylactic antiemetic regimens with PO rolapitant are appropriate for patients receiving MEC based on the FDA approval and clinical experience.

Oral rolapitant has an extended half-life and should not be administered at less than 2-week intervals. If a PO rolapitant is given on day 1 for either HEC or MEC, no further NK1 RA is needed on days 2 to 3. Similar to the other NK1 RAs, PO rolapitant improves control for delayed emesis compared with traditional antiemetic regimens. Rolapitant does not inhibit or induce CYP3A4; therefore, the dexamethasone dose does not need to be adjusted (see *Dexamethasone* in this Discussion). Rolapitant, however, inhibits CYP2D6, P-glycoprotein, and BCRP; therefore, caution is required when rolapitant is used concomitantly with drugs that are substrates of these enzymes, including thioridazine, pimozide, digoxin, irinotecan, topotecan, methotrexate, and rosuvastatin. The IV formulation of rolapitant was removed because of infusion-related hypersensitivity/anaphylaxis.

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Other Antiemetics

Before the advent of the 5-HT3 antagonists and NK1 RAs, the available antiemetics included phenothiazines,[149] substituted benzamides,[150,151] antihistamines,[152] butyrophenones,[153] corticosteroids,[154-156] benzodiazepines,[157,158] and cannabinoids.[159,160] Bright et al indicates that topical capsaicin ointment may be used as an adjunct option for delayed CINV; however, care should be taken to avoid application to sores, broken or irritated skin, infected skin, and sites of prior RT.[161] Currently, capsaicin ointment is not available in the United States; however, in other studies capsaicin cream was used to treat patients with nausea and vomiting not associated with cancer treatment.[162,163] Based on clinical trial data, the NCCN Panel includes olanzapine-containing regimens as another antiemetic option. Combination antiemetic therapy is generally more effective than single-agent therapy. Other agents such as gabapentin have also been evaluated as part of antiemetic regimens.

Dexamethasone

Dexamethasone has been used for many years in combination with other antiemetics. The antiemetic effects of dexamethasone may be due to interactions with the neurotransmitter serotonin and receptor proteins tachykinin NK1 and NK2; it may also act directly at the solitary tract nucleus in the medulla.[164] Before the mid-1990s, studies assessing dexamethasone as an antiemetic agent were characterized by small sample size and variations in efficacy outcomes between the studies. A meta-analysis of 32 studies (published from 1966–1999) was done in 5613 patients; the dose range was 8 to 100 mg of dexamethasone on day 1, and the mean total dose (acute and delayed) was 56 mg.[165] The authors concluded that dexamethasone offered a clear advantage over placebo for protection against anticancer agent–induced emesis in both the acute and delayed phases. There was incremental benefit when adding dexamethasone to both 5-HT3 antagonist-containing regimens and non–5-HT3 antagonist regimens. Although data suggested that dexamethasone was superior to 5-HT3 antagonists for protection against delayed emesis, there was a lack of a strong dose/response relationship. The authors could not rule out a subtle dose/response relationship for total doses <20 mg of dexamethasone, but even low doses showed clear efficacy.

The Italian Group for Antiemetic Research conducted two randomized, phase 3 multicenter trials to determine the dose of dexamethasone to be given on day 1 of an antiemetic regimen.[166,167] The first trial was conducted in patients who were chemo-naive who were receiving ≥50 mg/m^2 of cisplatin, which is considered HEC.[166] IV dexamethasone day 1 doses were 4, 8, 12, and 20 mg (approximately 130 patients/arm). All patients received the following: 1) ondansetron 8 mg IV on day 1; 2) metoclopramide 20 mg PO every 6 hours on days 2 to 4; and 3) dexamethasone 8 mg PO BID on days 2 to 3, followed by 4 mg PO BID on day 4. Complete protection from emesis and nausea was 69.2% and 60.9%; 69.1% and 61.0%; 78.5% and 66.9%; and 83.2% and 71.0% for the 4-, 8-, 12-, and 20-mg dexamethasone doses, respectively. For protection against acute emesis, the 20-mg dose of dexamethasone was statistically significant compared to the 4- and 8-mg doses. However, the 20- and 12-mg doses of dexamethasone were equivalent for protection against acute emesis. The 20-mg dose of dexamethasone was not significantly different from the other doses for protection against acute nausea. Adverse effects and control of delayed emesis and nausea were similar among the four groups.

The second Italian study compared three dosing regimens of dexamethasone on day 1 in patients receiving anthracyclines, cyclophosphamide, or carboplatin, either alone or in combination with other anticancer agents, which previously were considered to be MEC.[167] Note that AC regimens are now considered to be HEC by the NCCN Panel; likewise, carboplatin with an AUC of ≥4 is now considered to be HEC. For the prevention of acute emesis, during the first 24 hours, one of

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the following dexamethasone regimens was used in combination with 8 mg of IV ondansetron: 1) for arm A, 8 mg of IV dexamethasone before anticancer agents plus 4 mg PO dexamethasone every 6 hours for four doses, starting at the same time as the anticancer agents; 2) for arm B, 24 mg of IV single-dose dexamethasone before anticancer agents; or 3) for arm C, 8 mg of IV single-dose dexamethasone before anticancer agents. All patients received PO dexamethasone 4 mg BID on days 2 to 5. Complete protection from acute vomiting and nausea was 84.6% and 66.7%, 83.6% and 56.9%, and 89.2% and 61.0% for arms A, B, and C, respectively. The side effects and control of delayed vomiting and nausea were not significantly different between the three groups. The authors concluded that 8 mg of IV dexamethasone is the best dose when using dexamethasone in antiemetic regimens for patients receiving anticancer agents with these agents. Of note, 95% of the patients were being treated for breast cancer.

Information from early studies with PO aprepitant-containing regimens suggested that the dose of dexamethasone should be decreased from 20 mg to 12 mg because of a near doubling in the AUC of dexamethasone, presumably due to CYP3A4 inhibition (see *Drug Interactions* in this Discussion). The studies by the Italian Group for Antiemetic Research were done before the NK1 RAs were available, and dose-finding studies for dexamethasone on day 1 in combination with NK1 RAs and 5-HT3 antagonists have not been done.[166,167] This information, along with the previous data showing a lack of a dose/response correlation, was the basis of the NCCN Panel's recommendation of 12 mg of dexamethasone as the day 1 dose for all emetic categories when using any of the recommended antiemetic regimens to simplify the recommendations as described in the next paragraph.

Dexamethasone is associated with some side effects, such as insomnia, dyspepsia, hiccups, and increased serum glucose levels. Consider AM dosing to minimize insomnia. Dexamethasone can be taken with food to minimize dyspepsia; an acid-blocking therapy with H2 antagonist or proton pump inhibitor as clinically indicated can also be considered. Consider monitoring patients prior to therapy and as clinically indicated for increased serum glucose levels; dexamethasone should be used with caution in patients with diabetes mellitus.

The dosing for dexamethasone for the IV HEC and MEC regimens has been simplified. For the three-drug prophylactic olanzapine regimen—olanzapine plus palonosetron plus dexamethasone—for HEC and MEC, the dose of dexamethasone was decreased to 12 mg PO/IV for day 1, because all the other antiemetic regimens use this dexamethasone dose on day 1. Previously, the Panel had recommended a dexamethasone dose of 20 mg PO/IV on day 1 in the three-drug olanzapine regimen. For all the HEC regimens, the Panel also simplified the dosing for delayed dexamethasone to 8 mg PO/IV daily on days 2 to 4 (previously, some of the HEC regimens had used twice-daily dosing of dexamethasone). If dexamethasone is eliminated on subsequent days for the prevention of delayed CINV, the NCCN Panel recommends olanzapine be considered instead.

When dexamethasone is used with palonosetron for MEC, a randomized trial suggests that the dose of dexamethasone can be decreased to 8 mg on day 1 and also eliminated on days 2 to 3.[106] A similar phase 3 trial assessed palonosetron with dexamethasone on day 1 only versus palonosetron (day 1) with dexamethasone on days 1 to 3 in females receiving MEC regimens.[168] For females receiving dexamethasone on day 1 only (n = 166), the overall CR rates were 67.5% versus 71.1% for those receiving dexamethasone on days 1 to 3 (n = 166; difference -3.6% [95% CI, -13.5–6.3]). There was no difference in CR rates between the two regimens during the acute (0–24 hours post-chemotherapy; 88.6% vs.

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84.3%; *P* = .262) and delayed phases (days 2–5; 68.7% vs. 77.7%; *P* = .116).[168]

The doses and schedules for dexamethasone in the NCCN Guidelines are mainly based on the doses and schedules used in the clinical trials for each regimen. In a double-blind, noninferiority, phase III trial, Radhakrishnan et al explored the use of a dexamethasone-free regimen (olanzapine, palonosetron, and fosaprepitant [OPF]) for patients receiving HEC compared to a dexamethasone-containing regimen (olanzapine, palonosetron, and dexamethasone [OPD]).[169] Patients (n = 346) receiving single-day HEC were randomly assigned 1:1 to the OPF or OPD arms and monitored for CR rates for acute (0–24 hours), delayed (24–120 hours), or overall emesis (0–120 hours). The OPF arm had significantly higher vomiting CR rates for acute (94.7% vs. 85.6%), delayed (81.9% vs. 50.5%), and overall (79.6% vs. 48.8%) timeframes. While the nausea CR rate of the OPF arm was lower than that of OPD (77.9% vs. 81.6%), it was higher for delayed (53.4% vs. 39.6%) and overall (50.5% vs. 39.1%) timeframes. The results of this study provide support for a dexamethasone-sparing approach for certain patients receiving HEC.

The NCCN Panel believes that dexamethasone doses may be individualized; lower doses, frequency, or even elimination of dexamethasone on day 1 or subsequent days may be acceptable based on patient characteristics. Dexamethasone-sparing strategies may be appropriate for patients receiving MEC or HEC; eliminating dexamethasone on day 1 or subsequent days may be especially appropriate for patients with few identifiable risk factors for CINV or for those intolerant to corticosteroids (see *Pharmacologic Considerations for Antiemetic Prescribing* in the algorithm).[106,170-177] If dexamethasone is eliminated, clinicians should consider an antiemetic regimen combination consisting of a 5-HT3 RA, NK1 RA, and olanzapine on day 1 in addition to olanzapine for delayed CINV. In a prior version of the NCCN Guidelines for Antiemesis, dexamethasone-sparing strategies were not recommended for patients receiving cisplatin-containing HEC; however, this recommendation has been revised based on data from two studies.[170,171] Replacing dexamethasone with olanzapine may also be an option for those who cannot tolerate corticosteroids. Corticosteroid antiemetics should be avoided for 3 to 5 days before and 90 days after chimeric antigen receptor (CAR) and tumor-infiltrating lymphocyte (TIL) T-cell therapies. Upon disease progression, corticosteroids may be resumed if needed.

Olanzapine

Olanzapine is an atypical antipsychotic agent that is also useful as an antiemetic agent; it is an antagonist of multiple receptors involved in CINV, including dopamine, serotonin, histamine, and acetylcholine-muscarine.[22] A three-drug antiemetic regimen with olanzapine, dexamethasone, and palonosetron is effective for preventing acute and delayed emesis as described in the following sections.[22,178-186] An olanzapine-containing four-drug antiemetic regimen is also effective for preventing acute and delayed emesis.[178] A Cochrane analysis also reported that olanzapine regimens are effective.[187] The NCCN Panel recommends (category 1) olanzapine-containing three- or four-drug antiemetic regimens for both HEC and MEC based on the clinical trial data as described in the following sections.[38] Olanzapine can be substituted for dexamethasone if patients cannot tolerate dexamethasone (eg, patients with diabetes).

Common side effects with olanzapine include postural hypotension, anticholinergic side effects, fatigue, and sedation.[188] Therefore, the NCCN Panel recommends administering at bedtime whenever possible. Olanzapine should be used with caution in those who are older (see boxed warning/label indication regarding death in patients with dementia-related psychosis and additional warnings and precautions about type II diabetes and hyperglycemia).[189] Studies suggest that at low doses olanzapine is

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unlikely to have clinically relevant impact on QT interval; however, continued monitoring of ECG in patients with other significant risk factors for QT prolongation should be considered.[190,191] Previous data suggested that a 5-mg dose of olanzapine may be considered, especially for those who are older or who are oversedated.[192-198] Sedation is most notable on day 2 and improves over time; therefore, bedtime administration is recommended. A phase 3 study by Bajpai et al indicates that a 2.5 mg dose of olanzapine, when used as part of a 4-drug regimen, does not appear to have inferior antiemetic efficacy than a 10-mg dose, yet it has the benefit of lower sedation effects for patients expected to experience CINV from HEC agents. When not used as part of a 4-drug regimen, Olanzapine 5-mg or 10-mg doses should still be preferentially considered. Olanzapine may stimulate appetite and for patients experiencing concurrent chemotherapy-related anorexia a 2.5-mg dose may be considered.[199]

A single-arm, multicenter, phase 2 study evaluating the efficacy and safety of the 5-mg dose of olanzapine in combination with an NK1 RA, a 5-HT3 RA, and dexamethasone in patients with gynecologic cancer scheduled to be administered AUC ≥4 mg/mL/min carboplatin combination therapy (CBDCA), reported the combination to be an effective prophylactic antiemetic regimen.[200] Similarly, in a multicenter, randomized, double-blind, placebo-controlled, phase 3 trial (J-FORCE) evaluating the efficacy of the 5-mg PO dose of olanzapine with a three-drug regimen for the prevention of CINV, patients received either olanzapine or an PO placebo once daily on days 1 through 4 combined with aprepitant, palonosetron, and dexamethasone.[201] The primary endpoint of the study (ie, the proportion of individuals who achieved a CR [the absence of vomiting and lack of rescue medication use in the delayed phase]) was met, as 79% of patients (95% CI, 75–83) in the olanzapine group and 66% of those in the placebo group (95% CI, 61–71) exhibited a CR. It was concluded that to avoid unwanted sedation in select individuals, a 5-mg dose of olanzapine may be suitable for combination with HEC antiemetic prophylaxis. A prospective randomized controlled study assessed the efficacy of 5 mg versus 10 mg of olanzapine in the control of CINV as an add-on to ondansetron and dexamethasone in patients receiving HEC or MEC.[193] Dexamethasone was administered on day 1 prior to chemotherapy administration, continued to be administered for up to day 3 and up to day 4 for patients receiving MEC or HEC, respectively. Overall, both doses of olanzapine provided a similar level of emesis and nausea control when used for CINV prophylaxis. However, the severity of sedation (as measured by 5-day mean visual analog scale [VAS] score for sedation) with 10-mg olanzapine was 133% higher than with the 5-mg dosage.

A dose of 2.5 mg of olanzapine may be considered if patients have excessive sedation with the 5-mg dose. In contrast, the Panel recommends that increasing the dose of olanzapine to 10 mg can be considered if the previously administered 5-mg dose was ineffective.

Three-Drug Regimen

Several studies have reported favorable results following the addition of olanzapine to the antiemetic prophylaxis regimen for HEC agents.[202-204] A three-drug antiemetic regimen with olanzapine, dexamethasone, and palonosetron is effective for preventing acute and delayed emesis based on phase 3 trials, phase 2 trials, and meta-analyses.[22,178-186] A randomized phase 3 trial evaluated the effectiveness of an olanzapine (10 mg PO days 1–4) regimen versus an aprepitant (125 mg PO day 1, 80 mg days 2–3) regimen with dexamethasone 8 mg on days 2 to 4 for preventing acute and delayed emesis in patients (N = 251) receiving HEC (cisplatin or AC regimens); both treatment arms included palonosetron (0.25 mg IV) and dexamethasone administered on day 1.[185] The CR (no emesis and no rescue) rate was similar between the olanzapine and PO aprepitant regimens, both during the acute (97% vs. 87%) and delayed (77% vs. 73%) periods. The proportion of patients without nausea was similar for

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the acute period (87% in each study arm), but the olanzapine regimen was associated with a higher rate of nausea control during the delayed period (69% vs. 38%) compared with the PO aprepitant regimen.[185] A systematic review summarized the phase 1 and 2 studies of olanzapine for preventing acute and delayed emesis.[22] Across four studies (201 patients), the CR rate was 97.2%, 83.1%, and 82.8% for the acute, delayed, and overall phases, respectively. Other studies have also shown the value of olanzapine for delayed, refractory, and breakthrough emesis and nausea.[180-183,196]

In a randomized, double-blind, placebo-controlled study conducted by Jeon and colleagues, to evaluate the efficacy of olanzapine in patients receiving MEC regimens, patients were randomized to receive either olanzapine or placebo in combination with palonosetron and dexamethasone. The primary endpoint of the study was CR during the acute phase, while the secondary endpoints included CR during the delayed and overall (0–120 hours) phases, proportion of significant nausea, use of rescue therapy, and impact on QOL. It was found that although the CRs were relatively similar across both treatment arms, the percentage of patients with significant nausea (17.2% vs. 44%; *P* = .032) and the use of rescue therapy was lower in the olanzapine group associated with an overall better QOL.[205]

Finally, in a systematic review and meta-analysis of 27 RCTs evaluating 12 antiemetic regimens, Yokoe and colleagues found that olanzapine-containing regimens were both efficacious and cost-effective for CINV associated with HEC regimens. Although the combination of an NK1 RA, palonosetron, and dexamethasone was found to be the most efficacious of the three-drug regimens, olanzapine, when substituted for an NK1 RA, was reported to be a more cost-effective alternative. Overall, however, it was concluded that the four-drug olanzapine-containing regimen was the most effective to control CINV associated with HEC.[206] The NCCN Panel recommends (category 1) a prophylactic olanzapine-containing three-drug antiemetic regimen for both HEC and MEC based on the phase 3 and phase 2 trials.[22,178-186] As previously mentioned, the NCCN Panel recommends a dose of dexamethasone 12 mg PO/IV for day 1 for the three-drug antiemetic regimen with olanzapine, palonosetron, and dexamethasone. The Panel also agrees that palonosetron should be used in the three-drug olanzapine regimen; no data are available to support substituting any of the other 5-HT3 antagonists.

Four-Drug Regimen

A phase 3 randomized trial assessed adding olanzapine or placebo to an antiemetic regimen of PO aprepitant or fosaprepitant, a 5-HT3 antagonist, and dexamethasone for patients receiving HEC.[178] The four-drug regimen with olanzapine increased the CR rate (no emesis and no rescue) compared with placebo during three time periods (<24 hours after chemotherapy, 25–120 hours, and the overall 120 hours: 86% vs. 65% [*P* < .001], 67% vs. 52% [*P* = .007], and 64% vs. 41% [*P* < .001], respectively). In addition, more patients receiving the four-drug olanzapine regimen had no chemotherapy-induced nausea compared with placebo during the three time periods (<24 hours after chemotherapy, 25–120 hours, and 120 hours: 74% vs. 45% [*P* = .002], 42% vs. 25% [*P* = .002], and 37% vs. 22% [*P* = .002], respectively). FOND-O, a phase 3 randomized trial, assessed adding olanzapine to fosaprepitant, ondansetron, and dexamethasone in patients receiving HEC and HCT.[41] The CR was 55% in patients receiving the four-drug olanzapine regimen versus 26% in those receiving the three-drug regimen in the overall phase (*P* = .03); the CR was 60.8% versus 30%, respectively, in the delayed phase (*P* = .001).

Based on the Navari trial[178] and the NCCN Panel's clinical experience, the four-drug olanzapine antiemetic regimen is considered the preferred treatment option for HEC. In addition, clinicians can consider switching

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patients to the four-drug olanzapine regimen if patients have significant emesis after the first cycle of HEC when receiving other antiemetic regimens such as: 1) NK1 RA-containing regimens; or 2) the three-drug olanzapine regimen (olanzapine, dexamethasone, and palonosetron).[38] The Panel agreed that any NK1 RA (ie, not just fosaprepitant or PO aprepitant) could be used in the four-drug HEC regimen on day 1 (olanzapine, an NK1 RA, a 5-HT3, and dexamethasone), because all of the NK1 RAs are effective if the appropriate dose is used. Thus, aprepitant injectable emulsion, PO rolapitant, or NEPA may be used in the four-drug olanzapine regimen on day 1; however, none of these agents are continued on days 2 to 4.

Treatment Issues

These NCCN Guidelines include a section on pharmacologic considerations for the different antiemetics describing: 1) the major classes of antiemetics; 2) clinical pearls associated with the different types of agents; and 3) possible drug-drug or drug-disease interactions among the different antiemetics, although these drug interactions are less likely to occur with short-term use of antiemetic agents (see *Pharmacologic Considerations for Antiemetic Prescribing* in the algorithm).

Principles of Emesis Control

The goal of emesis control is to prevent nausea and/or vomiting (see *Principles of Emesis Control for the Patient with Cancer* in the algorithm). Prophylactic antiemetic regimens should be chosen based on the drug with the highest emetic risk in the anticancer agent regimen, previous experience with antiemetics, and patient-specific risk factors.[46] Patients need to be protected throughout the entire risk period, which lasts for at least 3 days for HEC and 2 days for MEC after the last dose of an anticancer agent. In addition to using antiemetic regimens, patients can adjust their eating habits and adopt other lifestyle measures that may alleviate nausea and/or vomiting (see *Eating Hints: Before, During, and After Cancer Treatment* from the National Cancer Institute). Suggestions include eating small frequent meals, food that is easy on the stomach, full liquid foods, and food at room temperature; patients can also avoid foods that make them feel nauseated.

When planning an antiemetic regimen, the health literacy of the patient must be considered, as well as sociocultural differences. Effective provider-patient communication may improve patient satisfaction, adherence, safety, and outcomes. Clinicians need to identify language and literacy barriers and provide appropriate resources (such as printed materials, medication calendars, and interpreter services) to help whenever possible. While many patients are interested in complementary and alternative medicines (CAM) to treat their cancer or cancer treatment-related side effects, there are currently no high-level, large, randomized clinical trials demonstrating superior benefit for these types of therapies in place of conventional antiemetic drug therapy. Patients should consult their provider or pharmacist to ensure safety and appropriateness.

Other situations for the clinician to consider in patient treatment include the use of antiemetics with HCT and the effect of steroids on immune checkpoint inhibitors (ICIs). High-dose chemotherapy in the setting of HCT is a unique and challenging situation where patients often have multiple risk factors for CINV in addition to intensive chemotherapy. As most conditioning regimens used in the HCT setting are highly emetogenic, clinicians are encouraged to use the corresponding recommended regimens to develop a personalized plan for CINV prevention. A segment of emerging clinical data suggests that dexamethasone, with its broad immunosuppressive effect, could blunt the effect of ICIs.[207-209] However, there are no randomized clinical trials that demonstrate the detriment of using dexamethasone as a prophylactic antiemetic when ICIs are used in combination with chemotherapy. Some clinical trials have not allowed dexamethasone when used solely as an antiemetic, and other publications

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suggest a negative impact on outcomes when steroids are used with ICIs.[210] Clinicians may wish to consider a dexamethasone-sparing approach (see *Pharmacologic Considerations for Antiemetic Prescribing* in the algorithm), or use alternative antiemetic agents, such as olanzapine for the purposes of CINV prophylaxis, until more definitive data become available.[209,211-213]

Prevention of Acute and Delayed Emesis

To prevent acute emesis, antiemetic therapy should start before the administration of anticancer agents and then should cover the first 24 hours. In the NCCN Guidelines for Antiemesis, the specific prophylactic antiemetic regimens are described for patients receiving highly emetogenic parenteral drugs, moderately emetogenic parenteral drugs, low emetogenic parenteral drugs, and minimally emetogenic parenteral drugs. Prophylactic antiemetic regimens for oral chemotherapeutic agents are also described in the NCCN Guidelines. This section discusses emesis prevention before and after anticancer agent administration rather than primary treatment for ongoing emesis.

Decreasing Acute Emesis

The NCCN Guidelines specify different prophylactic antiemetic regimens for patients with cancer who are receiving anticancer agents of different emetogenic potential (ie, high, moderate, low, minimal). Prophylactic antiemetics should be administered before anticancer agents. The recommendations for prophylactic antiemetic treatment include drug dosages. The guidelines reflect accumulating experience with the different antiemetic agents, demonstrating their effectiveness in a range of doses. As previously mentioned, if patients receive prophylactic antiemetic regimens, emesis will be decreased but not completely prevented. More than 90% of patients receiving HEC without antiemetic prophylaxis will have episodes of vomiting. However, if patients receive prophylactic (preventive) antiemetic regimens before treatment with HEC, then only approximately 30% of these patients will vomit.[7,9,10]

Highly emetogenic parenteral drugs in the NCCN Guidelines include carboplatin (AUC ≥4), carmustine (>250 mg/m^2), cisplatin (any dose), cyclophosphamide (>1500 mg/m^2), dacarbazine (any dose), datopotamab deruxtecan-dlnk (any dose), doxorubicin (≥60 mg/m^2), epirubicin (>90 mg/m^2), fam-trastuzumab deruxtecan-nxki (any dose), ifosfamide (≥2 g/m^2 per dose), mechlorethamine (any dose), melphalan (≥140 mg/m^2), sacituzumab govitecan-hziy (any dose), streptozocin (any dose), zolbetuximab-clzb (any dose), or AC combination regimens at any dose (eg, cyclophosphamide plus either doxorubicin or epirubicin).

With reference to carboplatin, when dosed at an AUC of ≥4, it is considered highly emetogenic; carboplatin at an AUC of <4, however, is considered moderately emetogenic.[214] Data show it is beneficial to add an NK1 RA to the two-drug regimen of a 5-HT3 antagonist and dexamethasone for the prevention of CINV associated with carboplatin-based regimens.[147,214-218] All of the commercially available NK1 RAs have an FDA-approved indication for MEC, but older versions of NCCN Guidelines supported the addition of an NK1 RA only for select patients receiving MEC with additional CINV risk factors or for those who progressed on previous therapy with a corticosteroid and a 5-HT3 antagonist alone.

Several parenteral anticancer drugs listed as moderately emetogenic in the NCCN Guidelines may be highly emetogenic in certain patients (eg, carboplatin [AUC < 4], carmustine [≤250 mg/m^2], cyclophosphamide [≤1500 mg/m^2], dactinomycin, daunorubicin, doxorubicin [<60 mg/m^2], epirubicin [≤90 mg/m^2], idarubicin, ifosfamide [<2 g/m^2], irinotecan, methotrexate [≥250 mg/m^2], oxaliplatin, trabectedin). AC-based regimens were reclassified in 2011 as highly emetogenic in the American Society of Clinical Oncology (ASCO) antiemetic guidelines.[91]

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Antiemetic Regimens for Parenteral HEC

The NCCN Guidelines recommend three- and four-drug antiemetic regimen options (all are category 1) for patients receiving HEC, including 1) an NK1 RA, a 5-HT3 RA, and dexamethasone; 2) olanzapine, palonosetron, and dexamethasone; or 3) olanzapine, an NK1 RA, a 5-HT3 RA, and dexamethasone.[178,185] The four-drug olanzapine regimen (olanzapine, an NK1 RA, a 5-HT3 RA, and dexamethasone) is the preferred regimen for patients at high risk for emesis from parenteral anticancer agents (see *Olanzapine* in this Discussion). If needed, lorazepam, an H2 blocker, or a proton pump inhibitor may also be added (alone or in any combination) to all of these regimens.[34,49,125] Note that the regimens and doses are often modified on days 2 to 4 after anticancer agents, for example where a dexamethasone-sparing approach may be appropriate.

Although it is not routinely recommended as a single agent, lorazepam may be useful for breakthrough nausea in certain circumstances, especially if anxiety is thought to be a contributing factor.[49,158,219] Lorazepam is also recommended for patients who are at risk for anticipatory nausea and/or vomiting (see *Anticipatory Emesis Prevention/Treatment* in the NCCN Guidelines for Antiemesis). If prescribing benzodiazepines, clinicians should assess the patient for potential risk for substance use disorder and addiction as continued use of benzodiazepines may lead to dependence.[220,221] Unless being treated with curative intent, benzodiazepines, such as lorazepam, should be avoided in patients ≤12 years of age.

Antacid therapy (eg, proton pump inhibitors, H2 blockers) should be considered if patients have dyspepsia, because patients sometimes have difficulty distinguishing heartburn from nausea. The NCCN Guidelines recommend lorazepam (0.5–1 mg PO) beginning on the night before treatment and then repeated the next day 1 to 2 hours before anticancer therapy begins for patients with anticipatory nausea. As a breakthrough agent, lorazepam 0.5–1 mg PO/IV/sublingual every 6 hours may be considered. The lowest effective dose and dosage interval possible should be used. Lorazepam may be administered with or without an H2 blocker or proton pump inhibitor if reflux symptoms occur.

For parenteral HEC, aprepitant is used at a dosage of 125 mg PO on day 1 and then 80 mg on days 2 to 3. When given with aprepitant, dexamethasone is used at a dosage of 12 mg on day 1; the dexamethasone dose can be PO/IV. Note that aprepitant injectable emulsion or IV fosaprepitant may be used instead of PO aprepitant on day 1 only. As previously discussed, a phase 3 randomized trial suggested that palonosetron is preferred over granisetron in combination with dexamethasone for HEC.[69] This trial has been criticized because: 1) the control arm was not adequately dosed; thus, the trial "stacked the deck" in favor of palonosetron; 2) a larger non–FDA-approved dose of palonosetron was used (ie, 0.75 mg IV); and 3) aprepitant was not used in this study. Therefore, the NCCN Guidelines do not recommend palonosetron as the preferred 5-HT3 antagonist for HEC. As previously noted, an alternative antiemetic regimen in the setting of parenteral HEC includes olanzapine (2.5–10 mg PO days 1–4), palonosetron (0.25 mg IV day 1 only), and dexamethasone (12 mg IV day 1 only).[185]

A Canadian meta-analysis suggested that the use of 5-HT3 antagonists (ie, ondansetron) on days 2 to 4 to prevent delayed emesis was not cost-effective; however, ondansetron (when used alone) protected against delayed emesis in this meta-analysis.[222] Palonosetron was not assessed in these studies. The NCCN Guidelines do not recommend a 5-HT3 antagonist for the prevention of CINV on days 2 to 4 for HEC.

Antiemetic Regimens for Parenteral MEC

The NCCN Guidelines recommend two- and three-drug antiemetic regimens for parenteral MEC, including: 1) dexamethasone and a 5-HT3

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antagonist with NK1 RAs, such as aprepitant (PO or injectable emulsion), fosaprepitant, netupitant, fosnetupitant, or PO rolapitant; 2) olanzapine, palonosetron, and dexamethasone; or 3) dexamethasone and a 5-HT3 antagonist (palonosetron or subcutaneous granisetron extended-release injection are preferred). If needed, lorazepam, an H2 blocker, or a proton pump inhibitor may be added (alone or in any combination) to these regimens.[7] Netupitant (or fosnetupitant) is only available combined with palonosetron (NEPA) and not as a single agent. As previously mentioned, an NK1 RA or olanzapine should be added (to dexamethasone and a 5-HT3 antagonist regimen) for select patients with additional risk factors or progression on previous therapy with a corticosteroid and 5-HT3 antagonist alone. These additional risk factors include younger age; female sex; anxiety and/or high pretreatment expectation of nausea and/or vomiting; and history of CINV, motion sickness, morning sickness during pregnancy, and little or no alcohol use.[122] IV fosaprepitant or aprepitant injectable emulsion may be substituted for PO aprepitant on day 1 only. The NCCN Guidelines recommend the use of 5-HT3 antagonists as one of several options to prevent delayed emesis for MEC. Any one of the 5-HT3 antagonists can be used in the first regimen for day 1; however, preferred 5-HT3s include palonosetron or subcutaneous granisetron extended-release injection when an NK1 RA is not included, as previously mentioned.[69,92]

Antiemetic Regimens for Parenteral Low Emetic Risk Anticancer Agents
Newly added agents under low emetic risk include mosunetuzumab-axgb, elranatamab-bcmm, epcoritamab-bysp, and lifileucel. Amivantamab-vmjw, azacitidine, and epcoritamab-bysp were previously categorized as MEC.

The single-agent antiemetic regimens for low emetogenic risk parenteral anticancer agents include dexamethasone, prochlorperazine, metoclopramide, or PO 5-HT3 antagonists such as granisetron, ondansetron, or dolasetron (see *Low and Minimal Emetic Risk Parenteral Anticancer Agents - Emesis Prevention* in the algorithm). Lorazepam, an H2 blocker, or a proton pump inhibitor may also be added (alone or in any combination) to all of these agents.[7] When using prochlorperazine or metoclopramide, patients should be monitored for dystonic reactions.[223,224] Diphenhydramine can be used for the treatment of dystonic reactions.[225,226] Benztropine may be used in patients who are allergic to diphenhydramine.[223] Amantadine is also an option for drug-induced dystonic reactions in patients who are intolerant of anticholinergic agents.[227-230]

Antiemetic Regimens for Oral Anticancer Agents
The emetogenic potential of oral anticancer agents is shown in the NCCN Guidelines, which are updated every year with the new agents. The Panel has also stratified this section further to provide additional guidance on prophylaxis for moderate- to high emetic risk agents. Antiemetic prophylaxis is required on days of PO administration for the following moderate to high emetic risk oral anticancer agents: azacitidine, busulfan (≥4 mg/day), ceritinib, cyclophosphamide (≥100 mg/m^2/day), fedratinib, lomustine (single day), midostaurin, mitotane, selinexor, and temozolomide alone (>75 mg/m^2/day) or with concurrent RT (≤75 mg/m^2/day). As-needed (PRN) dosing may be initially appropriate on days of PO administration for certain moderate to high emetic risk anticancer agents (abemaciclib, adagrasib, avapritinib, binimetinib, bosutinib [>400 mg/day], cabozantinib, crizotinib, dabrafenib, elacestrant, enasidenib, encorafenib, estramustine, etoposide, imatinib [>400 mg/day], lenvatinib [>12 mg/day], niraparib, olaparib, procarbazine, rucaparib, tovorafenib, and trifluridine/tipiracil); however, if the patient experiences CINV, then scheduled prophylaxis would be required. Newly added agents under the minimal to low emetic risk category include: capivasertib, eflornithine, fruquintinib, nirogacestat, and quizartinib. Abemaciclib and trifluridine/tipiracil were previously categorized as minimal to low emetic

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risk agents. Infigratinib and umbralisib were removed as they are no longer available in the United States.

For moderate to high emetic risk oral anticancer agents where prophylaxis is required, recommended prophylaxis includes single-agent antiemetic therapy with granisetron, ondansetron, dolasetron, or olanzapine. For oral anticancer agents that are minimal to low emetic risk or moderate to high emetic risk where PRN dosing is initially appropriate, recommended oral agents are given on a PRN basis only and include granisetron, ondansetron, dolasetron, metoclopramide, or prochlorperazine. Some patients receiving oral anticancer agents of low/minimal emetogenicity may experience nausea and/or vomiting; for these patients, follow the recommendations in the *Breakthrough Treatment for Anticancer Therapy-Induced Nausea/Vomiting* section of the NCCN Guidelines for Antiemesis, and consider escalating to a higher level of antiemetic prevention (ie, moderate-high, and consider prophylaxis) for the next cycle.

Antiemetic Regimens for Radiopharmaceutical Anticancer Agents
Due to the availability of an increasing number of radiopharmaceutical anticancer agents, the Panel consensus was to provide guidance on the emetogenic potential of this class of agents. Agents categorized as moderate emetic risk include lutetium Lu-177 dotatate and iobenguane iodine-131. Lutetium Lu-177 vipivotide tetraxetan was categorized as low emetic risk. Agents categorized as minimal emetic risk include radium-223 dichloride, sodium iodide I-131, yttrium-90 ibritumomab tiuxetan, strontium-89, and yttrium-90 microspheres. For most of these agents, clinicians should use the antiemetic regimens based on emetic risk specified for parenteral agents as described above.

There are special considerations for preventing antiemesis with lutetium Lu-177 dotatate. Nausea associated with lutetium Lu-177 dotatate is primarily derived from the accompanying amino acid infusion; therefore, variation in emetogenicity is reported between various compounded and commercially available amino acid products. Antiemetics should be administered 30 minutes prior to the initiation of amino acid infusion.

Since lutetium Lu-177 is a radiolabeled somatostatin analog, the manufacturer recommends avoiding repeated administration of “high doses” of glucocorticoids during treatment.[231] The rationale for this is that glucocorticoids can induce down-regulation of subtype 2 somatostatin receptors (SST2).[232] Since other antiemetic prophylaxis options are available (such as 5-HT3 RAs and NK1 RAs), the Panel recommends avoiding the use of glucocorticoids in those being treated with lutetium Lu-177.

Decreasing Delayed Nausea and/or Emesis

Delayed Nausea
Many antiemetic regimens are very useful for decreasing vomiting but are less useful for decreasing delayed nausea that many patients experience when taking emetogenic anticancer agents.[12,23,26,46] Patients rank nausea as more of a problem than vomiting.[12] Data suggest that PO rolapitant and netupitant are effective at decreasing delayed nausea.[141,143,147,148] Palonosetron and subcutaneous granisetron extended-release injection are the preferred 5-HT3 antagonists for preventing delayed nausea associated with MEC.

A phase 3 randomized trial assessed adding olanzapine or placebo to an antiemetic regimen of fosaprepitant or PO aprepitant, a 5-HT3 antagonist, and dexamethasone for patients receiving HEC.[178] More patients receiving the four-drug regimen with olanzapine had no anticancer agent–induced nausea compared with placebo during the delayed time period (ie, 25–120 hours, 42% vs. 25% [*P* = .002]). Nausea was also reduced with the four-drug regimen with olanzapine during the acute phase and the overall time period compared with placebo. The four-drug regimen with olanzapine increased the CR rate (no emesis and no rescue) during the

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delayed time period compared with placebo (67% vs. 52%; *P* = .007). The three-drug regimen with olanzapine, dexamethasone, and palonosetron was associated with a higher rate of nausea control during the delayed period (69% vs. 38%) compared with the three-drug PO aprepitant regimen (with palonosetron and dexamethasone).[185] However, the proportion of patients without nausea was similar for the acute period (87% in each study arm).[185] Therefore, olanzapine seems especially effective for decreasing nausea.

Delayed Emesis

The best management for delayed emesis is prevention.[233] A survey among oncology nurses found that there is low adherence (only 25%) to antiemetic guidelines for preventing delayed emesis.[234] For HEC, the prophylactic antiemetic treatment on days 2 to 4 depends on which antiemetics were used on day 1. Fosaprepitant, aprepitant injectable emulsion, PO rolapitant, granisetron extended-release injection, granisetron transdermal patch, palonosetron, or NEPA are used on day 1 only, because they are effective for an extended period of time. If PO olanzapine was used on day 1 for HEC, then PO olanzapine is continued on days 2 to 4. If PO aprepitant was used on day 1 for HEC, then PO aprepitant is continued on days 2 to 3. Dexamethasone may be continued on days 2 to 4 for HEC, depending on the regimen. For patients who cannot tolerate corticosteroids, a dexamethasone-sparing approach may be considered by replacing dexamethasone with olanzapine. However, 5-HT3 antagonists are given on day 1 only for HEC. There are several possible HEC antiemetic regimens on days 2 to 4, including: 1) PO aprepitant (if used on day 1) with dexamethasone and with or without olanzapine; or 2) olanzapine only. If needed, each of these regimens may also include lorazepam, an H2 blocker, or a proton pump inhibitor (alone or in any combination).[7]

The antiemetic regimens in the NCCN Guidelines include different options on days 2 to 3 for MEC, including single-agent therapy.[34,49,233] Antiemetic treatment on days 2 to 3 depends on which antiemetic regimens were used on day 1. If PO aprepitant or olanzapine was used on day 1, then PO aprepitant or olanzapine is continued on days 2 to 3. However, granisetron extended-release injection, granisetron transdermal patch, palonosetron, aprepitant injectable emulsion, fosaprepitant, PO rolapitant, or NEPA are not given on days 2 to 3.[71] There are several possible MEC antiemetic regimens for days 2 to 3, including: 1) PO aprepitant (if used on day 1) with or without dexamethasone; 2) dexamethasone only; 3) ondansetron, granisetron, or dolasetron only (if no NK1 RA, granisetron extended-release injection, granisetron transdermal patch, or palonosetron was given on day 1); or 4) olanzapine only.[233] If needed, each of these regimens may also include lorazepam, an H2 blocker, or a proton pump inhibitor (alone or in any combination).[7] The doses are decreased when used on days 2 to 3 for aprepitant (80 mg PO) and dexamethasone (8 mg PO/IV) compared with the doses given on day 1. However, the dose of olanzapine is not decreased on days 2 to 3.

Breakthrough Nausea and/or Vomiting Treatment

Breakthrough nausea or emesis presents a difficult situation because refractory ongoing nausea and/or vomiting is often challenging to reverse (see *Principles for Managing Breakthrough Emesis* in the algorithm). Generally, it is much easier to prevent nausea and/or vomiting than to treat it, which is why the NCCN Guidelines recommend prophylactic antiemesis regimens. Routine around-the-clock administration of antiemetics is recommended to prevent emesis, rather than PRN dosing. The general principle of breakthrough treatment is to add an additional agent PRN from a different drug class.[34] Some patients may require several agents each with a different mechanism of action. The oral route may not be feasible because of ongoing vomiting; therefore, rectal, topical, subcutaneous, or IV therapy is often required. Multiple concurrent agents,

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perhaps in alternating schedules or by alternating routes, may be necessary. Another option is to consider changing from the current NK1-containing regimen to an olanzapine-containing regimen, or vice versa, prior to the next cycle of anticancer agents. Olanzapine is possibly more effective than NK1-antagonist–containing regimens for preventing nausea.[22,185,186] Switching to a different 5-HT3 RA and/or NK1 RA with a different pharmacokinetic/pharmacodynamic profile is another option, although there is only anecdotal evidence that this may be helpful.

In a randomized phase 3 trial, the effectiveness of olanzapine (10 mg/day PO for 3 days) as treatment for breakthrough emesis was compared with metoclopramide in patients treated with HEC who developed breakthrough emesis or nausea despite antiemetic prophylaxis with palonosetron, dexamethasone, and fosaprepitant (n = 108 evaluable).[235] Patients were observed for emesis and nausea during the 72 hours after treatment with olanzapine or metoclopramide. During this observation period, more patients had no emesis (70% vs. 31%; $P < .01$) and no nausea (68% vs. 23%; $P < .01$) with olanzapine than with metoclopramide.[235] Thus, olanzapine was more effective in controlling breakthrough emesis and nausea compared with metoclopramide in this patient population. The Multinational Association of Supportive Care in Cancer (MASCC)/European Society for Medical Oncology (ESMO) and ASCO Guidelines recommend olanzapine for breakthrough emesis.[35,236] The NCCN Panel recommends olanzapine (category 1; preferred) for breakthrough emesis if olanzapine was not used on days 1 to 4 as part of a prophylactic regimen. This category 1 recommendation is based on the magnitude of superiority of olanzapine over metoclopramide in the randomized phase 3 trial.[235] Other recommended treatment options for breakthrough emesis may be added to the current antiemetic regimen such as metoclopramide, haloperidol, scopolamine transdermal patch, corticosteroids, cannabinoids, and lorazepam. It should be noted that haloperidol may prolong the QT interval on the ECG; however, some data suggest that lower oral doses do not contribute to QT interval prolongation.[237,238]

Dronabinol is a cannabinoid that has been FDA-approved for refractory nausea and vomiting when symptoms have not abated with conventional antiemetics. Note that dronabinol PO solution (5 mg/mL) and dronabinol capsules are not bioequivalent. Dronabinol PO solution has greater PO bioavailability than dronabinol capsules (2.1 mg PO solution = 2.5 mg capsules).[239] Recommended starting doses are dronabinol PO solution (4.2 mg/m^2) or dronabinol capsules (5 mg/m^2) both given three to four times daily. Lower doses are recommended in those who are older. Nabilone was removed from the guidelines as a treatment option as it is no longer available in the United States.

Before administering the next cycle of anticancer agents, the patient should be reassessed for other possible reasons for breakthrough emesis with the current cycle that are not related to anticancer agents, including brain metastases, electrolyte abnormalities, tumor infiltration of the bowel or other GI abnormality, and other comorbidities (see *Principles for Managing Breakthrough Emesis* and *Principles of Emesis Control for the Patient with Cancer* in the algorithm). Adequate hydration or fluid repletion should be ensured, and any possible electrolyte abnormalities should be assessed and corrected. If the antiemetic regimen (both on day 1 and days 2–4) did not protect the patient during the present cycle, the antiemetic regimen should be assessed, and alternatives should be considered before the next cycle of anticancer agents (see *Principles for Managing Breakthrough Emesis* in the algorithm). Because patients sometimes have difficulty distinguishing heartburn from nausea, the addition of antacid therapy should be considered, such as proton pump inhibitors and H2 blockers.

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Radiation-Induced Nausea and/or Vomiting

Antiemetic prophylaxis for RT-induced nausea and/or vomiting is based on the site of RT and whether it is combined with anticancer agents.[38,39,240,241] When RT is combined with anticancer agents, prophylaxis is dictated by the emetogenic potential of the anticancer agent regimen.[242] ASCO and MASCC/ESMO guidelines state that TBI is associated with the highest risk for emesis and that upper abdominal RT is associated with moderate risk.[38,39,243] A meta-analysis suggests that 5-HT3 antagonists are the preferred agents for preventing RT-induced vomiting.[244]

Patients undergoing RT to the upper abdomen may receive antiemetic prophylaxis with PO ondansetron or granisetron, with or without PO dexamethasone.[10,39] A randomized trial compared PO ondansetron with placebo in patients receiving daily fractionated radiotherapy including the abdomen. In this study, 67% of patients given ondansetron had complete control of emesis compared with 45% of patients who received placebo ($P < .05$).[245] A randomized trial showed that the addition of PO dexamethasone (4 mg daily) to the ondansetron regimen decreases emesis and nausea, although the effect was modest.[246] Patients receiving ondansetron/dexamethasone had better complete control of emesis (23% vs. 12%; $P = .02$) and a lower average nausea score (0.28 vs. 0.39; $P = .03$) compared with those receiving ondansetron alone. Another randomized trial in patients receiving radiotherapy to the upper abdomen found that PO granisetron decreased emesis and nausea compared with placebo.[247]

The NCCN Panel recommends that patients undergoing TBI or upper abdomen/localized RT receive antiemetic prophylaxis with either PO ondansetron or PO granisetron; either agent can be given with or without PO dexamethasone.[10,39,248] Treatment of breakthrough RT-induced emesis is similar to anticancer agent–induced emesis. Patients who experience breakthrough nausea and/or vomiting may be treated with a different class of agent, or with ondansetron or granisetron if they did not receive primary prophylaxis (see *Breakthrough Treatment for Anticancer Therapy-Induced Nausea/Vomiting* in the algorithm). For patients who receive both anticancer therapy and RT (including TBI), antiemetic prophylaxis should be based on the modality (anticancer therapy or radiation) with the highest emetic risk.

Anticipatory Nausea and/or Vomiting

Approximately 20% of patients develop anticipatory nausea and/or vomiting.[249] However, the rate of anticipatory nausea and/or vomiting appears to be decreasing with current use of more effective antiemetic regimens compared with older studies.[10] The most effective way to treat anticipatory nausea and/or vomiting is to prevent it by using optimal antiemetic therapy during every cycle of treatment.[34,38,250,251] Behavioral therapy has been used in patients with anticipatory nausea and/or vomiting.[38,252-257] Systematic desensitization may also be helpful.[253] Hypnosis with guided imagery is another behavioral technique that has shown some success in treating this condition.[254]

The antianxiety agent lorazepam has been combined with antiemetics for anticipatory nausea and/or vomiting.[251,258,259] The usual starting dose of lorazepam for anxiety is 0.5 to 1 mg PO, beginning on the night before treatment and then repeated the next day 1 to 2 hours before treatment begins with anticancer agents. The usual starting dose of lorazepam is 0.5 mg PO for treatment of anxiety in patients who are older, those with debilitating disease, and those with advanced liver disease (see prescribing information). This dose may be gradually increased if needed. Note that older patients are especially sensitive to the effects of benzodiazepines. The dose should be gradually reduced when decreasing or discontinuing lorazepam therapy.

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Olanzapine also has anxiolytic properties and may be considered for anticipatory nausea or vomiting given its use in the treatment of anxiety and other mood disorders.[260-262] Due to the currently available data on its effectiveness (see *Olanzapine* in this discussion), the NCCN Panel recommends that 2.5 to 5 mg of olanzapine be considered as a single PO dose the night before the start of an anticancer therapy.

The NCCN Panel recommends behavioral therapy options for anticipatory nausea and/or vomiting including yoga, cognitive distraction, relaxation exercises (eg, music therapy), and biofeedback (see *Anticipatory Emesis Prevention/Treatment* in the algorithm). The Panel also recommends lorazepam, olanzapine, and acupuncture. Lorazepam should be used with caution in patients receiving scheduled opioids because of the increased risk of respiratory depression. The NCCN Guidelines also recommend that patients avoid strong smells that may precipitate symptoms.

Multiday Emetogenic Anticancer Agent Regimens

Patients receiving multiday anticancer agents are at risk for both acute and delayed nausea and/or vomiting based on the emetogenic potential of the individual anticancer agents and their sequence.[34,263-267] It is difficult to recommend a specific antiemetic regimen for each day, especially because acute and delayed emesis may overlap after the initial day of the anticancer therapy until the last day. The period of risk for delayed emesis following completion of the anticancer agents also depends on the specific regimen and the emetogenic potential of the last anticancer agent administered in the regimen. For multidrug regimens, antiemetic therapy should be selected based on the drug with the highest emetic risk.[38] General principles for managing multiday emetogenic chemotherapy regimens recommended by the NCCN Panel are described in the algorithm (see *Principles of Managing Multiday Emetogenic Chemotherapy Regimens* in the algorithm). The Panel acknowledges that evidence is lacking to support every clinical scenario. Decisions should be individualized for each chemotherapy regimen and each patient. An extensive knowledge of the available clinical data, pharmacology, pharmacodynamics, and pharmacokinetics of the antiemetics and the chemotherapy and experience with patients (regarding tolerability and efficacy) are all paramount to successfully implementing these guidelines into clinical practice.

A meta-analysis reported that a three-drug regimen with a 5-HT3 RA, PO aprepitant, and dexamethasone was useful for decreasing emesis with multiday cisplatin regimens.[268] For antiemetic prophylaxis of multiday emetogenic anticancer agent regimens (eg, cisplatin-containing regimens), the combination of a 5-HT3 antagonist with dexamethasone was previously recommended in the 2011 MASCC/ESMO guidelines.[10,34] The NCCN Guidelines and 2017 MASCC/ESMO guidelines currently recommend a three-drug regimen. For single-day systemic therapy regimens, category 1 evidence is available for aprepitant, aprepitant injectable emulsion, fosaprepitant, netupitant, fosnetupitant, or rolapitant administered in combination with a 5-HT3 RA and corticosteroid. The clinical trial data to support these recommendations are described in the following sections.

Dexamethasone

Dexamethasone PO/IV should be administered once daily in the morning every day of MEC or HEC and continued for 2 to 3 days for anticancer agent regimens that are likely to cause significant delayed emesis. If the anticancer regimen already includes a corticosteroid, the dexamethasone dose may be modified or omitted. The NCCN Panel does not recommend the use of corticosteroids in antiemetic regimens for 3 to 5 days before and 90 days after CAR and TIL T-cell therapies, because corticosteroids may decrease the persistence of the CAR T-cell population.[269,270] However, corticosteroids may be resumed if needed following disease progression. All of the ICIs are of minimal emetic risk.

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5-HT3 Antagonists

For multiday anticancer regimens, a 5-HT3 antagonist should be administered each day before the first dose of MEC or HEC. IV palonosetron, subcutaneous granisetron, or transdermal granisetron may be used before the start of a 3-day anticancer regimen instead of multiple daily doses of PO/IV 5-HT3 antagonists.[271,272] It is not known whether repeat dosing with subcutaneous granisetron for multiday regimens would be effective. Repeat dosing of palonosetron (0.25 mg IV) is likely to be safe, based on the dose-ranging phase 2 trial and the three phase 3 trials using palonosetron as a single fixed dose (0.75 mg IV).[68,70,71,273] Compared to the approved dose of palonosetron of 0.25 mg IV, these higher doses were not associated with significantly different adverse events.

The optimal dosing with palonosetron, either daily or less frequently, in the setting of multiday anticancer regimens is not yet known. In one study, patients receiving highly emetogenic multiday cisplatin-based therapy for testicular cancer (N = 41) received multiday dosing of palonosetron (0.25 mg IV on days 1, 3, and 5) and dexamethasone, which prevented nausea and emesis in most patients on days 1 to 5 (51%) and on days 6 to 9 (83%); the most common adverse events were mild headache and constipation.[274] A study assessed palonosetron given for 1, 2, or 3 days in combination with dexamethasone for patients receiving multiday high-dose anticancer regimens prior to HCT for multiple myeloma (N = 73); during the 7-day emesis prevention period, approximately 40% to 45% of patients had no emesis (with no differences observed between palonosetron treatment groups), and no serious adverse events were reported. However, even among the patients who received either 2 or 3 days of palonosetron, only 20% had a CR (ie, emesis free without rescue medication).[119] Another study found that a palonosetron/dexamethasone regimen appeared to be more effective for multiday anticancer therapy than an ondansetron/dexamethasone regimen; patients received a second dose of palonosetron for breakthrough emesis, which was effective in 67% of patients who experienced nausea or vomiting.[271] Other publications have also cited the value of palonosetron for patients receiving multiday anticancer therapy.[275,276] Further studies are needed to define whether a need exists for repeat dosing of palonosetron in the setting of multiday anticancer therapy.

NK1 RAs

The potential role of NK1 RAs in the antiemetic management of multiday anticancer therapy has been investigated in several studies.[148,236,277-279] In a randomized phase 3 trial, the efficacy of adding PO aprepitant (vs. placebo) to an antiemetic regimen with a 5-HT3 antagonist and dexamethasone was evaluated in patients with germ cell cancers undergoing two cycles of a 5-day cisplatin combination therapy regimen (n = 69 evaluable).[277] Patients were randomized to receive PO aprepitant (125 mg PO day 3, 80 mg days 4–7) or placebo, combined with a 5-HT3 antagonist (days 1–5) and dexamethasone (20 mg days 1–2) during the first cycle, and then crossed over to the opposite antiemetic regimen during the second cycle of anticancer therapy. Thus, patients served as their own controls after receiving either PO aprepitant or placebo for cycle 1. Palonosetron was excluded from the options for 5-HT3 antagonists due to its longer half-life.[277] The primary endpoint of the study was CR (no emetic episodes and no rescue medication) during the overall study period (days 1–8). The CR rate for the overall study period was significantly higher with PO aprepitant compared with placebo (42% vs. 13%; $P < .001$). The CR rates were also higher with PO aprepitant during the acute phase (days 1–5; 47% vs. 15%; $P < .001$) and delayed phase (days 6–8; 63% vs. 35%; $P < .001$).[277] No statistically significant differences were observed between treatment regimens in terms of nausea (based on patient-reported VAS). Importantly, no increase in toxicity with PO aprepitant compared with placebo was reported.[277]

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The addition of PO aprepitant to granisetron and dexamethasone was evaluated in patients receiving multiday HEC and MEC (N = 78). In this study, the three-drug antiemetic regimen was given during anticancer therapy; PO aprepitant and dexamethasone were given for an additional 2 days following anticancer therapy.[279] A CR (during the time period from day 1 until 5 days after anticancer therapy) was observed in 58% and 73% of patients who received antiemetic regimens for HEC and MEC, respectively.[279] In a multicenter phase 2 study, an extended 7-day regimen with PO aprepitant (125 mg PO day 1, 80 mg days 2–7) combined with a 5-HT3 antagonist (days 1–5) and dexamethasone (8 mg PO days 1–8) was evaluated in patients with germ cell tumors undergoing anticancer therapy cycles with 5-day cisplatin-based regimens (N = 50).[278] During cycle 1 of anticancer therapy, 96% of patients had no emesis on day 1 and 82% had no emesis during days 1 to 7. In addition, 71% had no nausea on day 1 of cycle 1, and 27% had no nausea during days 1 to 7. More than 80% of patients had no emesis on any given day of any given anticancer therapy cycle. No unexpected or serious adverse events were reported.[278] In another study, aprepitant (125 mg day 1, 80 mg days 2–5) combined with palonosetron (0.75 mg day 1) and dexamethasone (9.9 mg day 1, 6.6 mg days 2–8) was evaluated in 30 patients with testicular germ cell tumors receiving 5-day cisplatin-based combination chemotherapy. Ninety percent of the patients achieved CR in the first round of chemotherapy; 82% and 78% also achieved CR in the second and third rounds of chemotherapy.[280]

Multiday dosing of netupitant/palonosetron fixed combination in association with BEAM (carmustine, etoposide, cytarabine, and melphalan)/FEAM (fotemustine, etoposide, cytarabine, and melphalan) prior to HCT was evaluated in two phase 2 studies. In a single-center, single-arm, phase 2 study investigating the efficacy and safety of NEPA and dexamethasone for the prevention of CINV in patients receiving BEAM therapy prior to HCT (N = 43), this antiemetic combination was deemed safe and effective.[281] A CR was achieved in 3 patients (30%) with an average of 10.6 (standard deviation [SD], 0.95) emesis-free days over an 11-day observation period, and few 1–2 grade adverse events (eg, headache, nausea, diarrhea). These findings are consistent with an Italian phase 2a multicenter study on NEPA without dexamethasone in patients with relapsed/refractory aggressive non-Hodgkin lymphoma (N = 70).[282] This study reported an overall CR value of 87.1% and complete control percentage of 85.7%. Less than 10% of patients experienced moderate or severe episodes of nausea and only one adverse event was reported (constipation).

An open-label, randomized controlled study compared the efficacy and safety of fosaprepitant (150 mg) given on days 1 to 3 (prolonged group) versus on day 1 (regular group) in patients receiving HECs.[283] Patients were randomly assigned to the prolonged group (N = 77) and regular group (N = 79). In the prolonged versus regular group, a statistically significant lower incidence of nausea (6.17% vs. 12.66%, *P* = .0056) and slightly lower incidence of grade 1 vomiting (1.62% vs. 3.80%, *P* = .0953) in the delayed phase was seen. This study concluded that prolonged use of fosaprepitant was safe and that the two groups did not have significant differences in adverse events within the delayed phase. NK1 antagonists may be used for multiday anticancer therapy likely to be moderately or highly emetogenic and associated with significant risk for delayed nausea and emesis. As per the labeled indication, 125 mg of PO aprepitant should be administered 1 hour prior to anticancer therapy on day 1, along with a 5-HT3 antagonist and dexamethasone. Aprepitant 80 mg PO should be administered daily on days 2 to 3 after the start of anticancer therapy along with dexamethasone.[263] Repeated dosing of PO aprepitant over multiple cycles of cisplatin-based therapy appears to be feasible and well tolerated. Importantly, protection from emesis and from significant nausea was maintained during the subsequent cycles of emetogenic anticancer therapy.[263,277] Based on smaller studies, aprepitant 80 mg PO may be safely administered beyond day 3 of initiating anticancer therapy.[133,278]

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Alternatively, for HEC regimens, aprepitant injectable emulsion 130 mg IV or fosaprepitant 150 mg IV with dexamethasone may be given on day 1, with no need for PO aprepitant on days 2 to 3, with recommended dosing of dexamethasone on days 2 to 4. Data are not available for repeat dosing of aprepitant injectable emulsion, or PO rolapitant.

Atypical Antipsychotics

If olanzapine is being used prophylactically as part of the antiemetic regimen, the Panel recommends that it may be used once daily (prior to chemotherapy or at bedtime) and continued for 2 to 3 days after chemotherapy for regimens that are likely to cause significant delayed emesis.[284]

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Summary

The NCCN Guidelines for Antiemesis provide an overview of the principles for preventing or substantially decreasing anticancer agent–induced or RT-induced nausea and/or vomiting and provide recommendations for prophylactic antiemetic regimens based on the emetogenic potential of anticancer agents. Prophylactic antiemetic regimens are recommended for patients who will receive emetogenic anticancer agents, because it is harder to control nausea and/or vomiting once it has started. Although vomiting can often be prevented or substantially decreased by using prophylactic antiemetic regimens, nausea is harder to control. This Discussion text for antiemesis describes these updates in greater detail, in particular the clinical trial data and references that support the NCCN Panel's recommendations in the algorithm.

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Discussion update in progress

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